

TFPT — Changelog

All changes and new results, by date

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This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (vN), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the introduction document. Module numbers vN refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

2026-07-21 (XVIII) (v508 — WP5e- γ of CELEST.SEAM.01 “the sphere-axion pairing check” promoted as CELEST.WP5E.GAMMA.01, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are EXACTLY $\text{span}\{s_5, s_3\}$ (dim 2) and the quartics EXACTLY $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into $\text{span}\{P_1, P_2, P_3\}$ — zero T_5/T_3 content, so Φ_{T_3} annihilates EVERY exchange image while $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$); the strict two-index $\mathbb{Z}_4$ rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with $\text{rank}([M | A_{\text{fix}}]) = 3$ and the annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$; side discovery $K^{(0)} = -15 K^{(2)}$; naturalness DISSOLVES (ch₂-natural and AB-weight couplings both certify (0,0) vs required (32,72) — scale-independent); controls: $SO(16)$ has no sphere partners AND uncanceled T_5 , D_8 no T_3 structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the $32 T_3$ residual MUST be carried by twisted BCOV contact terms (δ_1 binary: coefficient exactly (9, −30, −15, 0, 32), computed not fitted; escape δ_2 : the full-tensor ledger); Wolfram extension 444 $\rightarrow$ 450 (verified engine run); *NO marker moves — CELEST.SEAM.01 stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, SEAM.EQUIV.01 stays [O]*)

- **v508_celestial_wp5e_gamma_pairing.py** (CELEST.WP5E.GAMMA.01, 27 checks, ~ 4 s; sympy/Fraction exact, no floats; deterministic). Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on $H^2(\text{ALE})$, v505 S5.8) cancel the rigid $32 T_3$ residual of the Atiyah–Bott ledger ($A_{\text{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32T_3$, v505 S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem:* exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees (1,0,1)) and quartics dim 5 (bidegrees (2,0,1,0,2)) — the vertex basis is COMPLETE; *product theorem* (the master kill): $(as_5 + bs_3)(cs_5 + ds_3) = ac P_1 + (ad + bc)P_2 + bd P_3$ identically — every exchange image (any axion count, any rule, any couplings) is annihilated by Φ_{T_3} , and $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$. *Selection enumeration:* the strict two-index reading collapses ($C_{j,j'} \neq 0$ only for $j' = -j$, forcing $m = 0$: 16 raw triples, 4 nonzero, all $m = 0$); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$, $E_{00} = 225 E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15 K^{(2)}$ (the even-sector quadratics are PARALLEL — this collapse drives the Φ_P obstruction). *Unsolvability with certificate (the core):* M is 5×2 , rank 2, kernel 0;

$\text{rank}([M | A_{\text{fix}}]) = 3$; annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ — two independent obstructions ($32 T_3$ structural, rule-independent; $\Phi_P = 72$ selection-rule-driven); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are RIGIDLY SILENT); $\Phi_P(Q^{(0)}) = 0$ (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the P -block ($\det = -64$) but $T_5 = T_3 = 0$ persists. *Naturalness resolution*: ch_2 -natural couplings $(9/64, 1/4)$ give $(25/4, -11/2, 97/4, 0, 0)$ and AB-weight couplings $(1/2, 1/4)$ give $(12, -40, 76, 0, 0)$ — BOTH with certificate $(0, 0)$ vs required $(32, 72)$: the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”). *Negative controls*: invariant-only = v505 S3.9 replicated + sharpened; flipped rule $m+j-j'$: Φ_{T_3} kill flip-invariant, Φ_P rule-dependent; $SO(16)$: odd classes EMPTY and AB sum $(15, -18, -15, -4, 40)$ — T_5 does not even cancel (E_8 doubly special); D_8 : $16 T_8 + 12 s_8^2$, no T_3 structure — the question cannot even be posed. Number fences [C] only: $32 = (h, h) + (h', h') = 20 + 12$, $72 = 2\tilde{\lambda}_{e_8}^2$.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.GAMMA.01 ([E] vertex space/selection/certificate + [C] fences + [O] contact terms); CELEST.SEAM.01 stays Open with the WP5e targets UPDATED (“the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms (δ_1 , binary test) and the full-tensor ledger (δ_2)”). Papers: `tfpt_research_contracts` WP5e section extended by the γ block (γ done — negative result with certificate; the $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$ roadmap with success/kill criteria); `tfpt_3` celestial section becomes a TWELVE-step story with Step 12 “the exchange no-go and the contact-term criterion” and the “What remains” paragraph updated. Wolfram: six mirrors (the vertex-space dims $2/5$ by generator-wise nullspace arithmetic; the twisted quadratics with $K^{(0)} = -15 K^{(2)}$; the product theorem, symbolic; the rank-3 certificate $(0, 32, 72)$ with channels and naturalness images; the $SO(16)$ AB sum $(15, -18, -15, -4, 40)$; the D_8 quartic $16 T_8 + 12 s_8^2$) verified with the active engine the same day ($444 \rightarrow 450$, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag: v508 on the QFT/celestial node; registry + `run_all` (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

2026-07-21 (XVII) (v507 — the tautology attack on the v506 alignment bit, executed and refuted, promoted as SEAM.BIT.ORIGIN.01: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods $\Lambda/2\Lambda$ of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$ the free involutions are exactly the three σ_c : the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic λ): a marks-preserving root of σ_c exists $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$ (one per c ; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit $\iff j = 1728 \iff \tau = i$; the EQUIVALENCE TETRAD “Uhr existiert $\iff (\tau=i$ AND deck = CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift NONSPLIT”; the FERMION DICHOTOMY (the new physical face): central $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ nonsplit vs edge $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$ SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension $438 \rightarrow 444$ (verified engine run); *NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01 text precised only)*

- `v507_seam_bit_origin.py` (SEAM.BIT.ORIGIN.01, 26 checks, ~ 2.6 s; Part A sympy-exact torus/Möbius arithmetic + Part B exact $\text{Cl}(16)$ Fractions; deterministic). Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double $T_\tau^2 \rightarrow S^2$, v214/v260), so the common origin would force the central alignment automatically and the v506 bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of $w \mapsto -w$ on $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =$ the half-periods $\Lambda/2\Lambda$ (a $\frac{1}{3}$ -coefficient never qualifies); t_c descends iff $2c \in \Lambda$ — exactly the three half-period translations (control $c = \frac{1}{3}$ fails);

the descended maps are the EXPLICIT rational involutions $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$ with the exact curve-lift identity $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$, ψ_i RATIONAL (the $\wp$ -addition formula as pure algebra), closing to Klein V_4 ; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic V_4 (4, 3, 3), harmonic D_4 (8, 5, 3), hexagonal A_4 (12, 3, 3)) are EXACTLY the three $\sigma_c \Rightarrow$ every mark-free collar deck (v216) IS a half-period translation. *Core solve (symbolic λ):* σ_c admits a marks-preserving PGL_2 root $\iff \lambda = -1$ (σ_1), 2 (σ_2), $\frac{1}{2}$ (σ_3) — per half-period exactly ONE solution, intrinsically “the deck pairs separate harmonically” (pair cross-ratio -1); generic modulus: V_4 abelian, ALL σ_c central yet clockless (“central alone is empty” — the bit is the CONJUNCTION $D_4 \wedge$ central); at $\tau = i$ the stabiliser jumps to D_4 with σ_1 central (2 roots = the clock pair), σ_2/σ_3 edge-type (0 roots); the v506 seam deck lands on the central $\sigma_1 = -1/x$ under all 8 mark bijections; CM dictionary: mult-by- i fixes exactly $c^* = (1+\tau)/2$ (mult-by- ρ 3-cycles all three), the clock lift needs the CM unit i ($\psi = 2i/(x+1)^2$, t^2+1 irreducible over $\mathbb{Q}$) while half-period decks lift rationally. *Equivalence tetrad + circularity audit:* clock exists $\iff (\tau=i \text{ AND deck} = \text{the CM-fixed half-period}) \iff (\text{Stab} = D_4 \text{ AND deck central}) \iff \text{deck pairs harmonic} \iff \text{NS Fock lift nonsplit}$; the v493 converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B):* central arrangement (shift by 8): $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4-j$, fixed sites = non-marks = the silver $\{0, \infty\}$ between neighbour marks): the implementer EXISTS but $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2+1)^8$ vs $(\lambda^2-1)^8$; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement:* $\{\pm 1, \pm(3+2\sqrt{2})\} = “\tau = i \text{ with deck} = \text{a non-CM-fixed half-period translation}”$ (all 8 bijections $\rightarrow \sigma_2/\sigma_3$, never σ_1) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim):* the alignment bit remains a carrier datum; “no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION”. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.ORIGIN.01 ([E] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: SEAM.CLOCK.RIGIDITY.01 (“the bit is now fully characterised as a tetrad ... and carries a physical face: the non-split NS Fock extension class — v507”), QGEO.REALIZE.01 (“the class of the deck IS derived (half-period translation, v507); only its position stays the input”, stays [C]/[O]), P2.TYPING.01 (origin cross-reference on the common R1 bit). Papers: tfpt_3 celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, introduction/tfpt_1/tfpt_2/origin_theory one-sentence updates where v506 is cited. Wolfram: six mirrors (half-period descent $2c \in \Lambda$ with the rational σ_i lift identity; the core solve $\lambda \in \{-1, 2, \frac{1}{2}\}$; the stabiliser counts (4, 3, 3)/(8, 5, 3)/(12, 3, 3) via the complete 24-triple enumeration; the CM fixed point c^* + the lemniscatic $\psi = 2i/(x+1)^2$; the Cl(16) dichotomy 256γ -monomial vs scalar 2^7 on explicit 256×256 Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day (438 $\rightarrow$ 444, ALL WOLFRAM EXTENSION CHECKS PASSED); the v493 audit direction and the 28/21 arrangement census stay Python-side (flagged). VerificationDag: v507 on the P2/carrier and QGEO nodes; registry + run_all (501 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XVI) (v506 — seam clock rigidity, Route B of the v503 classification, promoted as SEAM.CLOCK.RIGIDITY.01: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck $z \mapsto -z$ has EXACTLY TWO square roots $z \mapsto \pm iz$ in $\mathrm{PGL}_2(\mathbb{C})$, both *automatically* order 4, and a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark- D_4 (counterexample $\{\pm 1, \pm(3+2\sqrt{2})\}$: harmonic with full D_4 but deck non-central $\Rightarrow$ 0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 — $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$, i.e. $U^2 = (-1)^F$ EXACTLY, a nonsplit $\mathbb{Z}_4$, with the $\mathbb{Z}_8$ tower $V^2 \propto U$, $V^4 \propto (-1)^F$ ($8 = 2|\mu_4|$ fermionically explained) and the R-sector control split ($U_R^2 = +1$); the one [C] rest: “the collar deck is the central involution of the mark- D_4 ” — the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment; Wolfram extension 434 $\rightarrow$ 438 (verified engine run); *NO marker moves* — QGEO.REALIZE.01 stays [C]/[O] (text precised), P2.TYPING.01 R1 mapped onto the same bit)

- **v506_seam_clock_rigidity.py** (SEAM.CLOCK.RIGIDITY.01, 34 checks, ~ 3.5 s; Part A sympy-exact Möbius/divisor arithmetic + Part B exact $\mathrm{Cl}(16)$ Fractions; deterministic). v503 localised the one remaining identification input (“why does the raw seam carry the order-4 clock”, QGEO-R1 $\equiv$ P2-R1); v506 reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give $g^2 = \text{deck} \Rightarrow \mathrm{tr} g \neq 0$ forces g diagonal with $(a/d)^2 = -1$, and $\mathrm{tr} g = 0$ forces $\det g = 0$ (impossible) — exactly two roots $z \mapsto \pm iz$, both projective order 4, mutually inverse ($\mathrm{Aut}(\mathbb{Z}_4)$), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set $\{u, -u, w, -w\}$ admits a marks-preserving root $\iff w = \pm iu \iff$ marks = one μ_4 orbit $\Rightarrow$ cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock², only the scale free; group form: $|\mathrm{Stab}(\mu_4)| = 8 = 2|\mu_4|$ with order profile $\{1:1, 2:5, 4:2\} = D_4$, center = $\{\mathrm{id}, \text{deck}\}$, exactly 2 order-4 elements = the clocks — “deck central in the mark- D_4 ” $\iff$ “clock exists”; the counterexample $\{\pm 1, \pm(3+2\sqrt{2})\}$ is deck-free AND harmonic ($j = 1728$, cross-ratio $\frac{1}{2}$) with full D_4 , but the deck sits *non-centrally* (the central involution crosses the $\pm$ pairs) $\Rightarrow$ 0 roots; the RP filter passes BOTH roots (the $\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity is irreducible, v492 NC3). *B3/B4 (n-arithmetic + controls)*: free orbit $\Rightarrow n \mid 4$; transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$ (kernel argument through S_4); $n = 3$ fixes a mark; $n = 6$ dies independently at the cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); $n = 8$ impossible on 4 marks; hexagonal stabiliser A_4 (0 order-4 elements, squares only orders $\{1, 3\}$); generic stabiliser V_4 (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. *Theorem B (Fock side, the core result)*: S^2 has $b_1 = 0 \Rightarrow$ unique spin structure $\Rightarrow$ NS/bounding on every circle (established P1 datum); one-particle: $D^2 = -1$ (charpoly $(\lambda^2+1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4+1)^4$, ζ_8 spectrum); Fock (exact $\mathrm{Cl}(16)$): $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$ implements the deck on all 16 generators and $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ EXACTLY; $(cU)^2 = c^2(-1)^F$ is *never* 1 (center of $\mathrm{Cl}(16)$ = scalars): the $\mathbb{Z}_4$ lift is FORCED, $1 \rightarrow \mathbb{Z}_2^{(-1)^F} \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2^{\text{deck}} \rightarrow 1$ nonsplit; the quarter-shift lift V (unique even block implementer) has $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2|\mu_4|$ — the v492 $\mathbb{Z}_8$ spin bridge realised on the 16-Majorana Fock space; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_\nu = 1$ at $\nu = 16$; robustness: the forcing already holds at 8 sites ($16 =$ the carrier count $2^{g_{\text{car}}-1}$, not a tuning). *The one [C] rest (verbatim)*: “the collar deck is the central involution of the mark- D_4 ” ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim.
- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.CLOCK.RIGIDITY.01 ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: QGEO.REALIZE.01 (“the rest is now ONE bit: the central position of the deck in the mark- D_4 ; order/uniqueness/orbit/ $\tau=i$ follow — v506”, stays [C]/[O]), P2.TYPING.01 (R1 mapped onto the same bit, fermionic halving), QGEO.EMERGE.LIGHT.01 context sharpened.

Papers: `tfpt_3` celestial Step 1 (fermionic $U^2 = (-1)^F$ sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, introduction/`tfpt_1/tfpt_2/origin_theory` one-sentence updates at the v499/v503 citation sites. Wolfram: four mirrors (the complete square-root solve; D_4 center + order-4 counts + the silver counterexample; the $Cl(16)$ identity $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$ on explicit 256×256 Jordan–Wigner matrices + the split R control; the n -arithmetic + generic kill) verified with the active engine the same day ($434 \rightarrow 438$, ALL WOLFRAM EXTENSION CHECKS PASSED); the 24-triple stabiliser enumeration and the A_4 profile stay Python-side (flagged). VerificationTag: v506 on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (v505 — WP5e- β of CELEST.SEAM.01 “the equivariant anomaly ledger on twistor space” promoted as CELEST.WP5E.BETA.01: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators $(2, 4, 2)$ with Dedekind sum $\frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$, equivariant characters $(248, 0, -8, 0)$ by two routes, invariant Okubo $36\langle x, x \rangle^2$ with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual $32T_3$, the index bridge $f(m) = \text{ch}_2(T_m)$ exact with glue defect -78 both routes, the level dials $k = 1$ geometric (current count $240/0/0/0$), and the honest E3 REFUTATION: a_0 fills the GRAVITON slot $O(2)$, NOT the axion slot $O(-2)$ — the three sphere classes fill the three twisted axion slots instead; the preregistered kill (“inflow demands a level $\neq 1$ ”) does NOT fire on the equivariant skeleton; WP5e: α (CFT side, v502) + β (equivariant twistor skeleton, v505) executed, the global BCOV quantisation stays [O]; Wolfram extension $429 \rightarrow 434$ (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- `v505_celestial_wp5e_beta_equivariant_ledger.py` (CELEST.WP5E.BETA.01, 47 checks, ~ 1.3 s; `sympy/Fraction` exact, no floats; deterministic). The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on $\mathbb{C}^2/\mathbb{Z}_4$ with the E_8 μ_4 -glue monodromy. *AB ledger*: $\det(1 - g_j^{-1}) = (2, 4, 2)$; $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$; $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$ two routes; invariant average $60 = \text{carrier}$; Frobenius 61568 ; $\text{tr}_{g_2} = -8 = 120 - 128$. *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x, x \rangle^2$, $36 = \tilde{\chi}_{e_8}^2$ — v495 re-derived through the glue grading); the twisted sectors do NOT reduce (T_5 : $(-8, 32, -8)$, T_3 : $(48, -64, 48)$); the AB sum cancels the D_5 quartic exactly and leaves the rigid $32T_3$ (the only quartic-free weighting is the $\mathbb{Z}_4$ projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities ($\text{Tr}_{120}, \text{Tr}_{128}, \text{Tr}_{E_8} = 9(\text{tr}^2)^2$) exact. *Index bridge*: $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACT (E1 = E2 arithmetic); glue ch_2 defect -78 both routes (classes $(-24, -30, -24)$); Killing sums $(320, 320, 240, 320)$, total $1200 = 2h^\vee(h, h)$. $k = 1$ *geometric*: lattice current count 240 at $k=1$, exactly 0 at $k = 2, 3, 4$; embedding residual $47k^2 + 219k - 266 = (0, 360, 814, 1362)$; integrality alone fixes nothing (honest, as v502); one scale $\Rightarrow$ one level (v493 periods). *E3 refutation (prominent)*: $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, NOT the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$ as a [C] coincidence) — “the theory brings its GS axion as a_0 ” is REFUTED; instead the three $H^2(\text{ALE})$ classes carry exactly the Coxeter eigencharacters $\{i, -1, -i\}$ = the three twisted slots (bijection); slot count $1+3 = 4 = |\mu_4|$ soft. *Negative controls*: $\text{diag}(i, i)$ breaks the ledger four ways ($\sum 1/\det = \frac{1}{4}$, ω -weight -1 , 5 generators); $SO(16)$ glue: characters $(120, 0, +120, 0)$, defect $-30 \neq -78$, bulk Okubo fails (independent 4th Casimir); $k = 2$: current count 0 , c -prefactor $-31/48 \neq -\frac{1}{3}$. [C] dictionaries ($\text{CS}(\rho) \bmod 1 = \text{discriminant weight}$; $32 = (h, h) + (h', h')$; $-30 = -h^\vee$; Ω -contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the $O(-2)$ bulk axion, CPS level-from-flux, the pairing $32T_3 \leftrightarrow$ sphere axions. VERDICT B; the kill test does not fire on

the equivariant skeleton.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.BETA.01; CELEST.SEAM.01 updated to “WP5e: α (CFT side, v502) + β (equivariant twistor skeleton, v505) executed; the global BCOV quantisation stays [O] — with the rigid $32T_3$ twisted residual and the sphere-axion pairing as the named remaining targets” at *unchanged* Open status. Papers: contracts WP5e bullet $\rightarrow$ “ $\alpha + \beta$ done” with the a_0 refutation as an honest finding (kill-test and honest-success paragraphs consistent); **tfpt_3** celestial section becomes the *eleven*-step story (new Step 11 “the equivariant anomaly ledger on twistor space”, “what remains” = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge + -78 defect; the per-sector Okubo sums over the 240 glue roots with the $32T_3$ residual; the lattice current counts + embedding residuals; the axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day ($429 \rightarrow 434$, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag qft node gains v505; registry + **run_all** (499 modules at this step; 500 with v506, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

2026-07-21 (XIV) (v504 — WP5d- β of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ($\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$), GF2 spans full, rank $32/32$; disjoint exactly index $2 =$ the v501 $\ln 2$ bit), the entropic touching defect BOUNDED $< \ln 2$ with the Ising $\frac{1}{4}$ -exponent approach ($p = 0.2444$) while the preregistered $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate $\pi K(1-x)/K(x)$ ($1.3\text{--}2.0\%$) with EXACT orbifold inheritance ($C \rightarrow -C$ + the $\Lambda^2 C$ compound — Longo heredity), and Pimsner–Popa $E(a) - a/2 = PaP/2$ identically ($\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$, $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$ — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension $424 \rightarrow 429$ (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)

- **v504_celestial_wp5d_klm_completeness.py** (CELEST.WP5DB.01, 37 checks, ~ 12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval μ -index; v501 (WP5d- α) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ EXACTLY (GF2 spans full $64/64$, $256/256$, $512/512$, $1024/1024$; literal matrix rank $32/32$); disjoint intervals give exactly HALF (rank $16/32$, index 2 — the missing sector odd $\otimes$ odd is the v501 $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps $2/10/52$, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law ($(c/3) \ln 2$ increments, 0.115525 , dev 1.2×10^{-6} — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the $\mathbb{Z}_2$ /orbifold deficit rises $0.4045 \rightarrow 0.5899$ ($N = 64 \rightarrow 4096$), stays $< \ln 2$, and $(\ln 2 - \Delta_2) \sim N^{-p}$ with $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder exponent); split-point scan reflection-symmetric to 3.9×10^{-14} ; ED vN same direction; gap continuity to the v501 $x \rightarrow 1$ endpoint; honest note: “defect $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index (Longo–Xu) is the correct statement; the $U(1)$ control bursts $\ln 2$ from $L = 128$ and grows as $\frac{1}{2} \ln \text{Var } Q_A$

(Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$). *Split/nuclearity*: coupling-block σ_k fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at $d = 16/64/256/512$); $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*: P_A flips $C \rightarrow -C$ (σ_k identical, 8.3×10^{-17}) and the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ (singular values $\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*: $E(a) - a/2 = PaP/2$ identically; attainment in integers (all 16384 (even, odd) monomial pairs at $N = 8$: 8192 extremal with $M^2 = 4$, $\text{Tr } M = 0$, $PMP = -M$, 0 violations; 2048/2048 at $N = 12$; pencil $\min \lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$; index consistency $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m+1) \rightarrow 0$. *Controls + honesty*: dimerised phase trivial (drift 3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit exactly 0; $\mathbb{Z}_2$ finite-size honesty (14.9% below $\ln 2$ at $N = 4096$ — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu’s* (cited, not claimed), the concrete seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d: α (μ -index) + β (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (Xu’s theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet $\rightarrow$ “both stages done” with the β -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); **tfpt_3** celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); **tfpt_4** keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group $\lambda_{E_4} = \frac{1}{4}$ + index consistency; the Λ^2/Minors compound with Cauchy–Binet and $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$; the $U(1)$ dephasing $\lambda = 1/(m+1)$ family) verified with the active engine the same day (424 $\rightarrow$ 429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + **run_all** (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field’s global $\det'$ winner is the *hexagonal* torus, $\tau=i$ is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff* $\tau=i$, an isolated zero); and the residual *converges*: QGEO-R1 $\equiv$ P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420 $\rightarrow$ 424 (verified engine run); *NO marker moves* — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays a declared axiom)

- **v503_qgeo_emergence_light.py** (QGEO.EMERGE.LIGHT.01, 17 checks; mpmath 40 digits + v280-style lattice DtN + sympy exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*: $\log F = \log(\tau_2|\eta|^4)$ (Osgood–Phillips–Sarnak) is exactly S -invariant (dev $< 10^{-35}$), so $t=1$ is symmetry-forced critical on the rectangular family $\tau = it$; there $g''(1) = -1.298 < 0$ (strict maximum) but the τ_1 -direction has $h''(0) = +0.298 > 0$: $\tau=i$ is a *saddle* of the full moduli space; the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ at 40 digits, generic $0.163 \neq 0$); the *global* $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to 8×10^{-15} exactly at $\tau=i$ and rises strictly monotonically

both ways (min 8.4×10^{-2} away) — an isolated zero; $R^2 = \mathbb{Z}_2$ exact as permutation matrices (the v492 $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t=1$. *The $\mathbb{Z}_2$ chain is modulus-blind*: 4 marks + $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (v216 confirmed without mark input); pillowcase $\det' = \frac{1}{2}$ torus $\det'$ exactly ($\pm$ -pair structure) — only the order-4 clock distinguishes $\tau=i$; $\text{Aut}(E_\rho) = \mathbb{Z}_6$ has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ($j = 1728$ identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3 + \lambda^2 + \lambda + 1$, order 4 = $|\mu_4| = N_{\text{fam}} + 1$) = the P2 cusp-class carrier datum of v499 R1 $\Rightarrow$ QGEO-R1 $\equiv$ P2-R1, *one common residual*; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input $[[\mathbf{C}]]$, rest = its dynamical realisation $[\mathbf{O}]$; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data $\Rightarrow$ order-4 clock” is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker $[\mathbf{C}]/[\mathbf{O}]$; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt_3 Step 2 (light-converse sentence), origin_theory necessity paragraph, introduction/tfpt_1/tfpt_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities $E_2(i) = 3/\pi$, $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$ exact, $j/\text{DedekindEta}$ special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock $\Rightarrow$ square + Coxeter) verified with the active engine the same day ($420 \rightarrow 424$, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run_all (497 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XII) (v502 — WP5e- α of CELEST.SEAM.01 “prefactor and level book-keeping” promoted as CELEST.WP5E.ALPHA.01: the $q^{-1/3}$ prefactor of E_4/η^8 IS exact μ_4 vacuum-energy bookkeeping (the clock is inner $\Rightarrow$ shift orbifold $\theta = 0^8$, common $-c/24 = -\frac{1}{3}$; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and $k = 1$ is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial $31124 - 27000 = 4124$); honest sharpening: glue- h integrality holds for ALL k and fixes nothing; the BCOV/Costello–Li twistor inflow stays $[\mathbf{O}]$ — WP5e is subdivided (α done, WP5e proper open); Wolfram extension $416 \rightarrow 420$ (verified engine run); SEAM.EQUIV.01 stays $[\mathbf{O}]$, no marker moves)

- **v502_celestial_wp5e_prefactor_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor*: Hurwitz- ζ exact ($\zeta(-1, \theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$; Ising shift $\frac{1}{16}$ per Majorana); the clock is *inner* ($h = (2, 2, 2, 2; 0^4)$ and $h' = (0^5; 1, 1, 1, -3)$ both read the class mod 4 on all 240 roots; $(h, h) = 20$, $(h', h') = 12$, sum 32) $\Rightarrow$ twist $\theta = 0^8$ in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so $E_{\text{osc}} = -\frac{1}{3} = -c/24$ is sector-independent ($c = 8 = \text{rank} = \text{Sugawara } 248/31$; one common η^8); sector ground exponents $-\frac{1}{3} + (0, 1, 1, 1)$; *discriminant form = Casimir energy* two ways (spectral flow $j^2(h, h)/32 \bmod 1$; free fermions exact: $D_5 = 10$ Majorana $\rightarrow \frac{5}{8}$, $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$, glue diagonal = $16 = 2^{g_{\text{car}}-1}$ = the WP5d seam carrier $\rightarrow 1$); the rotation reading *fails* (deck $\frac{3}{16} \neq \frac{3}{8}$ — “clock $\neq$ deck” at the Casimir level); the four sector characters sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. *Part B — the level*: $k = 1$ forced three independent ways — (i) the current condition $h(J) = k = 1$ (adjoint closure $60+64+60+64 = 248$ at spin 1); (ii) the conformal embedding ($47k^2 + 219k - 266 = 47(k-1)(k+266/47) = 0$; D_8 : $128k(1-k) = 0$); (iii) $c = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor); plus the v498 singular-vector dial ($\|(E_{-1}^\theta)^m 0\|^2 =$

$m!k(k-1)\cdots(k-m+1)$, singular vector at grade $k+1$; $31124 - 27000 = 4124$ at $k=1$ only, $\langle s|s \rangle = +4$ at $k=2$). *Honest sharpening (mandatory)*: glue- h integrality $h(J^j; k) = k(0, 1, 1, 1)$ is integer for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls*: $D_8/SO(16)_1$ (same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue- h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$ — the common prefactor breaks); wrong $k \in \{2, 3, 4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s|s \rangle = 4/12/24$; $31124 \neq 4124$). **[C]** Kac generation for $k \geq 3$ ($k = 1, 2$ machine-anchored by WP5b/WP5c). **[O]** the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{P}T$, the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level $\neq 1$ ”) is not triggered on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- α (done) **[E]/[C]**: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper **[O]**: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- α landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status $\rightarrow$ “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays **[O]**”. Wolfram: four exact mirrors (Hurwitz- ζ vacuum energies; discriminant = Casimir with the deck failure; the k -fixing equations incl. the “fixes nothing” integrality table; the character sums + D_8 control) verified with the active engine the same day ($416 \rightarrow 420$, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters ($116/116 + 424/424$ after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (XI) (v501 — WP5d- α of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the $\ln 2$ offset at machine precision, $\mu_{\text{gauged}} = 4 =$ the v490 census, and the condensation arithmetic $16 \rightarrow 4 \rightarrow 1$ closes with $\mu = 1$; the preregistered KILL (“ μ -offset $\neq 0$ after condensation”) does NOT fire; contract now “WP1–WP5d executed; *WP5e the single remaining milestone*”; Wolfram extension $412 \rightarrow 416$ (verified engine run); SEAM.EQUIV.01 stays **[O], no marker moves)**

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator — $(E_8)_1 \Rightarrow \mu=1$ (Haag-dual two-interval net, one sector) vs $SO(16)_1 \Rightarrow \mu=4$ — executed at the α (lattice-index) stage on the v367 seam edge (16 critical Majorana layers, $c_- = 8$, v450-style ring; Peschel-exact, every Gaussian formula ED-validated to $\leq 1.5 \times 10^{-15}$; the superselection identity $[\rho, P_{AB}] = 0$ machine-exact). *The $\mu = 1$ reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* — c fit 0.5000, $\max |F(x)| = 5.97 \times 10^{-5}$ at $N=512$, falling monotonically to 1.3×10^{-6} at $N=1024$ (Casini–Huerta extensivity as a lattice witness). *The μ -offset*: Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at $N=512$; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = 2.000000$, $\mu_{\text{gauged}} = 4.000000 =$ the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs $16 \ln 2$ (the offset measures $\ln[\text{index}]$, not a blob). *Haag-duality indicator*: fermionic $S(E) = S(E')$ to 8.2×10^{-12} ; the orbifold *breaks* complementarity ($D(E) - D(E') = 0.427$ at $a=64$); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with max 1.2637 at the self-dual $x = \frac{1}{2}$ — a *double-limit* statement

(the first naive fixed- x check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*: $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$; KLM/Longo–Rehren $16/4^2 = 1$ and $16/2^2 = 4 \rightarrow 4/2^2 = 1$ (both routes exact); $\Sigma d_i^2 = (4, 4, 1)$; $\theta_v = 1$ exactly at $\nu = 2c_- = 16$, rivals $\neq 1$ — so $\mu = 1$ after condensation ([**C**] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*: $\nu=1$ single Majorana (offset non-removable, $\theta_v = e^{i\pi/8} \neq 1$ — the discriminator has teeth); trivial phase (all $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset exactly 0). [**O**] WP5d- β (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet $\rightarrow$ “ α stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor uplift) is the remaining open piece”. `tfpt_3` celestial section: new Step 8 (one paragraph, conservative); `tfpt_4` keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + [**C**] dictionary); CELEST.SEAM.01 status $\rightarrow$ “WP1–WP5d executed; WP5e [**O**] (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients; Σd^2 + index-reading arithmetic; 16-fold-way θ_v) verified with the active engine the same day (412 $\rightarrow$ 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). `wolfram/README.md`, GATE.WOLFRAM.02, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + `run_all` (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family ω_w exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 = $V(2\theta)$ weight by weight; $|s\rangle$ IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 $\rightarrow$ 412 (verified engine run); SEAM.EQUIV.01 stays [O**], no marker moves)**

- `v500_celestial_wp5c_gns_limit_state.py` (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*. ω_w is quasi-free: loop sector = the affine $k=1$ vacuum n -point functions via the *machine-determined* compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$ (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x -adic). *Convergence sharp*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N+1$ ($N=8$), not yet at $w = N$ — the WP5a threshold $w = n+1$ holds *identically at the state level*; positivity of every ω_w checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the E_8 Cartan matrix, $\det 1$), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel = $V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_j F_i|s\rangle$ norm 0. μ_4 -equivariance: $\omega_w \circ \text{clock} = \omega_w$ for all w ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) = Θ_{C_j}/η^8 at q^2 — the two-routes identity at the *state* level. *Resolution of WP5b S4.10*: $|s\rangle$ is the zero vector of $\text{GNS}(\omega_\infty)$ — the twisted two- C_1 -mode floor (6 quarters = $q^{3/2}$) concerns only pre-limit objects (GNS has no state there, $6 \bmod 4 \neq 0$). *Negative controls*: $k=2$ ($\langle s|s\rangle = +4$, dominant blocks full rank — no ideal in the kernel); $k=0$ (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); D_8 (rank 2076 = 7380 – 5304, kernel = $V(2\theta_{D_8})$ PSD, but $h = (0, \frac{1}{2}, 1, 1)$: one block of four — one-block completeness stays E_8/μ_4 -specific); wrong family $E'_w = mw+r$

(the 248 layer erased, $710955 \neq 248$); no damping ($897266 \neq 248$); *CCR obstruction*: at fixed radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — no w -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. [C] quasi-free extension to the full $*$ -algebra; Kac positivity beyond level 2. [O] local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5c bullet $\rightarrow$ done with the core numbers (state-level threshold $w = N+1$, the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing `v500_celestial_wp5c_gns_limit_state.py`; the WP5b handover marked answered. `tfpt_3` celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity [E] + [C] identifications + [O] net questions). Wolfram: four exact mirrors (level-2 block census from the root data; rank/kernel arithmetic with $\dim V(\omega) = 3875$ by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial + D_8 census) verified with the active engine the same day ($408 \rightarrow 412$); the exact Gram/PSD computation stays Python-side (flagged). `VerificationDag` qft node gains v500; registry + `run_all` (494 modules at this step, suite green). SEAM.EQUIV.01 untouched; no marker moves.

2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor $a = (1, 1, 2)$ typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the `tfpt_2` “balanced” selection typed as stability; Wolfram extension $403 \rightarrow 408$ (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])

- `v499_p2_weights_deligne_bg.py` (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank $3 = \# \text{marks} - 1$, v137/v228; cusp-class local monodromy $\lambda^3 - 1$ and exact irreducibility $|\langle U, M_0 \rangle| = 24 = |S_4|$, $\sum_g |\chi|^2 = 24$, v117; unitarity (U)), the Deligne canonical extension E of the flavor connection satisfies: $\deg E = -4$ as a theorem (residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ the unique lift to $[0, 1)$, trace 1 per mark; par-deg 0; ∞ apparent, monodromy product = 1 exactly); $h^0(E) = 0$ as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary $h^0=0 \Leftrightarrow a_i \geq 1$); integrality = Birkhoff–Grothendieck. Hence $\{a\} = \{1, 1, 2\}$ unique and $g_{\text{car}} = e_2 = 5$. *Selector typed*: the Schur–Horn permutohedron has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the `tfpt_2` “Global computation” theorem) and $h^0=0$ kills $\{0, 2, 2\}$ — the “balanced/minimal-variance” heuristic is now stability/positivity. *The h^1 discrepancy, real and resolved*: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$, so “generations = H^1 of the rank-3 bundle” is FALSE (negative control: forcing it demands $\deg -6$ and loses uniqueness, $e_2 \in \{9, 11, 12\}$ never 5); correct two-object typing: $L = \mathcal{O}(-D)$ carries the generations ($h^1(L) = 3 = N_{\text{fam}}$, the v137 basis = $H^0(\Omega^1(\log D))$), E carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator $3 = N_{\text{fam}}$ forces the $\mathbb{Z}_3$ cusp cover $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$ (genus 2; NOT the pillowcase double, genus 1, $j = 1728$ — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$; the regular-representation model lands on $\{0, 2, 2\} =$ the unstable Schur–Horn companion ($h^0 = 1$): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity $\rightarrow$ 4/31 solutions; $\deg -3/-5$ breaks; n -mark scan $e_2 = (n-2)(n+1)/2 = 5$ ONLY at $n = 4$ (the value, not the uniqueness, is the $n=4$ content); MS-rational $e_2 = \frac{16}{3}, \frac{44}{9}$. *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s $\text{diag } A_0 = a/4$ input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499”

(v491 stays active, no supersede). Registry + `run_all` (493 modules, suite green).

- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control + h^1 discrepancy; Deligne residue-trace degree with the A_0^* witness char poly; Schur–Horn lattice points; Biswas $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double $j = 1728$; the n -mark formula), verified with the active engine the same day ($403 \rightarrow 408$, ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + `GATE.WOLFRAM.02` + `tfpt_safeguards` Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): `introduction` anchor paragraph, `tfpt_1` (5,3) keybox, `tfpt_2` H2 winbox *and* the “Global computation” theorem (the balanced selection typed as stability), `origin_theory` (5,3) keybox — all citing `v499_p2_weights_deligne_bg.py`; no status card changes. Website: `VerificationDag P2` node gains v499; generated indices via build; no marker moved anywhere.

2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)

- **tfpt_3_e8_audit_bootstrap: new section “The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow”.** The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible π ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the E_8 μ_4 -glue as a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space, $\text{clock}^2 = \text{deck}$ (spin $\mathbb{Z}_8$, $8 = 2|\mu_4|$) [E] (v492); (2) deformation — $XY = Z^4 + a_0$ pure scale, $j = 1728$ frozen, $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$ [E] (v493); (3) consistency honestly demoted — $(\kappa/c_3)^2 = 12$ exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the $(E_8)_1$ character is not a jet-grading conformal block (growth $n^{2/3}$ vs $n^{1/2}$) but the μ_4 period sum 248 holds exactly: a boundary-limit shadow of `SEAM.EQUIV.01/MMST` shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold $w = n+1$, $27000 = (h^\vee)^3$ derived, quotient = sector sum = 4124, $SO(16)_1$ contrast [E] (v497); (6) the deleting operator explicit — $|s\rangle = (E_{-1}^\theta)^2|0\rangle$, $J_1^a|s\rangle = 0$ on all 248 (tally 190/57/1), level dial $2(1-k)$, one-block closure E_8/μ_4 -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close `SEAM.EQUIV.01`; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. `tfpt_research_contracts`: a reader’s note at the head of the `CELEST.SEAM.01` contract cross-references the new Paper-3 section (the contract stays the full technical reference). `tfpt_4_frontier`: one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)

- **v498_celestial_wp5b_singular_vector.py** (CELEST.WP5B.01, 53 checks, exact integer/Fraction, deterministic, no floats). Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction*: $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ (weight 2θ , level $2 = 8$ quarter units $= q^2$, an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the $[e_\alpha, e_{-\alpha}]$ sign *forced* by the Jacobi identity (SGN = -1 , not chosen), the invariant form κ derived ($\kappa(\theta^\vee, \theta^\vee) = 2$), Jacobi exact on all 57600 $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496 θ -adjacent, all $N_{\alpha\beta} = \pm 1$. *Singularity (machine)*: $J_1^a|s\rangle = 0$ for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the *only* case that sees the level k); plus $J_2^a|s\rangle = 0$, $E_0^a|s\rangle = 0$, exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial*: the F_1^θ coefficient is $2(1-k) - 2/0/-2$ at $k = 0/1/2$: without the central extension the deletion operator does not exist. μ_4 data: glue class $j(\theta) = 1$ (machine-built h -adapted chamber, $\langle\theta, h\rangle = 5$, height $29 = h^\vee - 1$), clock phase $i^{2j} = -1$, class(2θ) = 2 sheet-even; the deleting θ - $\mathfrak{sl}_2$ crosses the sheet-odd classes (1, 3) while its square is sheet-even; cross-table θ -grading $\times$ glue class consistent (rows (1, 56, 126, 56, 1), columns (52, 64, 60, 64)); 8 quarters $= q^2$ via the per-period dictionary. *Level-2 match*: weight 2θ has multiplicity 1 in the level-2 Fock, so $U(\mathfrak{g})|s\rangle$ is the $27000 = (h^\vee)^3$; the direct $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2$: $|s\rangle$ is the kernel generator of $\text{Fock} \rightarrow E_4/\eta^8$. *Negative controls (honest separation)*: the level-1 current state $E_{-1}^\theta|0\rangle$ (one witness, $\rightarrow -k|0\rangle$ — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at $k=2$ the *cube* is (generic Kac $(E^\theta)^{k+1}$ mechanics, honestly typed); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion; $5304 \neq 14^3$) vs E_8 (comarks all ≥ 2 , glue weights (0, 1, 1, 1) integer) — the *one-block closure* is the E_8/μ_4 -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-period dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2 : exactly the WP5c GNS/limit-state question; WP5c–e stay open; SEAM.EQUIV.01 untouched.
- **Contract + surfaces synced. tfpt_research_contracts**: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial $2(1-k)$; $j(\theta) = 1$, clock phase -1 ; orbit BFS through depth 4; the D_8 contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c–e [O]”). **tfpt_3_e8_audit_bootstrap**: the celestial-contract paragraph extended by the WP5b sentence. **Ledger**: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c–e [O] (roadmap)”). **Wolfram**: 5 exact v498 mirrors added to **tfpt_readouts_extension.wl** and *verified* with the active engine the same day (398 → 403, ALL WOLFRAM EXTENSION CHECKS PASSED); **wolfram/README.md** + GATE.WOLFRAM.02 counters advanced; **tfpt_safeguards** Layer-6 count + website mirrors 116/116 + 403/403. **Website**: **papers.ts** contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c–e roadmap”), DAG **qft** node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b–e [O] (roadmap); Wolfram extension 394 → 398 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)

- **v497_celestial_wp5a_null_ideal.py** (CELEST.WP5A.01, **34 checks, exact integer/Fraction, no floats**). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m , radial number r ; $E_w = m + wr$ in quarter units $u = q^{1/4}$) contains the chiral jet grading as its $w=2$ member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n+1$, strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit $\varepsilon = 1/w \rightarrow 0$ with explicit threshold $w = n+1$, not a slice (convention clarified: 897266 = the quarter-moded loop Fock at $u^4 = \text{integer level 1}$). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96)$, Dynkin indices 6750/750; level-1 integrability ($\langle 2\theta, \theta^\vee \rangle = 4$, $\langle \omega, \theta^\vee \rangle = 2$) leaves the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3 : $34752 = \chi_3$). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304 + 1820 + 135 + 1$), $5304 \neq 14^3$ (the $h^{\vee 3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ break one-block fusion (vs E_8 ’s integer $(0, 1, 1, 1)$); only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ($64 \neq 60$). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.
- **Contract + surfaces synced. tfpt_research_contracts**: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold $w = n+1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel $\supseteq$ ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function = E_4/η^8 incl. the $q^{-1/3}$ prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to **tfpt_readouts_extension.wl** and *verified* with the active engine the same day (394 → 398, ALL WOLFRAM EXTENSION CHECKS PASSED); **wolfram/README.md** + GATE.WOLFRAM.02 counters advanced; **tfpt_safeguards** Layer-6 count + website mirrors 116/116 + 398/398. Website: **papers.ts** contract mirror (“WP5a done, WP5b–e roadmap”), DAG **qft** node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak $F_0 = 11.19275$ Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute ε), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs $\rightarrow$ 491 usable $\rightarrow$ 119 segment-coherent $\nu(t)$ points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection” $F_0 = 11.19275$ Hz is corrected on record (true Vela $F_0 = 11.1861692$ Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear* $\delta\nu(\tau)$ residual (μ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches ε_{90} below $\varepsilon = 1$ ($\varepsilon_{50} = 0.55$ in 2019 only) $\Rightarrow$ no absolute per-channel ε -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude $\sqrt{2g} = 0.40$ (leg gains 0.046/0.114, min $p = 0.014$ — the 2019 audit-level chance feature: Bonferroni $\times 10 = 0.14$, shuffle $p = 0.40$).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute ε); all-channel universal-DSI 95% UL $\varepsilon < 0.11954$ ($\approx 12.0\%$; HKSJ $t_2 = 2.92$, $\rho = 0.3$ -inflated 0.137; $6.91\times$ the predicted $\varepsilon = 0.0173$), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5% $\rightarrow$ recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list, PG.06b FULL bullet), `results/results.json`, scorecard row `data_value/source` via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 $\rightarrow$ 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` $\rightarrow$ 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; `tfpt_readouts_extension.wl` $\rightarrow$ 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 $\rightarrow$ 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); v479 registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)

- **v495_celestial_wp3_gs_alignment.py** (CELEST.WP3.01, **25 checks, exact Fraction/sympy**). The pre-registered WP3 “alignment test only”: the Okubo coefficient $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$ is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ($\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$; $\mathfrak{e}_6/\mathfrak{e}_7$ carved out of the v492 E_8 root set; negative control: $\mathfrak{sl}_4$ fails, $5/32$ vs $3/32$); the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim + 2) = h^{\vee} + 6$ holds exactly across the Deligne series; E_8 : $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\text{fund}} = 1/10$ (adjoint-trace) exact, so $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$ resp. $1/300$ — with κ/c_3 itself *irrational* ($2\sqrt{3}$). Byproduct: the printed $\lambda_{\text{so}_8}^2 = 3/2$ in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence)**: the alignment format passes 8/8 across the whole list (zero selectivity), $\tilde{\lambda}$ -integrality $2/8$ (shared with $\mathfrak{sl}_3$), only $g_{\text{car}}=5 \mid h^{\vee}$ is $\mathfrak{e}_8$ -selective ($1/8$), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not $\mathfrak{e}_8$ -selective evidence. Verdict B.
- **v496_celestial_wp4_character_shadow.py** (CELEST.WP4.01, **25 checks, exact integer/Fraction/sympy**). WP4, the type mismatch head-on: the $(E_8)_1$ character $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$ is *not* a conformal block of the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra in its own jet grading — localised three ways: (a) spin $1 - d/2$ unbounded below and 248 never a tower dimension ($d \leq 100$); (b) growth $n^{2/3}$ vs $n^{1/2}$ (quadratic cumulative count $31s^2 + 92s + 60$, $31 = k + h^{\vee}$; f_n/χ_n strictly increasing $n = 1..12$); (c) the level-2 null ideal deletes exactly $27000 = 30^3 = (h^{\vee})^3$ out of $\text{Sym}^2(248) = 1 + 3875 + 27000$ — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes = $60 = C_0$ currents, jet slice cycles $(60, 64, 60, 64)$, one μ_4 period sums to 248, glue weights $(0, 1, 1, 1)$ integer; free loop Fock $897266 \gg 248$, so the truncation must come from a limit) — the same MMST scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).
- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` ($3 + 4$ exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray $\nu(t)$ leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (tfpt-pulsar vela-full)**. The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) $\rightarrow$ 491 usable $\rightarrow$ 240 per-obs detections $\rightarrow$ **119 segment-coherent $\nu(t)$ points** (108 fully coherent), median $\sigma_{\nu} \approx 0.6 \mu\text{Hz}$; barycentring validated against PINT to $3.4 \mu\text{s}$ — the *first X-ray $\nu(t)$ probe* of the search program. Sanity gate PASS: $\Delta\nu/\nu$ 2019 = 2.39×10^{-6} vs 2.5012×10^{-6} published, 2021 = 1.272×10^{-6} vs 1.26×10^{-6} ; the 2024 glitch

is not reducible (only ~ 150 -s snapshots, 9.7 ks — documented).

- **Frozen-kernel comb** ($\omega = 2.5827$, $\varepsilon_{\text{pred}} = 0.0173$): **verdict data_limited (range- and amplitude-limited)**. Reach 2019: 2.17 comb periods ($<$ the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw $p = 0.014$ has the kernel as smallest- p battery member BUT Bonferroni $\times 10 = 0.14$ and shuffle $p = 0.395 \Rightarrow$ chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021 $p = 0.731$; $\ln \tau$ stack $p = 0.049/\text{shuffle}$ 0.307. Surrogate-calibrated injection at the real sampling and noise: $\varepsilon_{50} \approx 0.55$ (2019 only), ε_{90} never reached below $\varepsilon = 1$, 0% power at the predicted $\varepsilon = 0.0173$ — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag: $\lambda = e$ in the 2021 leg (raw $p = 0.002$, Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider: <3 -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record**. The old PG.06b single-obs “pipeline proven” detection at $F_0 = 11.19275$ Hz ($H = 18.4$) was a noise peak; the true Vela frequency at that epoch is $F_0 = 11.1861692$ Hz ($H = 99.7$, consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray nu(t) (PG.06b FULL)” (status stays **data_limited**; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); **experiments/README.md** catalog + §1 prose, pulsar README, **experiments/next.txt** synced. No claim; firewall intact.

2026-07-21 (the two 2026-07-20 prediction-of-record rows promoted to the suite: v494 — α_s running + μ -distortion branch discriminator machine-checked, ledger rows COSMO.RUNNING.01/COSMO.MUDIST.01, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494_cosmo_running_mu_record.py (23 checks, all green)**. The α_s -running and μ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact): $\alpha_s = dn_s/d \ln k = -2/N_\star^2 = -r/6$, $\beta_s = -4/N_\star^3$ (NOT $-8/N_\star^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check $< 1\%$): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$. [C] data: Planck 2018 (-0.0045 ± 0.0067) $+0.57\sigma$; record leg P-ACT-LB ($+0.0062 \pm 0.0052$) -1.33σ (watch channel: positive central value; kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch). [C] μ -distortion (Chluba window, $+21\%$ calibrated on the Λ CDM anchor): sharp branch $N_\star=51.4$ $\mu = 1.6 \times 10^{-8}$, profiled record $N_\star=56.1$ $\mu = 1.94 \times 10^{-8}$, band $1.5\text{--}2.3 \times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; PIXIE baseline 9×10^{-8} does not reach; kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$). [E] comb blindness (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window ($|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050). Fences: $N_\star$ stays an external reheating input (GATE.NSTAR.01); the frozen band $[50, 60]$ stays the surface of record (v84/v86). Ledger rows COSMO.RUNNING.01 and COSMO.MUDIST.01 added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. **predictions.txt** gains the two rows (α_s [C] with the -1.33σ P-ACT-LB pull and the positive-running kill; μ -distortion [C] data-limited with the Voyage-2050 branch split); website **lib/predictions.ts** gains the matching cards **alpha-s-running** and **mu-distortion** (confrontation + experiment blocks pointing at **experiments/tfpt-discovery**); the cosmology DAG node lists v494 and states the running/ μ readouts.
- **Paper bodies**. **tfpt_1** (Inflation interface) gains the running consistency-relation

paragraph ($\alpha_s = -2/N_\star^2 = -r/6$, exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; `tfpt_2` (inflation-pivot keybox) gains the μ -distortion branch-decider sentence (sharp 1.6×10^{-8} vs profiled 2.0×10^{-8} , 3σ at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

2026-07-20 (VIII) (E8 mass-ladder blind-recovery bed EXECUTED: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov-E8 spectra of $\text{BaCo}_2\text{V}_2\text{O}_8$ (8/8) and CoNb_2O_6 (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8_ladder_bed_v1.yaml).** The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset: $\text{BaCo}_2\text{V}_2\text{O}_8$ (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**, $\chi^2/\text{dof} = 0.35$, MC $p = 0.0056$ (singles-only run $p = 0.0004$; σ -variant robust); CoNb_2O_6 (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**, $\chi^2/\text{dof} = 0.52$, $p = 0.0038$ (m_7/m_8 lie outside the 0.6-THz low-pass window; the broad 0.43-THz (m_1+m_2) -continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of $2I - A(E_8)$ equals the Zamolodchikov ladder to 1.5×10^{-15} ($m_2/m_1 = \varphi$ to 2.2×10^{-16} ; lowest Cartan eigenvalue $2 - 2\cos(\pi/30)$ to 4.5×10^{-14}).
- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset $\pm\{0.3, 0.5\}m_1$: 0/8 detections; jitter dose-response $\leq 5\%$ at the destructive 30% level; wrong ladders on the real data (harmonic 1..8, A_3 -PF, D_8 -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates, χ^2 -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked $\rightarrow$ consistent (stage `not_applicable`, tier `analog_positive_control`, leakage `detector_control` — the executed-analog basket of `qc-recovery-kernel` and `qgeo-eit-soff`); counters 120 rows: 49 consistent / 3 parked; `experiments/README.md` catalog row + stats (generator); website `predictions.ts` EXPERIMENTS_AUDIT counters; `experiments/next.txt`. This validates the *ladder-detector pipeline* — it confirms Zamolodchikov E8, *not* the TFPT axioms: no `[E]`, no `\veri`, no ledger row.

2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation $XY=Z^4+a_0$ is a PURE SCALE with the $\tau=i$ pillowcase frozen, the μ_4 clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ with no sector leak; *verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O]*)

- **v493_celestial_wp2_clock_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the `experiments/tfpt-discovery/probe celestial_seam_wp2_clock_deformation_probe.py`. *Geometry*: $P(iZ)=P(Z)$ forces $a_3=a_2=a_1=0$ sharply and two-sided ($\prod_k (Z - i^k w) = Z^4 - w^4$); $XY=Z^4+a_0$ is smooth iff $a_0 \neq 0$ (disc = $256 a_0^3$); the branch points $Z^4 = -a_0$ are one free μ_4 orbit in *every* fibre with cross-ratio 2;

the binary-quartic invariants of $w^2=Z^4+a_0$ are $I=12a_0$, $J=0$ *identically*, so $j=1728$ and the $\tau=i$ pillowcase shape (v168/v214) is *frozen* for all a_0 — a_0 is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy*: $\Pi = t(i-1)(1, i, i^2)$, all three vanishing spheres carry the same volume $\sqrt{2}t$ (lockstep); the clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (char x^3+x^2+x+1 , eigenvalues $\{i, -1, -i\}$ = the A_3 exponents, $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$), fixes no cycle ($\det(A-1)=-4$), and *is* the Picard–Lefschetz monodromy of the family ($a_0 \rightarrow e^{2\pi i}a_0$ gives the 4-cycle); the surviving direction is the weight-1 χ_1 -Fourier diagonal $(1, i, i^2)$, *not* the naive invariant diagonal; the versal weights $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$ are the regular μ_4 representation. *Algebra* ($BHS \mathbb{Z}_2 \rightarrow \mathbb{Z}_4$, [arXiv:2305.09451](#)): mode dictionary $\{w[m, n] : m \equiv n \pmod{4}\}$; the fibre bracket $-4\text{Nambu}(XY - Z^4 - a_0)$ anchored at the $a_0=0$ orbifold ($\{X, Y\}=16Z^3$, Jacobi + Casimir exact, uniform clock-degree -1); S-algebra corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade — no sector leak, the v492 equivariant sector deforms consistently; $a_0 \leftrightarrow BHS$ ’s weight-0 c^2 slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/ $\mathbb{Z}_2$ middle node, $4=2 \times 2$). *Negative controls*: $a_1 Z \Rightarrow j=0$ (hexagonal); pure $a_2 Z^2$ stays singular (A_1); $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications (-4Nambu = the $k=4$ CCA bracket; period = root difference; seam-scale reading via $\text{clock}^2=\text{deck}$); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).

- **Contract updated to WP2 done.** `tfpt_research_contracts CELEST.SEAM.01` section: WP2 bullet “finite, next” $\rightarrow$ *done* (*verdict B*) with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction = χ_1 -diagonal, a_0 pure scale); kill-test paragraph K1 “survived by WP1 and WP2”; status line `CELEST.WP2.01 [E]/[C]` executed, WP3–WP5 stay [O]. Ledger rows `CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02` (four exact v493 mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as v491); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). `SEAM.EQUIV.01` stays [O]; no marker moves anywhere. Exact (sympy).

2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data_limited→null, no marker moves)

- **H3 stage 2 executed** (`experiments/hfqpo-ladder/results/archival/`). The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms 4.4σ , 280.1 Hz at 3.71% 9.9σ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth $\nu_3 = 1.5\nu_u$ (661.5/414/252/363 Hz) nor the integer control line $4\nu_0$ (588/368/224/322.7 Hz) is detected in any source ($\sim 0\sigma$ single-trial; trials correction $N=8$ as preregistered). 3σ rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record $92=184/2$, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (`hfqpo-ladder`) `data_limited→null` (120 rows: 26 null / 33 `data_limited`), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); web-

site predictions.ts (hfqpo-tooth entry + EXPERIMENTS_AUDIT counters); predictions.txt; experiments/README.md; experiments/next.txt. No verification/ module, no ledger row, no marker move (the search target stays [O]).

2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the E_8 μ_4 -glue is the flat $\mathbb{Z}_4$ monodromy of the equivariant celestial chiral algebra on the A_3 ALE space, with the exact correction clock²=deck (spin bridge $\mathbb{Z}_8$, $8=2|\mu_4|$); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])

- **v492_celestial_z4_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128 μ_4 -glue grading of E_8 ($240 = 52+64+60+64$, additive on all 6720 root pairs) is *inner* — $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\text{Ad}(\exp(2\pi i h/4))$, an order-4 inner torus automorphism = a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense; the A_3 -side detector $h' = (0^5; 1, 1, 1, -3)$ reads the same class = the diagonal (1, 1) of the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ (the v92/v125 Lagrangian glue). $\mathbb{C}^2/\mathbb{Z}_4$ is verified as the A_3 singularity $XY=Z^4$ (Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$, Molien exact); the glue-equivariant SDYM(E_8) sector *closes* with graded dimensions $60(d+1)/64(d+1)$ — possible only because $\dim \mathfrak{g}_j = (60, 64, 60, 64)$ — with zero modes = the carrier $D_5 \oplus A_3 + \text{Cartan} = 60$ and density $1/4 = 1/|\mathbb{Z}_4|$; the four glue-sector characters sum exactly to $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$ (v377), with sector weights = the discriminant form $(5x^2+3y^2)/8$ and integer glue-diagonal $h = (0, 1, 1, 1)$ (= locality of the $(E_8)_1$ extension). *Critical correction (verdict B)*: the A_3 deck acts on the celestial sphere as $z \mapsto -z$ (order 2), *not* as the order-4 clock; the clock is the $U(2)$ phase diag($i, 1$) (normalising the deck, $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator), with the exact spin bridge (spin clock)² = deck ($\mathbb{Z}_8$, $8 = 2|\mu_4|$, the $c_3 = 1/(8\pi)$ winding integer). Clock-invariance selects the 1-parameter A_3 deformation $XY=Z^4+a_0$ whose four branch points are one μ_4 orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action diag(i, i) killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity = $\text{Aut}(\mathbb{Z}_4)$.
- **New research contract CELEST.SEAM.01 in tfpt_research_contracts** (own section, the fourth contract alongside $U_{\text{wall}}/G_{\text{metric}}/\text{CONTRACT.F.01}$): hypothesis (seam = glue-equivariant $\mathbb{Z}_4$ -orbifold sector of the CCA on the A_3 ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum $(E_8)_1$ existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (papers.ts section, script index). SEAM.EQUIV.01 stays [O]; no marker moves. Exact (sympy).

2026-07-20 (IV) (evidence scorecard: five new typed rows — the α_s running prediction of record, the μ -distortion A_s/N_* branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); experiments level only — no verification module, no gate closes, no marker moves)

- **α_s running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index: $\alpha_s = -2/N_*^2 = -7.6 \times 10^{-4}$ at $N_*=51.4$ (exact slow roll -7.09×10^{-4} ; band $N_* \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$) and $\beta_s = -4/N_*^3$ (leading order; NOT $-8/N_*^3$) — the identity $\alpha_s = -r/6$ holds only at leading order ($\sim 7\%$ off exact). Data: Planck 2018 -0.0045 ± 0.0067 ($+0.57\sigma$), P-ACT $+0.0060 \pm 0.0055$ (-1.22σ), record leg P-ACT-LB $+0.0062 \pm 0.0052$ (-1.33σ , Calabrese et al. 2025 JCAP); β_s (Planck run+runrun) 0.010 ± 0.013 (-0.77σ). Forecast: LiteBIRD+CMB-S4 $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$ (0.14σ); CMB-S4+DESI+Euclid statistical $\sim 3.3 \times 10^{-4}$

(2.2σ , systematics-limited). Same `N_star_reheating` independence group as the $n_s/r/A_s$ rows (anti double-counting: never an independent hit); kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.

- **μ -distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch $N_\star=51.4$ ($A_s=1.76 \times 10^{-9}$) $\mu \approx 1.6 \times 10^{-8}$, profiled record $N_\star=56.1$ ($A_s=2.10 \times 10^{-9}$) $\mu \approx 2.0 \times 10^{-8}$, band $1.5\text{--}2.3 \times 10^{-8}$; the 23% branch split ($\Delta\mu = 3.0 \times 10^{-9}$) gives a 3σ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline $|\mu| < 9 \times 10^{-8}$ does NOT reach — factor 4–7 above the band; FIRAS $|\mu| < 9 \times 10^{-5}$). Honest S15 typing: the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is STRUCTURALLY BLIND in the μ window — window attenuation $|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two A_s rows (one branch question, several readings). Kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (`parked_analog`; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall: $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$ with no charge Hall (seam edge $c_-=8$, v450/v451/v489); no engineered material yet ($\alpha\text{-RuCl}_3$ $c_-=1/2$ contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated κ_{xy} , PRX 12, 021025). (ii) E8 mass-ladder validation bed ($\text{CoNb}_2\text{O}_6/\text{BaCo}_2\text{V}_2\text{O}_8$): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201, NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz $m_1\text{--}m_6$; Coldea et al. 2010 Science $m_2/m_1=\varphi$) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning: $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$ at $M/m = 8.67$ (2 resonant pairs); best candidate $^{52}\text{Cr}_2\text{--}^6\text{Li}$ ($M/m=8.635$, $s_0=1.2890$, $\lambda=11.44$, +0.44% vs $(3/2)^6$, gap deviation +0.18%); alternatives $^{39}\text{K}_2/^{41}\text{K}_2\text{--}^6\text{Li}$ ($\lambda = 11.59/11.20$, needs double Feshbach control), $^{176}\text{Yb}_2\text{--}^{23}\text{Na}$ (+5.9%); solver validated on the Naidon–Endo anchors (homonuclear 22.69, Cs_2Li 4.877); first end-to-end HARDWARE injection of the comb detector at tunable $\lambda \approx 11.39$; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.
- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

2026-07-20 (III) (v491 — $g_{\text{car}}=5$ as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely $\{1, 1, 2\}$, sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)

- **v491_p2_partition_corollary.py (P2.PARTITION.01).** Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE — $\{1, 1, 2\}$ — so $e = (e_1, e_2, e_3) = (4, 5, 2)$ and hence $g_{\text{car}} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\text{fam}} = e_2 - e_3 = 3$ are *corollaries*, not inputs (consistency fixed point $(2^4-1)/5 = 3 = \text{rank } H^1$). [E] negative controls, each assumption load-bearing: drop positivity $\rightarrow 4$ solutions, e_2 ambiguous $\{0, 3, 4, 5\}$; sum $3 \rightarrow e_2=3$; sum $5 \rightarrow$ not unique; $2/4$ parts $\rightarrow e_2 \in \{3, 4\}/\{6\}$; Mehta–Seshadri RATIONAL weights $\rightarrow e_2 = \frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins $(1, 1, 2)$ with ALL THREE targets $(4, 5, 2)$; the partition needs only $e_1=4$. [E] sum rule: $4 \times \frac{N-1}{2} = 4$ iff $N=3$; the v115 route $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$ agrees. Independence chain: #marks = 4 from v216 (P1-side, no g_{car} input); $\text{rank } H^1 = 3$ pure topology (v137/v228); $|a|_1 = 4$ g_{car} -

free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity $h^0=0/\text{sum}$ “given (U)”); AX.P2.01 stays a declared axiom, and the N_{fam} direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ((5,3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5,3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

2026-07-20 (II) (v490 — the T^2 spin-structure parity census: $\mu_{\text{pre}}=4$ witnessed on the lattice, the 16-fold-way bridge $\theta_v=1$ on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)

- **v490_seam_parity_census.py (SEAM.MU.01).** The T^2 ground-state fermion parity of the v367 $p+ip$ collar, resolved per spin structure (AA, AP, PA, PP) and determined TWICE — Read-Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial $M=3$ phase, with the twisted real-space spectra matching the Bloch bands to 9×10^{-15} . [E] per layer $(+, +, +, -)$ (the $\nu=1$ Ising signature, TRIM $\equiv$ Pfaffian in every sector); [E] the 16-layer carrier has parity¹⁶ = + in ALL FOUR sectors $\Rightarrow$ gauged torus GSD = 4 = $\mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$ — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$ before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers: $\theta_v = e^{2\pi i \nu / 16} = 1$ exactly at $\nu = 2c_- = 16$ (v489), rivals $\nu = 1, 2, 8$ not holomorphically condensable; [E] honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH $M=1$ and $M=3$ ($\gamma_f=0.000001/0.000000$) — it does NOT discriminate $(E_8)_1$ vs $SO(16)_1$. [C] the KLM step $\mu : 4 \rightarrow 4/2^2 = 1$ stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$. [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node qft). Python (numpy + scipy Schur + sympy exact θ_v).

2026-07-20 (v489 — the chiral central charge $c_-=8$ measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); *no gate closes, no marker moves*)

- **v489_seam_modular_commutator.py (SEAM.CCC.01).** The Kim–Shehab–Kim modular commutator $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$ (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367 $p+ip$ seam collar model, entirely from the Peschel Majorana covariance ($W_X = -2 \text{artanh } \Gamma_X$, $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$), on a disk about the torus centre cut into three 120° wedges. [E] ground-state purity $\text{eig}(i\Gamma) = \pm 1$ to 6×10^{-15} ; [E] per-layer $c_- = 0.499998$ at $L=32$ (convergence 0.499793 at $L=16 \rightarrow 0.500000$ at $L=48$, deviation -1.8×10^{-7}); [E] controls: trivial $M=3 \rightarrow 0.000000$, orientation flip $M=-1 \rightarrow -0.499986$; [E] layer additivity exact (1.4×10^{-14}); [E] the 16-layer carrier lands on $c_-=8=g_{\text{car}}+N_{\text{fam}}$ (direct 7.9967 at $L=16$, additivity route 8.0000); [E] clip invariance 5×10^{-8} . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ($c_-=8$) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node qft, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; no result, no status change)

- **introduction.tex** — new section “TFPT in the light of algorithmic physics”. Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.

2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$ Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)

- **v488_majorana_clebsch_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung $M_R = c_3^3 \bar{M}_{PI}$: a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{car}/N_{fam} = 5/3$ — declined for lack of mechanism. The mechanism search is now EXECUTED: [E] NO RENORMALISABLE CHANNEL — $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$ and the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247), so M_R only via the $d=5$ operator $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$; channels $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ with only singlet + $\mathbf{45}$ admissible. [E] CHARGE TABLE: $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$, $\text{Tr}_{16} Y^2 = \frac{10}{3}$ (v485). [E] THE $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2, 3\}$ -smooth, and the multiplicative closure of $\{2, 3\}$ -smooth rationals never contains $\frac{5}{3}$ (no factor 5); the UNIQUE natural 5 of the embedding, $k_Y = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: Clebsches are family-space scalars; $\text{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory by construction. [O] compiler-flavor inventory typed and declined: exact $\frac{3}{5}$ ’s only in wrong slots (K lepton gen2/gen1 ; $R + L$ lepton gen3/gen2 ; never gen3 vs a degenerate base). NET: the rung + 5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate (M_R free inside the PS window, the $\Sigma m_\nu = 0.059$ eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk’s one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)

- **v487_transfer_clock_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$ vs uniform $(\frac{1}{3}, \frac{1}{3})$ is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: [E] RUNG READING — the lazy rule’s one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124: $\lambda_n^{1/p_2} = 1 - n/N_{fam}$, pole at $n=N_{fam}$): $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$, while v327’s uniform

rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. [E] DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ -even pair mode carries rung $n=1$ (survival $\frac{2}{3}$), the σ -odd mode rung $n=2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel) that `experiments/frb-parity-comb` tests. [E] FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + non-negative rates forces UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs $h = -\frac{1}{6} < 0$); the “sheet-symmetric choice” candidate is superseded by this proof. [E] BEND COROLLARY: $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$ — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] what remains open is exactly what was open before v486: the arity $\{2, 3\}$ (anchor input) and the clock’s semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (v486 — generator hunt: the FULL transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ from ONE forced local rule — the unique lazy $\mathbb{Z}_2$ -pair walk (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$; the previously generator-less $(1/3)^6$ mode gets its origin; v327’s uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)

- **v486_transfer_full_rule.py** (HYP.REWRITE.02). The verified seam transfer spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (v54/v221), but the only local generator in the suite (v327) produces $\{1, \frac{2}{3}, 0\}$ — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUENESS: for the symmetric 3-channel rule $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$ (one absorbing family channel + $\mathbb{Z}_2$ pair) the survival spectrum is $\{s+h, s-h\}$; demanding the PHYSICAL pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$ — the lazy $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum $\rightarrow$ rates). [E] SIX-HAND GENERATION: over the order-6 = $2N_{\text{fam}} = |R^+(A_3)|$ hand, $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly — both decay gaps now have one recursive generator: $6 \ln \frac{3}{2}$ (recovery, v76) and $6 \ln 3$ (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule’s zero mode is power-persistent; its λ_2 arity story ($\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$) is unchanged. [O] the arity $\{2, 3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = $1/|\mathbb{Z}_2|$ as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits $a^T K a=41$, Koide 53, and the recorded-only $1^T Q a=16$ — and the lepton-exponent observation $(5, 3, 2) = K\text{-row} = \text{atom triple}$; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-14 (IV) (v485 — the merged analytic target’s computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)

- **v485_contact_diagonal_closed.py** (SEAM.CONTACT.UNIT.02). The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of $|D|$ on circumference ℓ is $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$

EXACTLY (symbolic limit + 30-digit checks); it vanishes iff $\ell = 2\pi = 1/(4c_3)$, the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure: $F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$, while the φ_0^{ret} split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly — $\det(I - uC) = (1 - 4u)(1 + 2u)^2$, $u = \varepsilon c_3 \ln 2$, all orders, the linear term ABSENT precisely because the diagonal vanishes ($\text{Tr } C = 0$); the series matches v484's $\text{Tr}(C^k) = 4^k + 2(-2)^k$ term by term. [E] WEIGHT DICHOTOMY: the SAME $48 = 3 \times 16$ admissible states carry BOTH multiplicities — counted flat they give the φ_0^{ret} puncture count $48 = \Omega_{\text{adm}}$; $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget $41 = 10b_1 = 40 + 1$ ($\text{Tr}_{16} Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHIO.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam ζ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ c_3 per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHIO.PUNCTURE.01 is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$, derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)

- **v484_seam_contact_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight c_3 per boundary contact insertion”: the $F_{U(1)}$ ladder $\{0, 3, 6\}$, the φ_0^{ret} puncture $\{1, 4\}$). This module shows the shared rule is NOT a new unknown: [E] the unit chain $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$ — one *bare* boundary insertion divided by the orbit order $|\mu_4|=4$ is exactly c_3 ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the μ_4 separations ($G(\pi/2) = -4c_3 \ln 2$, $G(\pi) = -8c_3 \ln 2$; mark matrix $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$, spectrum $\{4, -2, 0, -2\}$), so the μ_4 -invariant (orbit-averaged) insertion carries $-(c_3 \ln 2)C$ exactly and the log-det contact expansion is exactly graded in c_3 per insertion ($\text{Tr } C^k = 4^k + 2(-2)^k$). [E] the grammar is suite-wide: the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$ (v60) carries the SAME multiplicity $\Omega_{\text{adm}}=48$ as the φ_0^{ret} puncture at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\text{adm}}$, $41 = 10b_1$); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` (φ_0^{ret} split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

2026-07-14 (II) (v482–v483 — the deep-dive round: the ν -scale decision computation EXECUTED (the integer c_3^3 -rung excluded unstructured, the $5/3$ -proximate structure escape recorded and declined) and the φ_0^{ret} puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are t -independent rationals, so the π^{-4} must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)

- **v482_seesaw_rung_decision.py** (FLAV.NUSCALE.03). The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung $M_R = c_3^3 \bar{M}_{\text{PI}}$ needs a rescue factor $\times 1.670$ in m_3 (0.0301 vs 0.0503 eV), while $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$ ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and a PS-leg Yukawa-beta envelope $\times [0.5, 1.5]$ move it by at most $\times 1.10$ singly and $\times 1.165$ ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise $y_\nu = y_t$ (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor $r \sim 1.67$; its post-hoc 0.18% proximity to $g_{\text{car}}/N_{\text{fam}} = 5/3$ is recorded and DECLINED per v354/v355 (no forcing mechanism; the $t_0 \sim 30$ discipline of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu = 0.059 \text{ eV}$) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` (ν -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483_pillowcase_exact_traces.py** (HYP.PHI0.GEOM.01). The GEOMETRIC side of the φ_0^{ret} puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat $\tau=i$ pillowcase is a t -INDEPENDENT RATIONAL — $\text{Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma, \rho, \rho^3$ at every tested t (position-space Gaussian image sums $< 10^{-9}$; mode-space proof: only $n=0$ is γ -diagonal), matching the Atiyah–Bott fixed-point counts $4 \times \frac{1}{4}$ ($\det = 4 = |\mu_4|$) and $2 \times \frac{1}{2}$ ($\det = 2 = |\mathbb{Z}_2|$); the pillowcase contact term is EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark and the clock-equivariant trace exactly 1, for all t . THEREFORE the π^{-4} in $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ($4 = |\mu_4|$ marks of weight c_3 give c_3^4 , multiplicity $\Omega_{\text{adm}} = 48$; the bare $1/(2\pi)$ -propagator 4-cycle differs by the exact rational $\frac{3}{16}$). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight = c_3 ” (the same residual class as the EM-Ward c_3 -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` (φ_0^{ret} split paragraph); ledger rows HYP.PHI0.GEOM.01 + HYP.PHI0.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

2026-07-14 (website + figure refresh — the /verification journey views become *live* HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“ ~ 295 -script journey”, “260 scripts” $\rightarrow$ the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).

- **Prediction surface: the empirical-audit narrative restructured.** The single $\sim 13\,000$ -character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (`AuditRound`: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long `ALPHA.QUILLEN` residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~ 475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the `SEAM.EQUIV.01` bedrock and certification round; QFT-skeleton caption now names the one `[O]` residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\ctthree`, ...) rendered as red KaTeX errors on the site (e.g. the φ_0^{tree} icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + “Show more” toggle — applied to the paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.
- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four μ_4 marks realised as the exactly solvable *four-interval* multilocal modular geometry with $\omega \circ \rho = \omega$ manifest; and the absolute ν -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation $y_\nu=y_t$ — *no gate closes, no marker moves*)

- **v479_kronheimer_quiver_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the $\det K=1$ keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ are the null vector of the affine- E_8 Cartan matrix AND the Perron vector $A\delta=2\delta$ of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the $\mathbb{C}^2/\Gamma$ geometry; $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$ forced by the null identity); the gauge datum is $|2I|=120=|\mu_4|h(E_8)$, $h(E_8)=30=2 \cdot 3 \cdot 5$; the resolution carries the unimodular $-E_8$ form ($\det=1$, SNF = identity — the Poincaré link) with $3 \cdot 8=24$ deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2, 3, 5)$ cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in *origin_theory* (R3 keybox + new Kronheimer paragraph) and *tfpt_research_contracts* (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in *TfptCarrier/KronheimerMarks.lean* (marks = null/Perron vector over $\mathbb{Z}$, $\sum \delta_i^2=120=4 \cdot 30$, $30=2 \cdot 3 \cdot 5$, dimension count = 4; lake build clean, #print axioms = the three standard kernel axioms only).
- **v480_multilocal_four_interval.py (QGEO.MULTILOCAL.01).** The seam circle minus its four marks IS the $n=4$ multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, $\mu=1$ — the same one bit as $\det K=1$); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with $\rho^4=-1$ EXACTLY (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$ -lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} , each sector a single-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C \rho^{-1}=C$ at 10^{-14} , so $\omega \circ \rho = \omega$ holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at 10^{-2} (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGEO.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in *tfpt_research_contracts* (marks keybox + rigidity contract + v478 leg A), *origin_theory* (BW paragraph), *tfpt_4_frontier* (v358 companion note); Python-only by nature.
- **v481_seesaw_carrier_ladder.py (FLAV.NUSCALE.02, CANDIDATE class, genre v467/v468).** FLAV.NUSCALE.01’s $y_\nu=1$ probe was not the carrier normalisation: one $SO(10)$ 16 per family + the minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1, 2, 2)$) forces $y_\nu=y_t$ at the matching scale (named premise). With explicit 1-loop RG (gauge/ y_t/λ up; ADKLR Weinberg-operator running down, hep-ph/0108005) the (y_ν, M_R) trade-off collapses to M_R alone: the observed $m_3=0.0503$ eV demands $M_R=9.3 \times 10^{13}$ GeV — INSIDE the PS window $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$ (v249) at $\log_{c_3}(\bar{M}_{\text{Pl}}/M_R)=3.15$ (the $y_\nu=1$ inversion gives the structureless 2.58). [X] honesty gate: the integer rung $M_R=c_3^3 \bar{M}_{\text{Pl}}$ predicts $m_3=0.030$ eV — 40% low at 1-loop (κ run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named

(2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band $m_3 \in [0.002, 0.115]$ eV and the NO floor $\Sigma m_\nu = 0.059$ eV; frozen record untouched. Integrated in `tfpt_2_standard_model` (new paragraph after v272), introduction status card (third named candidate), `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.

- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the ϕ_0 puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$ verified) but does NOT force the value $48c_3^4$ (pure rationals vs π^{-4}) — recorded in `experiments/theory-contracts/phi0_donnelly_equivariant_probe.py`; the target stays [O].

2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy $c_{\text{est}} \rightarrow 1$, CHM parabola, even bands exactly zero) — the bridge’s modular side meets TFPT’s Einstein-derivation input; and the measure condition reduced to one exact KMS time $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$ with the $h(E_8)=30$ near-miss explicitly declined; both legs stay [O], *no status moves*)

- `v478_entropic_continuum_legs.py` (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has $S(L) = (c/3) \ln L + k$ with $c_{\text{est}} = 1.00089/1.00038/1.00021$ (monotone $\rightarrow$ the Dirac $c=1=2 \times \frac{1}{2}$, the v450 reading; gapped control $\sim 6 \times 10^{-8}$, area law); the Peschel modular Hamiltonian $K = \ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x) = x(L-x)/L$ (corr $0.87 \rightarrow 0.99$, shrinking plateau drift; constants deliberately not asserted), the even-range bands vanish identically (particle–hole, $\sim 10^{-36}$) and the odd tail is 1.8% with parabolic envelope — the state-side Δ_Σ (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary — $d\mu = \delta(t-t_0)dt$ gives $\mu_2/\mu_1^2 = e^{t_0}$ exactly, so the v477 one-moment condition forces $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$ (one dimensionless KMS time, closed form); the near-miss to $h(E_8)=30$ (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row GRAV.ENTROPIC.LEGS.01; both legs stay [O], no marker moves.

2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving R^2 route typed as *one moment condition* $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ with the exact closure $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)

- `v477_entropic_scaleflow.py` (GRAV.ENTROPIC.SCALEFLOW.01). [E] The Frullani identity upgrades to a scale-flow representation: $-\ln a = \int \frac{dt}{t} (e^{-ta} - e^{-t}) = 2 \int \frac{dx}{x} (e^{-a/x^2} - e^{-1/x^2})$ (symbolic + 10^{-20} quadrature) — Bianconi’s log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT’s $S_{\text{rel},\chi}$ evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight $w(t) dt/t$ gives $3\beta R \mu_1 + \frac{17}{24} \beta^2 R^2 \mu_2$; the flat weight has $\mu_1 = \mu_2 = 1$ and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding $m^2 = c_3^7 \bar{M}_{\text{Pl}}^2$ forces

exactly $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ — the v475 kill-test number IS a moment ratio — and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency: $f_0 = 6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row GRAV.ENTROPIC.SCALEFLOW.01; the R^2 gate and the v359 typing stay [O], no marker moves.

2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)

- **v476_entropic_compression_toy.py** (GRAV.ENTROPIC.COMPRESS.01). The last open work package of the v473 round, exercised on a gapped staggered-mass chain ($N=16$). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to 10^{-15} ; thermal boundary: $\text{spec}(PCP) \subset (0,1)$, Peschel $K = \ln((1-C_A)/C_A)$ finite, $S>0$ — v155 in the bridge context). **The reading is decided:** on the pure bulk $f(x)=x/(1-x)$ is singular on $\text{spec}(C)=\{0,1\}$, so the literal operator-side reading $Pf(C)P$ of the v473 conjecture is *ill-posed* while the state-side $f(PCP)$ is finite — the compression identity can only mean “build Δ_Σ from the *compressed* relative metric”, which matches Bianconi’s own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic): $[C^2]_{AA} - (C_{AA})^2 = BB^t$, $[C^3]_{AA} - (C_{AA})^3 = ABB^t + BB^tA + BDB^t$ — first order in the cross block vanishes identically, so for any analytic f the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity $f(C_T) = e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \leq x^2$ ($d/x^2 \in [0.4, 1.0]$) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\text{eff}}>0$, v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays [O], no marker moves.

2026-07-13 (v475 — the R^2 kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ ($\approx 51.7\bar{M}_{\text{Pl}}$, trans-Planckian) $\Rightarrow$ the light-trace-mode reading is dead, KMS renormalisation the only surviving R^2 route; plus the explicit Lorentzian-positivity witness; *no status moves*)

- **v475_entropic_scalaron_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. [E] On a maximally symmetric background the relative curvature operator $\tilde{\mathcal{R}}\tilde{g}^{-1}$ has eigenvalues *exactly* $\{R; R/4 \times 4; R/6 \times 6\}$ (Bianconi’s Appendix-B flattening verified symbolically, incl. the 6×6 two-form block), giving the exact vacuum action $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the v473 pin), $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$ is the exact tensorial R^2 factor. [E] With the pinned $\beta' = c_3/6$ both mass routes (Starobinsky matching and $f'/3f''$) agree: the raw entropic scalaron is $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$, i.e. $m_{\text{raw}} \approx 51.7\bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$ GeV — TRANS-PLANCKIAN, not light; v473’s interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$ -trace mode)

is *dead in naive form*; a viable mechanism must supply exactly $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$ in mass² (the kill-test number now WITH tensorial factors). [E] Domain: first log pole at $R=1/\beta=384\pi^2 \bar{M}_{\text{Pl}}^2$ (trans-Planckian) — the obstruction is the mass, not the domain. [E] Lorentzian-positivity witness: a timelike gradient gives the Gg^{-1} eigenvalue $1 - \alpha v^2 \leq 0$ for $v^2 \geq 1/\alpha$ (ln undefined), spacelike/Euclidean stay positive — the v473 [C] caveat exhibited, the OS route sturdier. [C] Verdict: interpretation 1 survives (Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

2026-07-13 (v474 — the operator level of the entropic-action bridge: the D_5 Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the $5 \rightarrow \bar{5}$ conjugation, and the v473 Q -target *decided* — supports exactly $\{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$, the naive pair-block reading killed; *no status moves*)

- **v474_entropic_hodge_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4-algebra of the v473 bridge, machine-checked on the finite carrier Fock space $\Lambda^\bullet \mathbb{C}^5$ ($\dim 2^{g_{\text{car}}}=32$). [E] CAR $\Rightarrow$ Clifford: the ten Γ 's from $(a_k^\dagger, a_k)$ satisfy $\{\Gamma_a, \Gamma_b\}=2\delta_{ab}$ exactly (all 100 pairs, integer matrices); the 45 bilinears $[\Gamma_a, \Gamma_b]/4$ commute with $\gamma=(-1)^N$, span exactly $45=\dim \mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). [E] the Hodge–Dirac symbol: $c(\xi)=a^\dagger(\xi)-a(\xi)$ satisfies $c(\xi)^2=-|\xi|^2 \text{Id}$ for symbolic ξ — Bianconi's $D=d+\delta$ is Clifford multiplication at symbol level. [E] the fold is the $5 \rightarrow \bar{5}$ conjugation: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1 , so her fiber $1+5+10$ becomes the GUT half-spinor $16=1+\bar{5}+10$ (the SM carrier decomposition). [E] honest obstruction: $\Omega^{\leq 2}$ is *not* Clifford-invariant ($\Lambda^2 \rightarrow \Lambda^3$ leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). [E] the Q -target enumeration (the v473 [C] conjecture decided at the algebraic level): $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$ with integer k has exactly the supports $(d, k)=(2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8, 2^{g_{\text{car}}}=\dim \Lambda^\bullet\}$ — minimal uniform solution $q=c_3^2=1/(64\pi^2)$ per Fock mode, bonus identity $\text{Tr } Q^2/\delta_{\text{top}}=32/48=2/3=\dim \Lambda^\bullet/\Omega_{\text{adm}}$; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers $\sqrt{16/5}, \sqrt{2}$ — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the R^2 mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

2026-07-13 (v473 — the entropic-action bridge: Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned $\beta'_B = c_3/6 = 1/(48\pi)$, the R^2 gap $3(8\pi)^9$ pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])

- **v473_entropic_action_bridge.py** (GRAV.ENTROPIC.ACTION.01). TFPT's classical field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi's entropic action $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$ (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$) is a local covariant action whose variation gives Einstein + an emergent nonnegative Λ at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her $\Omega^{0,1,2}$ fiber on the five-slot carrier $\mathbb{C}^5$ counts $1+5+10=16=2^{g_{\text{car}}-1}=\dim S_{D_5}^+$, while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling $3\beta R$ to $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*,

$\beta'_B = c_3/6 = 1/(48\pi)$; [E] her $\Lambda_G = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$ is nonnegative and quadratic at the fixed point, and with $\varepsilon_* = e^{-\alpha^{-1}}Q$ it reproduces the closed v60 branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ with the exact target $\text{Tr} Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$ ([C] until Q is constructed); [E] the R^2 kill test — the raw log expansion sits *exactly* $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$ below the TFPT Starobinsky coefficient $1/(12c_3^7)$, so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT's OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient $g_{a\gamma\gamma} = -4c_3$, the free-seam Higgs band $m_H \approx 129\text{--}134\text{ GeV}$, and the never-run BH HFQPO $\times 1.5$ ladder-tooth search; no new vN, no status moves)

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$, $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free $g_{a\gamma\gamma}$ in GeV^{-1} — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction $\lambda(\bar{M}_{\text{P1}}) = \beta_\lambda(\bar{M}_{\text{P1}}) = 0 \Rightarrow m_H = 133.5\text{ GeV}$ (band 129–134, m_t/α_s -sensitive; measured 125.25 sits a few GeV below — the known slight metastability, Buttazzo NNLO $\lambda(M_{\text{P1}}) = -0.0143 \pm 0.0057$, 2.5σ) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- (m_t, α_s) RGE pull off the surface at $> 5\sigma$). (iii) **BH HFQPO ladder tooth:** the one discriminating HFQPO test — a third tooth at $\nu_3 = (3/2)\nu_u$ (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (`experiments/hfqpo-ladder`, preregistered): added as an [O] search-target bullet to the `tfpt_horizon_readouts` “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 $+9.2\sigma$, GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (`predictions.ts` 23 $\rightarrow$ 25 cards, `papers.ts` frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

2026-07-03 (v472 — the determinant line over the $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whispot probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)

- **v472_quillen_detline_moduli.py (ALPHA.QUILLEN.DETLINE.01).** ALPHA.QUILLEN.EXACT.01 names the α^3 level as the curvature of the *determinant line over the $U(1)$ seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ($h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$, v367/v470) in real space on an $L \times L$

torus with *twisted* boundary conditions — the twist torus (θ_x, θ_y) is the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus: $\oint = 2\pi \cdot 1$ exactly (integer to 10^{-9}) at $M=1$ for $L=4$ AND $L=6$ (size-independent); controls: trivial collar $M=3 \rightarrow 0$, orientation flip $M=-1 \rightarrow -1$; the twist-moduli integer equals the Bloch-BZ integer for all three M (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap ($= 2.0$) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the $U(1)$ moduli = inflow level $k_0 = |C| = 1$ — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) =$ the inflow response” holds *verbatim at the finite/model level*, where $\log \det$ of the occupied frame replaces $\log \det_\zeta$. **Not closed [O]:** the *continuum* ζ -determinant identification on the abstract seam ($=$ the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: `tfpt_1` (sixth honest step in the EM closure), `tfpt_research_contracts` (residual certification), `tfpt_safeguards` (α -form lever), root README, `zenodo_description.html`. Exploration original: `experiments/tfpt-discovery/quillen_detline_twist_moduli.py` (same day).

- **Bird’s-eye whitespot rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted).** (i) `electron_sector_cabibbo_probe.py`: the α metrology triangle (TFPT $\alpha^{-1} = 137.0359992168$ vs Rb-2020 $+0.99\sigma$ / CODATA22 $+1.90\sigma$ / a_e -route $+3.39\sigma$ / Cs-2018 $+6.33\sigma$ — TFPT takes the Rb side of the 5.5σ metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$ is excluded by the electron at $\sim 2 \times 10^4\sigma$; linear scaling at $\sim 10^2\sigma$; quadratic survives — the [C] bridge must scale at least quadratically in m_ℓ); the Cabibbo-angle anomaly as a third dissolution watchdog ($\lambda_C = 0.2243762$ exact sits $+0.08\sigma$ on the PDG-2026 rescaled kaon average, between $K_{\ell 3}$ and $K_{\mu 2}$; exact unitarity puts $V_{ud} = 0.97450$ — superallowed is the outlier at $+2.6\sigma$, all neutron/pion routes agree). (ii) `kernel_calderon_uv_universality.py`: the raw-collar Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol $mI'_\nu(mR)/I_\nu(mR)$, Richardson residual 3×10^{-4} ; N_θ -independent; $|k|$ principal symbol; μ_4 block structure at machine precision; both spec- T masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays [O]). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.
- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$ [frozen], $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$ [the 2026-07-02 numerology-flagged Nebenbefund], $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ [frozen], $\Sigma m_\nu = 0.0588 \text{ eV}$ [v468 route]] + spatial flatness determines the whole flat Λ CDM budget: $H_0 = 67.15 \text{ km/s/Mpc}$ (-0.39σ Planck18, -5.67σ SH0ES22 — the parameter-free Planck side of the Hubble tension), $\Omega_m = 0.3133$ (-0.27σ), $t_0 = 13.86 \text{ Gyr}$; honest stresses recorded: $\omega_b -2.02\sigma$ (BBN-only -0.73σ), $\omega_c -1.19\sigma$, DESI-DR2 $H_0 -2.16\sigma$ (one correlated low- h direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the ρ_Λ leg the α engine), Ω_c leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)

- **v471_seam_horizon_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope $2C(\gamma)$ on deficit cones (0.01–0.05%) and replica sheets $n=2$ (0.8%), $n=3$ (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ -scale μ_{lat} drifts — the v152 anchor exhibited on data, with *no* canonical $\ln(m/\mu)=3/4$ emerging (recorded honestly); the BFK/Schur split is exact ($\sim 10^{-16}$) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying $2C_{\text{cone}}$ via Kac doubling; the v302 transfer masses $6\ln(3/2)$, $6\ln 3$ sit on the calibrated EH line ($\leq 0.05\%$); and the Perron/attractor mode ($m=0$) is IR-divergent — **the recovery gap $\Delta=6\ln(3/2)$ is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes $1/G$ finite). SEAM.THEOREM.01 stays [O]; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row data_limited→consistent).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at $f_0/(1+z)$ — all templates moved to detector-frame $M(1+z)$ (before the fix GW190521, $z=0.56$, was filtered $\sim 1.6\times$ too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked $p=0.262$, kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni $\times 12$): ratio semantics (amplitude $(2/3)^6$ vs *energy reading* $(2/3)^3$ vs step $2/3 \times \mu_4$ per-bounce phases $\{0, \pi/2, \pi, 3\pi/2\}$ (the boundary-birefringence analogue — path birefringence from parity violation $c_- = 8$ cancels in the inter-echo ratio; only a per-reflection phase does not) $\times$ lags 0.5–350 ms (ECO through Planckian window), with joint $(2, 2, 0) + (2, 2, 1)$ subtraction and a template-agnosticism control: NO_VARIANT_ECHO on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

2026-07-02 (v295 robustness patch — external-review follow-up: the numeric a_2 -kernel tolerance must dominate its own finite-difference floor)

- **v295_flataway_a2_exact.py (FLATAWAY.A2.01).** Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference a_2 kernel is compared against an $\varepsilon=10^{-3}$ second finite difference whose *own* truncation error is $O(\varepsilon^2) \sim 10^{-6}$, yet the tolerance was 10^{-6} — on his machine $k=3$ landed at 1.1×10^{-6} (ours: 2.9×10^{-7}). Fix: tolerance widened to 5×10^{-6} so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F -normalisation = the embedding index $k_Y=5/3$; both targets stay [O])

- **v469_seam_crossedproduct_route.py** (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the $SO(16)_1$ spinor has $h_s=N/16=16/16=1\in\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1\rtimes\{1,s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)=4/2^2=1\Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4 μ_4 glue runs on the same integer: $h(J^k)=\{1,1,1\}$ for $k=1,2,3$ (the v125 isotropy $q(k(1,1))=k^2$ at net level), $\mu=16/4^2=1$, same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D + $c_-=8$ [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern $|C|=1$ vs 0 (control), $\nu=16\times|C|=16$, $c_-=\nu/2=8$ — and $\nu\equiv 0 \pmod{16}$ is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening.** TfftCarrier/SeamResidualAxiom.lean gains the *parallel* peer-reviewed derivation seamResidualClosed’: the invariant-level axiom collar_invariants (R1’) + *one* combined 1995–2001 package crossedproduct_route_theorem; the new joints are kernel facts with *no* axioms — lr_locality_integer ($h_s=16/16\in\mathbb{Z}$), glue_locality_integers ($\mu_4 h(J^k)\in\mathbb{Z}$), klm_holomorphic ($4/2^2=1$, $16/4^2=1$), sixteenfold_pin ($\nu=16\equiv 0 \pmod{16}$, $c_-=8$), bundled as crossedproduct_arithmetic ([propext] only); #print axioms seamResidualClosed’ = exactly four; the original seamResidualClosed (AGT/AMT witness) is kept unchanged as the independent second route; lake build OK (3285 jobs).
- **v470_alpha_inflow_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ($\alpha^{-1}=137.0359992168$; Quillen split $<10^{-35}$). [C] *the level is computed, not minimal*: the α^3 coefficient $k_0=1$ *equals* the FHS bulk Chern invariant $|C|=1$ of the *same* p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez–Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) $\eta=CS$ part of $\delta\log\det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), *replacing* v435’s minimality assumption. [E] $k_Y=\text{tr}(Y^2)/\text{tr}(T_3^2)=(5/6)/(1/2)=5/3$ (Ginsparg, Phys. Lett. B197 (1987)): $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the “GUT 3/5” is the inverse embedding index; $\frac{41}{6}=\frac{20}{3}+\frac{1}{6}$ (carrier decomposition). [C] the seam F -normalisation (the one [C] of v434) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ\rangle=k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses — $c_-=8$ (gravitational, feeds SEAM.EQUIV.01/S3) and $C=1$ ($U(1)$, feeds EM1) — so closing R1’ feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta\log\det_\zeta(\text{seam})$ = the inflow response” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E].
- **Integration.** Both registered in run_all.py + script_registry.csv (cluster

frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed' route); the exact joints Wolfram-mirrored (365 $\rightarrow$ 368: the LR/glue locality integers, the KLM μ -arithmetic, the sixteen-fold pin, the $k_Y=5/3/b_1$ identities); cited in `tfpt_1` (α fixed point + EM-Ward residual prose), `tfpt_research_contracts` (seam residual + Lean section), `introduction` (closure ledger), `tfpt_5_redteam` (Target A residual wording) and the README/website status surfaces; exploration provenance `experiments/tfpt-discovery/closure_route_.py` (2026-07-02 note in `experiments/next.txt`). Suite ALL CHECKS PASSED through v470. Curated version bumped to **v5.4** (Zenodo deposit updated).

2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$)

- **v467_thirdgen_mixing_pattern.py (FLAV.THIRDGEN.PATTERN.01).** The suite's three $\sim 2\sigma$ mixing tensions ($\sin^2 \theta_{13} + 2.0\sigma$, $\sin^2 \theta_{23} + 1.76\sigma$, $|V_{cb}| + 1.79\sigma$; v307) are exactly the three *third-generation* channels and share *one* relative deviation $-5.36\% \pm 1.67\%$ — non-zero at 3.2σ , -0.03σ from $-\varphi_0$. [E] vocabulary identities: $\lambda_C^2 = \varphi_0(1-\varphi_0)$ (the frozen Cabibbo); $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$; the suppression size $\varphi_0 = 5.3\%$ sits inside the seam-gapped band $(2/3)^6 = 8.8\%$ (v393). [N] the ladder-base forms $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2/(1+\lambda_C) = 0.041119$ drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1-2 channels stay untouched and correct (dressing λ_C would fail at -24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2 = 4.3$). [C] **honest:** post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0)/(1-c_3)$ /resolvent degenerate $< 1.4\%$). [O] no seam-transport derivation (the Bernoulli-variance reading $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$ is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at ~ 0.0219 (sub-1%) kills the record's bare- φ_0 reading, exclusive $|V_{cb}| \sim 0.0395$ kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468_dm2_ratio_jarlskog.py (FLAV.DM2RATIO.01).** The splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270, $\delta = 4\pi/3$) has one canonical derived CP invariant, and [N] $|J_{\text{PMNS}}| = 0.029653$ meets the NuFIT 6.0 measured ratio 0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3 = \sqrt{|J|} = 0.17220$ vs $0.17264(229)$) — zero dials. [N] the informal `tfpt_2` table heuristic $m_2/m_3 \sim \pi\varphi_0 = 0.16704$ (a “ $\sim$ ” entry, never frozen) is now -2.44σ and is *superseded as the comparator*. [C] spectrum consistency: $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$ eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the $\mu\tau$ texture gives $\theta_{13} = 0$ at leading order). [X] JUNO shrinks the window $\sim 4\times$; adopting the v467 pattern would shift $|J| \rightarrow 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.
- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows FLAV.THIRDGEN.PATTERN.01 and FLAV.DM2RATIO.01 (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328's “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu=\frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11% , extending the empirical over-determination to *five* sectors)

- **v466_seed_leptonmass_channel.py (SEED.LEPTONMASS.01).** The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu=\frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0=1/(6\pi)+48c_3^4$. [E] plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu=\frac{12}{7}\varphi_0^2=0.00484673$ to 10^{-9} . [N] **new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu=0.004836$ (PDG 2024) gives $\varphi_0=0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6=8.78\%$, v393) but largely cancels for m_e/m_μ , so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306's five + this) has $\chi^2/\text{dof}<1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster χ^2 by $>100\times$, and the scheme-ambiguous $m_c/m_s=(34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in run_all.py and script_registry.csv (cluster registry); ledger row SEED.LEPTONMASS.01; cited in tfpt_5_redteam (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ , the θ_{13} -independent externally-runnable slice of v306)

- **v465_seed_crosssector_joint.py (SEED.CROSSECTOR.01).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB *EB/TB* polarisation observable) and the Cabibbo angle $\lambda_C=|V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0=1/(6\pi)+48c_3^4$. [E] plumbing: the axiom φ_0 reproduces the frozen λ_C , $\sin^2\theta_{12}$, β to 10^{-9} . [N] **cross-sector (headline):** back-solving φ_0 *independently* from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\text{comb}}$. [N] **one-parameter fit:** the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06% ; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007 , $\chi^2/\text{dof}<1$, combined pull 0.08σ . [N] **θ_{13} exclusion declared:** adding $\sin^2\theta_{13}$ raises χ^2 by ~ 4 (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is the honest θ_{13} -independent statement, not a hidden dilution. [N] **power:** a data↔formula shuffle explodes the cluster χ^2 by $>5\times 10^4$. A consistency / out-of-sample statistic (a subset of v306); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in run_all.py and script_registry.csv (cluster registry); ledger row SEED.CROSSECTOR.01; cited in tfpt_5_redteam (the seed-cross-validation firewall winbox, alongside v305/v306) and in the origin_theory birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (v463/v464 + Lean + Wolfram — the $c=8$ holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a $c=8$ holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463_seam_c8_holomorphic_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The $c=8$ holomorphic-uniqueness pin. [E] $c=8$ is NOT unique at level 1: $c=\dim \mathfrak{g}/(1+h^\vee)=8$ has THREE simple solutions A_8 ($\dim 80$), $D_8=\mathfrak{so}(16)$ ($\dim 120$), E_8 ($\dim 248$), so the $\det K=4$ $SO(16)$ rival is a genuine same- c competitor. [E] HOLOMORPHY FORCES $\dim V_1=248$: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, q^1 coefficient 248 (the 1-dim candidate space, vacuum $a_0=1$). [E] $\dim 248+$ level-1 $c=8 \Rightarrow E_8$ uniquely ($248/(1+30)=8$); tower $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$. [C] Dong–Mason/Schellekens: the holomorphic $c=8$ VOA is unique $= V_{E_8}$ (even unimodular rank-8 lattice unique $= E_8$, Mordell–Witt) — so the identification half is classification-forced.
- **v464_seam_oneparticle_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking R_1 . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P=$ projection onto $K<0$ ($P^2=P$ to $< 10^{-12}$), K fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ($\max |P_{2N}-P_N| \sim 1/N \rightarrow 0$). [E] the limit is $c=1$ Dirac: entanglement $S(L)=(c/3) \ln L$ gives $c \rightarrow 1$, 16 copies $\Rightarrow c_-=8$ (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each $\dim=8(1+h^\vee)$) and `holomorphy_selects_e8` (forced $\dim V_1=248=E_8 \neq 80, 120$) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)-(xxxi) the SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **v461_seam_strict_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367 $p+ip$ collar. [E] CHERN INTEGER: the FHS Chern is $C=+1$ (chiral $M=1$), 0 (trivial $M=3$), -1 (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|=1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless),

so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = $|C|=1 \neq 0$ and $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$ (the same 8 as c_3 's one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0, $C=0$; since $C=1 \neq 0$ NONE exists, so v424 sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.

- **v462_seam_spinor_continuum.py** (SEAM.EQUIV.SPINOR.01). The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension $SO(16)_1 \rightarrow (E_8)_1$ exhibited three ways. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/ E_8 identity $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$ gives $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL $120 \rightarrow 248$: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1, 248, 4124, 34752\}$ ($248=120+128$). [E] FINITE 16-MAJORANA FOCK: $C(16, 2)=120$ NS bilinear currents and $2^{16/2}=256$ Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$ CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_- \rightarrow 8$ and the lowest scaled NS gap $x_1 \rightarrow 1$ with $|x_1 - 1| \sim 1/L^2$ ($L=64..1024$). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact (θ identity + tower + $120/128/256$, Wolfram-mirrored) + numerical (finite- L , Python-only).
- **Lean** (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01). The two former cited axioms `mmst_existence` $\circ$ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate ChiralCFT_c8), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding = $|C|=1 \neq 0$, $c_- \neq 0$) and `fock_sectors` (v462: $C(16, 2)=120$, $2^8=256$, spinor 128, $120+128=248$); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram**. `tfpt_readouts_extension.wl` adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462 θ -identity + $120+128=248$ sector / Fock counts.

2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **v460_seam_s3_lto.py** (SEAM.S3.LT0.01). NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- N tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367 $p+ip$ collar. [E] FLAT-BAND PARENT: the flattened Bloch $Q=h/|h|$ has $Q^2=I$ (flat ± 1 bands) and $P=(I-Q)/2$ is an exact projector EQUAL to v367's occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n, 0)|$ decays rapidly ($\xi \approx 1.14$ at $M=1$, $\exp R^2=0.995$, $|P(20, 0)| < 10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ ($R^2=0.999$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED:

the flattened modular Hamiltonian $K_{\text{flat}} = \text{sign}(K)$ inherits $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$ ($u_{\Theta} = J$) and $[\rho, K_{\text{flat}}] = 0$ EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off- μ_4 neg controls firing. [C][O] the chiral $c_- = 8 \neq 0$ (FHS Chern $|C| = 1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756; $C \neq 0$ forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in run_all.py, the registry, the ledger and cited in tfpt_research_contracts (route (xxvii) of the convergent attacks).

2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- N Tomita–Takesaki tower, chirality forced by P_1 , an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **v454_seam_edge_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators $\rightarrow$ continuum Virasoro with the correct c ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits $c_{\text{Dirac}} = 1.000$ ($\frac{1}{2}$ per Majorana, $L = 64..1024$); [E] the lattice $U(1)$ current closes its affine level exactly ($\langle 0 | J_n J_{-n} | 0 \rangle = n$, $n = 1..8$; a gapped control vanishes); [E] the level-1 Sugawara $c = \dim G / (1 + h^\vee) = 8$ for both $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). [C] $16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$ is the level-1 current-algebra c_- ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455_seam_bw_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_{\Theta} = J$ — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E] $\|\Theta K \Theta + K\| = 0$ exactly at every N ($N = 8..128$); [E] an N -independent gap $6 \log \frac{3}{2}$; [E] $[\rho, K] = 0$ exactly at every N ; [E] an N -independent neg-control margin $2\|K\|$. [C][O] the inductive limit inherits $u_{\Theta} = J$; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456_seam_chirality_from_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ($c_- \neq 0$, the S3 phase) is *forced* by the one-sidedness that defines $c_3 = 1/(8\pi)$ (P1) (G6). [E] c_3 ’s “8” is the one-sided count $8 = |\mathbb{Z}_2| \cdot (f K / \pi) = 2 \cdot 4$; [E] a reflection sends the Chern integer $C \rightarrow -C$ ($+1 \rightarrow -1$ on the $p+ip$ model); [E] a two-sided boundary forces $C = -C = 0$. [C] the one-sided seam (no reflection) forces $C \neq 0$, magnitude $c_- = 8 = \text{rank } E_8$ — S3 is a consequence of P_1 ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.
- **v457_seam_character_killtest.py (SEAM.EQUIV.KILLTEST.01).** An explicit character/sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). [E] the $(E_8)_1$ character $\chi = E_4 / \eta^8$ with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; [E] the weight-1 count $248 = 120 + 128$ (not the free 120); [E] the sector count $|\det \text{Cartan } E_8| = 1$ (not $|\det \text{Cartan } D_8| = 4$). [X] a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **v458_seam_mmst_citation_audit.py (SEAM.EQUIV.MMST.AUDIT.01).** The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy $c = \frac{1}{2}$ (Thm 4.11), [C] H4 decoupled additivity to $c = 8$, H5 the WZW range $8 \leq 8 \leq 16$ with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension $SO(16)_1 \rightarrow (E_8)_1$ ($\det K \ 4 \rightarrow 1$), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459_seam_lattice_voa_route.py (SEAM.EQUIV.LATTICEVOA.01).** The second, independent route to $(E_8)_1$ supplying exactly the residual of v458 (G5). [E] E_8 even unimodular ($\det = 1$); [E] the weight-1 space $248 = 8 + 240$; [E] the 240 roots split 112 integer +

128 half-integer (the spinor, $248=120+128$). [C] the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar = A_{E_8} , tied to P_1 by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.

- **Lean (FORM.SEAM.RESIDUAL.01, TftptCarrier/SeamResidualAxiom.lean).** Kernel-checks the whole G-block arithmetic (the one-sided $8=|\mathbb{Z}_2|\int K/\pi$, the reflection $C\rightarrow -C$, $c_-=8$, the MMST range $8\leq 8\leq 16$, decoupled additivity, $248=8+240=120+128$, $240=112+128$, $\det K$ 1 vs 4) by **decide**, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (CollarRealizedAsLatticePhase) plus two cited external theorems (MMST v458, AGT v459); lake build OK.
- **Wolfram + integration.** Six exact mirrors added to **tftpt_readouts_extension.wl** (Sugawara $c=8$ for $120/15$ and $248/31$; the one-sided $8=2\cdot 4$ and reflection $C=-C\Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with $248=120+128$ and $|\det \text{Cartan}|$ 1 vs 4; the root split $112+128$ and $248=8+240$) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- N convergence, three independent reads of $c_-=8$, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4 -symmetry derived from the four marks)

- **v449_seam_mmst_uniform.py (SEAM.EQUIV.MMST.UNIFORM.01).** Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A $Q(1,2,4,7)/12 \rightarrow 2+\sqrt{3}$, the deviation shrinking monotonically; [E] an *independent* cross-ratio B $Q(1,3,5,9)/12 \rightarrow$ its own value 2, so convergence is not a tuned point; [E] a uniform $1/N$ rate $|Q(N)-Q_\infty|\cdot N \leq 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450_seam_edge_entanglement.py (SEAM.EQUIV.EDGE.ENTANGLEMENT.01).** A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives $c_{\text{Dirac}}=1.000$ for the critical free-fermion chain; [E] so c per Majorana = $1/2$; [E] a gapped control saturates ($c=0$). [C] bulk–edge: $c_-=N_{\text{Maj}}\cdot \frac{1}{2}=16\cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$. SEAM.EQUIV.01 stays [O].
- **v451_seam_edge_cardy_tower.py (SEAM.EQUIV.EDGE.CARDY.01).** Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension $h_\sigma=1/16$ (the periodic–antiperiodic ground-energy split) and [E] the energy dimension $h_\epsilon=1/2$ (the lowest NS pair excitation), both [E] L -independent (conformal $1/L$ scaling). [C] $\{c, h_\sigma, h_\epsilon\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3,4)$; 16 copies give $c_-=8$. SEAM.EQUIV.01 stays [O].
- **v452_seam_e8_modular.py (SEAM.EQUIV.E8.MODULAR.01).** The $(E_8)_1$ torus modular signature, extending v156’s q -expansion to the transformation data. [E] the character $\chi=E_4/\eta^8$ is S -invariant $\chi(-1/\tau)=\chi(\tau)$ (one primary, $S=[1]$); [E] T -phase $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$ ($c=8$); [E] leading power $\chi q^{1/3} \rightarrow 1$ so $q^{-c/24}=q^{-1/3} \Rightarrow c=8$; [E] holomorphic $c=8 \equiv 0 \pmod{8}$ with T -phase order $24/\gcd(8,24)=3$. [C] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453_seam_mu4_from_marks.py (SEAM.RIGIDITY.MU4FROMMARKS.01).** Folds the last structural premise of the rigidity chain (v446’s residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are* $\mu_4=$ roots of z^4-1 ; [E] the order-4 clock $z \rightarrow iz$ is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio $\lambda=2$ (in the D_4 stabiliser, v168); [E] the form basis $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$ is μ_4 -graded

(weights $(1, 2, 3)=A_3$ exponents). [C] so QGEO.SYM.01 follows from marks = μ_4 + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].

- **Lean: TftCarrier/SeamEdgeChern.lean (FORM.SEAM.EDGE.CHERN.01).** Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk-edge channel count $|C|=1$, $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$, the assembly $2c_-=N_{\text{Maj}} \cdot |C|$ (so $c_-=8$), $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$, holomorphic $c \equiv 0 \pmod{8}$ and the order-3 modular T -phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror. tfpt_readouts_extension.wl** gains nine exact checks (v452 modular data $\times 4$, v450 entanglement assembly, v451 Ising Kac weights, v453 marks = $\mu_4 \times 3$): 345 → 354 checks, all passing (GATE.WOLFRAM.02). E_4 is evaluated via its q -series since `EisensteinE` stays symbolic for complex τ .
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge $c_-=8$ from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the $(E_8)_1$ torus modular data (v452), and the μ_4 premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from μ_4 -symmetry, an independent topological reading of the edge $c_-=8$, and pinning the one external MMST fact at four-point order)

- **v446_seam_clock_invariance.py (SEAM.RIGIDITY.CLOCKINV.01).** Closes v445's single residual one notch further: the operator condition $[\rho, \Lambda_\Sigma]=0$ (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (fixed by RP+reflection, v54) of a μ_4 -invariant RP covariance satisfies $[\rho, \Lambda]=0$ (swept $N=4..32$); [E] a μ_4 -breaking covariance gives $[\rho, \Lambda] \neq 0$ (iff-sharp neg control); [E] Λ is a positive contraction (spectrum in $(0, 1]$); [E] the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (v258) of the invariant state has $[\rho, K]=0$ (Tomita–Takesaki), broken control $\neq 0$. [C] so with v445 (commute $\Leftrightarrow$ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ ; the residual reduces from an operator commutation to “the seam RP data is μ_4 -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].
- **v447_seam_edge_chern.py (SEAM.EQUIV.EDGE.CHERN.01).** An *independent* (topological) reading of the edge central charge, corroborating v444's correlator route from a different observable. [E] the $M=1$ collar has integer Chern number $C=+1$ (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E] $C=0$ in the trivial phase and $C=-1$ at $M=-1$ (opposite chirality); [E] the open strip carries exactly $|C|=1$ chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at $1/2$ only in the topological phase. [C] bulk-edge: $c_-=N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C|=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ — the same $c_-=8$ as v444. SEAM.EQUIV.01 stays [O].
- **v448_seam_edge_fourpoint.py (SEAM.EQUIV.EDGE.FOURPOINT.01).** Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector $G^2=G$ ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; [E] Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (v444) and four-point order. SEAM.EQUIV.01 stays [O].

- **Lean:** `TfptCarrier/SeamRigidityForcing.lean` (`FORM.SEAM.RIGIDITY.FORCING.01`). Kernel-hardens v445 (and v442’s uniformity): the entrywise forcing iff $\text{clock}_4 a = \text{clock}_4 b \Leftrightarrow a = b$ (i.e. $i^a = i^b \Leftrightarrow a \equiv b \pmod{4}$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2 = 64 < 256$ and the order discriminator ($128 > 64$) are all proved by kernel `decide` (`forcing_kernel_facts`, no domain axioms); only the operator-level lift to the full L^2 is the named premise. `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains three exact v445 checks (entrywise forcing for all $a, b \in 0..31$; commutant dimension $64 < 256$; order-2 commutant $128 > 64$ swept $N=4..64$): 342 $\rightarrow$ 345 checks, all passing (`GATE.WOLFRAM.02`).
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) records the five-direction narrowing of the `SEAM.EQUIV.01` keystone (edge scaling v444, forcing converse v445 + Lean, derived clock-invariance v446, independent topological $c_- = 8$ v447, four-point pinning v448) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (v336). The research-contracts note lists the same as “eleven convergent attacks”.

2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4 -block-diagonality, upgrading the rigidity step from “permit” to “force”)

- `v445_seam_rigidity_forcing.py` (`SEAM.RIGIDITY.FORCING.01`). The rigidity route (v398/v442) showed a μ_4 -block-diagonal K commutes with the order-4 clock $\rho = \text{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff): $[\rho, K] = 0$ iff $K_{ij} = 0$ whenever $i \not\equiv j \pmod{4}$, verified entrywise and swept $N=4..128$ — commuting *forces* μ_4 -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K] = 0\} = \sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra ($N=16$: $64 < 256$) — a genuine quantitative restriction. [E] OFF-BLOCK $\Leftrightarrow$ NON-COMMUTING: every off-block matrix unit has $\|[\rho, E_{ij}]\| \geq \sqrt{2} > 0$ (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ($N=16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4 = |\mu_4| = h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det = 1$; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N \geq 4$. [C] SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4 -block-diagonality, so RP-definability (v54/v398) + clock-invariance *force* the block-diagonal Λ_Σ — v442’s “permit” upgraded to “force”. [O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = `QGE0.SYM.01`, v323/v335); `SEAM.RIGIDITY.01/SEAM.EQUIV.01` stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven `SEAM.EQUIV.01` routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake`

build OK; #print axioms clean. Tracked as two new FORM.* ledger rows; \veri-context in tfpt_research_contracts; Lean manifest refreshed.

- **FORM.SEAM.BW.HSMI.01** (TfptCarrier/SeamStandardPair.lean) — **standard-pair relations, kernel-checked (hardens v438)**. The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel decide (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom): $[K, P]=P$, $J^2=1$, $JKJ=-K$, $JPJ=P_+$ (and $\text{tr } K=0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of **SEAM.EQUIV.01** to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01** (TfptCarrier/SeamApplicabilityLedger.lean) — **the audit count, kernel-checked (hardens v441)**. The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ($8 \leq 8 \leq 16$) and `npw_outside_bucket` ($\det K=1 \neq 4$) — ALL with NO axioms (pure kernel decide). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full lake build succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; **SEAM.EQUIV.01** stays [O].

2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c=1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368 $p+ip$ collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), \veri-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444_seam_edge_scaling.py** (**SEAM.EQUIV.CONTINUUM.EDGE.01**) — **chiral-edge motor**. The lattice edge carries the kinematic and conformal data of a chiral $c=1/2$ Majorana CFT: [E] a linear, chiral in-gap dispersion (velocity $v \approx 0.98$, $R^2=0.9999$, with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta \approx 1$, versus the *exponential* bulk row and the exponential trivial-phase ($M=3$, no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ($\eta=1.001 \pm 0.007$, successive profile spread $0.028 \rightarrow 0.013$ across $N_x=36..60$) — the numerical hallmark of an existing scaling limit. [C] $\eta=1 \Leftrightarrow$ one chiral Majorana ($c=1/2$) per edge, so the 16-copy carrier has a chiral $c_-=8$ $(E_8)_1$ edge CFT ($c_-=g_{\text{car}}+N_{\text{fam}}$). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.
- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; **SEAM.EQUIV.01** stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends ALL CHECKS PASSED), and are \veri-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438_seam_hsmi_borchers.py (SEAM.EQUIV.BW.HSMI.01) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P \geq 0$, boost K with $[K, P]=P$, reflection J with $JKJ=-K$, $JPJ=P_-$), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact, $sl(2, \mathbb{R})$ positivity (min eig > 0); [C] synthesis; [O] the continuum standard pair.
- **v439_seam_lieb_robinson.py (SEAM.EQUIV.CONTINUUM.LR.01) — continuum motor.** The explicit gapped $p+ip$ collar (v367) has a uniform-in- N gap, exponential clustering with N -independent ξ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440_seam_lto_rp_beta.py (SEAM.EQUIV.LTORP.BETA.01) — β -homotopy.** The (LTO-RP)/BW data ($\Theta K \Theta = -K$, $[\rho, K]=0$) are β -independent, and along the whole path $C_\beta=1/(1+e^{\beta K})$ from the NPW26-proven *trace* state ($\beta=0$) to the seam KMS state ($\beta=1$) every covariance is reflection-related, μ_4 -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441_seam_applicability_ledger.py (SEAM.EQUIV.APPLICABILITY.01) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.
- **v442_seam_rigidity_uniform.py (SEAM.RIGIDITY.UNIFORM.01) — uniform rigidity.** The band-limited μ_4 -character rigidity of v398 is uniform in N : exact 4-sector commutation to $N=256$ (not merely the $N \leq 64$ sweep), a uniform gap, and an N -independent clock-breaking lower bound $\|[\rho, V]\|=2$ — so it is not a small- N artifact (the full- L^2 forcing stays [O]).
- **v443_seam_e8_discriminator.py (SEAM.E8.DISCRI-MINATOR.01) — kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest $c=8$ rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; $c=8$ alone cannot decide (v277), only $\det K=1$ (the genus-1 statement, v90) selects E_8 . A seam with $\text{GSD}>1$ would falsify $(E_8)_1$. [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the `tfpt5-shadow` mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5-shadow.zip` the independent surfaces reproduced (`build.sh notes` → 11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **`make_docs_map.py` is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **`audit_sync.py` is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean `AUDIT OK` on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says `ALL CHECKS PASSED` while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or `vN` module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.
- **`robots.txt`** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.

- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (E8.DEGREE.JOINT.01) — a consolidation of v6/v66/v355/v431 that derives more from the E_8 Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
 - **[E] all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: $\text{rank} = \# \deg = 8$, $h = \max \deg = 30$, $\# \text{positive roots} = \sum \exp = 120$, $\# \text{roots} = 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod \deg = 696,729,600$, $\sum \deg = 128 = \dim S^+$.
 - **[E] the joint quadratic (the new lock).** The two TFPT structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the *unique* root pair of $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: sum of roots = $\text{rank } E_8 = 8$ (the rank-fill, v6), product = $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}} = h$, v431); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\text{car}} = 5$ and $N_{\text{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \text{rank}, \max\}$.
 - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (E8.DEGREE.JOINT.01); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 $\rightarrow$ 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (v436 — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module v436 (OVERDET.FLOOR.02) — it *hardens* OVERDET.FLOOR.01 (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432’s nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-“8” independence 0.1^3 , seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.
 - **[E] assumption-minimal floor.** The α cubic-class census (v100 layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.
 - **[E] monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$ scorecard-only $\sim 25.8 \rightarrow \alpha$ -census-only ~ 4.98 ; each rung uses strictly less evidence and the “not chance ($> 4\sigma$)” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
 - **[C] honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. [C] Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of v435 in named theorems — Dai–Freed (η -invariants and determinant lines, the exponentiated η -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\text{Tr } F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that

the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains [O] external math (no fabricated closure).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (v435 — a fourth honest step on the α^{-1} Quillen target: a π -power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining [O] left after v434 (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
 - [E] **π -power / metric-independence test.** The three EM-closure coefficients carry π -powers exactly $\{0, 3, 6\}$, equal to their c_3 -orders: Maxwell $k_0 = 1 \rightarrow \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-independent) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming v342/v434’s “ a^3 is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
 - [C] **conditional level quantisation.** If the a^3 term is the seam $U(1)$ Chern–Simons/ η boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
 - [O] **not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\text{Maxwell}} = a^3$ — stays external math (type A, v384); the seam F -normalisation stays the one [C] (residuals 1 & 3, v434); α^{-1} stays [E].
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside v433/v434; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (v434 — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03) — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
 - [E] **b_1 is the $U(1)$ a_4 heat-kernel coefficient (the $\beta = a_4$ theorem).** The one-loop β function of a gauge coupling is the a_4 (Seeley–DeWitt a_2 in $d=4$) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5) \cdot \text{that} = 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β , v159/v246), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (v342) — not a fitted multiplicity.
 - [E] **the geometry factors off:** $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$, with $\text{rank } E_8 = 8$ the $\chi=2$ seam boundary count (v216) and c_3^6 the six boundary insertions (the π -power test, v342).

- **[C] residual (1) $\equiv$ residual (3).** With b_1 and the c_3 -powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F -normalisation. So “the actual seam $a_4 = 8b_1c_3^6$ ” and “the seam F -normalisation” are *one* [C] obligation, not two.
- **[E] residual (2) located (not closed):** the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0 , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
- **[E] net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside v433; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (v433 — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting v391 and v342)

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02) — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (v382); it *grounds and connects*, it does *not* close it.** The target stays [O] (external math, type A, v384); the value $\alpha^{-1} = 137.0359992$ stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose ζ -determinant variation reached only the low Seeley–DeWitt orders a_0, a_1 ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}, \Omega^2/12$). This module supplies the missing link between the two.
 - **[E] a solvable 4D model reaches the a_4 order:** the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order v391’s 1D circle toy could not reach.
 - **[E] it is the order v342 uses:** so v391 (“a ζ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey a_4 ”) are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.
 - **[E] the measure is c_3 -built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam’s one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the v382 split is re-verified at $\alpha^{-1} = 137.0359992$.
 - **[O] not closed:** computing the *actual* seam $U(1)$ ζ -determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam F -normalisation fixing the coefficient *value* stays [C] (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.02); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

2026-06-27 (v431, v432 — the “unmapped” E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped” E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430): $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \text{det-ladder}\{8, 14, 20\})$. [E] the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \pmod 6$, so the degrees occupy only the residue classes $\{0, 2\} \pmod 6$. [E] the $6k$ family $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$ is the v91 spine; the $6k+2$ family $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavor diamond (v135). [E] “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). [C] **honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$ Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays [C] (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the L1 $\leftrightarrow$ L2 bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. [E] compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. [C] multiplying only across disjoint pieces (input over-determination $\times$ the α census $1/94500 \times$ the φ_0^{ret} -seed cluster counted once $\times$ one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. [E] that floor sits ~ 20 orders *above* the v100 conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. [C] **honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from E_8 ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five E_8 Casimir degrees $\{12, 14, 18, 20, 24\}$ with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min \deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped degrees. [E] the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ ($128 = 2 \cdot 64$), not a spare singlet. [E] forced-disjoint: the sheet/deck set $\{2, 16, 32, 128\}$ has empty intersection with $\{12, 14, 18, 20, 24\}$, its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the v355 discriminator), so there is no forced sheet $\rightarrow$ unmapped-degree identity.

- **A structural negative, not a status change.** It consolidates the S^- -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (tfpt_4_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run_all.py, script_registry.csv, status_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt_5_redteam, the magnitude/phase keybox of tfpt_3_e8_audit_bootstrap, the WIMP-no-go passage of tfpt_4_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (v429 — the ‘unmapped’ golden E_8 structure is the geometry of the one external input θ_i)

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle E_8 structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine- E_8 attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude φ cannot be numerology. [E] this sharpens that statement: because $N_{\text{fam}}=g_{\text{car}}-2$, the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=(g_{\text{car}}-2)\pi/g_{\text{car}}$, which is *exactly* the interior angle of the regular g_{car} -gon — for $g_{\text{car}}=5$ the pentagon, $\theta_i=3\pi/5=108^\circ$. [E] its cosine is golden, $\cos(3\pi/5)=(1-\sqrt{5})/4=-1/(2\varphi)$ (partner $2\cos(2\pi/5)=1/\varphi=\varphi-1$, $\varphi=2\cos(\pi/5)$). [E] the golden character is *unique* to $g_{\text{car}}=5$: among the small regular n -gons only $n=5$ has an irrational interior-angle cosine ($n=3, 4, 6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\text{car}}=5$ signature (v354/v313) and $\theta_i=3\pi/5$ is the one external cosmological input (v211) — and *refines* v354 from “ φ in no physical readout” (it is the unmapped $g_{\text{car}}=5$ signature, v313) to “ φ touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i\approx 170^\circ$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the finite- T solver that decides the branch (v185/v211). It does *not* derive θ_i , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.
- **Surfaces synced.** run_all.py, script_registry.csv, status_ledger.csv, the dark-matter section of tfpt_4_frontier; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix tex-artefacts/tfpt_references.tex** (“External works built upon”), \input into the six math-heavy documents (tfpt_1, tfpt_2, tfpt_4, tfpt_horizon_readouts, origin_theory, tfpt_research_contracts). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto;

Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold μ_4 identity made explicit (v422).** v422 already proved that the seam deck/divisor μ_4 , the carrier Galois group $\text{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\text{disc}(D_5)=\text{disc}(A_3)=\mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses — in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the $(2, 3, 5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$, $30=2\cdot 3\cdot 5=h(E_8)$, $\varphi(30)=8$), so each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or E_8): all seven collapse to one generator $\Rightarrow$ *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in $c_3=1/(8\pi)$ from rank E_8 , $h(D_5)=2(5-1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$. [C] refines and partly walks back v427, consistent with v236/v305/v313.
- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), `tfpt_safeguards` Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (`papers.ts`, `Safeguards.tsx`, `release.ts`, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures (make_figures.py, in manifest.sha256):** the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq 5/13$).

- **tfpt_safeguards expanded.** Layer 5 now states the null model explicitly ($\prod p_i=10^{-25.8}$, $2 \times 200k$ pseudo-theories $\leq 5/13$, $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$, shuffle 1.2) and the α uniqueness ($\leq 10^{-30.7}$, ~ 102 bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$ MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of F_{transfer} .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap $6 \ln(3/2)$) restricted to each sector. [E] the v99 Koide generator linearised at $q=2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, v320); QCD rate $b_3=-7$. [C] only the anchors (v_{geo} , C_p , M_R) stay external; F_{relic} 's cosmological IC θ_i is the one wall. A reduction (extends v383 static→dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric μ_4 -even collar at $\beta=1$ KMS satisfies $\Theta K \Theta^{-1} = -K$ ($u_\Theta = J$) and $[\rho, K]=0$, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det \text{Cartan}(E_8)|=1$ (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3+2i)=13$, Eisenstein $N(3+2\omega)=7$, cyclotomy $N(3+2\zeta_5)=55$, Galois $|(\mathbb{Z}/5)^\times|=4$, $|\det \text{Cartan}(E_8)|=1$, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=11$ ($2^4=16$), Coxeter $h(E_8)=30=2 \cdot 3 \cdot 5$ with $\varphi(30)=8$ — and the anchor $a=(1, 1, 2)$ is the compression generator. Extends v305.
- **New paper tfpt_safeguards (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (v427), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in build.sh, make_docs_map.py, make_manifest.py and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** run_all.py (421 modules), the registry, the ledger, tfpt_research_contracts (the F_{transfer} and keystone sections), tfpt_safeguards; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone SEAM.EQUIV.01: v424)

- **New module v424 (SEAM.EQUIV.LTORP.01).** An honest MMST-style reduction of the one open keystone — SEAM.EQUIV.01's intrinsic Bisognano–Wichmann lemma (the v308/v309 residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick

(arXiv:2605.10693, [NPW26]), with the central mapping *corrected*. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]=0$, v309/v199), not the flow $-\sigma^\psi=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26’s axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta=-K$, *distinct* from $\omega\circ\rho=\omega$: a positive character-block-diagonal K has $[\rho, K]=0$ yet never $\theta K\theta=-K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|=4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \text{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; SEAM.EQUIV.01 stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via v89/v281).

- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (arXiv:2605.10693), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (arXiv:2203.07421).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from QG.AMB.01: v423)

- **New module v423 (QGAMB.NODEP.01).** Sharpens the v369/v384 prose redundancy (“QG.AMB.01 is a [C] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ ($31 = 2^{g_{\text{car}}} - 1$), susceptibility $\chi = 729/665$ finite (negative control gap $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$); [E] QG.AMB.01 is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1} , the mixing angles and cosmology are each *computed* by a compiler script, v3/v9/v268/v7); [E] the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (PS.DIRAC $\rightarrow$ CONTRACT.QFT4D $\rightarrow$ QG.G2 $\rightarrow$ GATE.METRIC), a coherence edge, not a numerical input. [C] verdict: QG.AMB.01 is gap-decoupled certification — v369 as a DAG/script fact; honest scope — this does *not* discharge SEAM.EQUIV.01, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via v337).
- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open SEAM.EQUIV.01 premise is the *holomorphic* $|\det K|=1$ point that selects $(E_8)_1$ — the central charge $c=8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois $\leftrightarrow$ Net bridge: v422)

- **New module v422 (SEAM.GALOIS.NET.01).** The seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to G_{net} /SEAM.EQUIV.01. [E] $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor $4 = \text{cyclic, det } 4$); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2 \times \mathbb{Z}_2$ for n even, so the carrier D_5 (rank $5 = g_{\text{car}}$, odd) is cyclic while D_8 (rank 8, even) is Klein; [E] $(\mathbb{Z}/5)^\times = \langle 2 \rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathbb{Z}_4 \times \mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1, 1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (v89/v154); [E] negative control:

the Klein / order-2 $\langle(2, 2)\rangle$ gives $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16)$ (det 4), NOT E_8 (det 1) — cyclic $\mathbb{Z}/4 \rightarrow E_8$, Klein $\rightarrow D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\text{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored (324 $\rightarrow$ 327).

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide Q is rescaling-invariant ($Q(cm) = Q(m)$), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit $\frac{2}{3} - Q_{\text{src}} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to $\sim 5\%$), the PDG pole sits at $2/3$ (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$ per-lepton-log estimate shifts Q by EM-loop order toward $2/3$ (the v82 direction); [O] the exact flow time $t \approx 2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] verdict: F_{pole} is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (v327), external EM supplies the flow; F_pole stays [C], firewalled (v187), so F_{transfer} is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, RES.COXETER.SYMMETRY.01): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. [E] $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1, 2, 3, 5, 6, 10, 15, 30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; [E] by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \text{rank } E_8$), and the μ_4 ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, v316); [E] so the seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5 (v418) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. [C] this is the positive content of v409: the $2^2=4$ is the order of $(\mathbb{Z}/5)^\times$, and “no contraction rate” is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic $h=30$, Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321 $\rightarrow$ 324).
- **New figure coxeter_galois.pdf.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \text{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split $(3, 2)$ read in the three atom-rings gives $(7, 13, 55) = (\text{scalaron}, \Delta_Q, \text{quark numerator})$, and the carrier-5 clock missing

from **v417** is found as the 4×4 Φ_5 -companion C_5 . **[E]** $C_5^5 = I$, $\text{tr } C_5 = -1$, $\det C_5 = 1$ (the four primitive 5th roots), with the golden ratio as its real-part data ($2 \cos 72^\circ = 1/\varphi$, $2 \cos 144^\circ = -\varphi$; output-side, **v313/v349**; $\dim 4$, not 3 — this refines **v417**'s gap from “no operator” to “no 3×3 operator”); **[E]** $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5) = \det(3I+2C_5) = 55 = 5 \cdot 11 = g_{\text{car}} \| \text{Pl}(K) \|_1$, the c_u/c_d numerator (**v410/v411**); **[E]** the triple $\det(3I+2\text{Comp}(\Phi_n)) = (7, 13, 55)$ for $n=3, 4, 5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, **v416**); **[E]** NEG control: the anchor $(5, 4) \rightarrow (21, 41, 461)$ reaches named values only in the ω - and i -rings ($21 = N_\omega(5, 4)$, $41 = 10b_1$, **v222**) but 461 (prime) in the carrier ring, so the carrier ring separates $(3, 2) \rightarrow 55$ from $(5, 4) \rightarrow 461$. **[C]** honest: $(3, 2)$ is the *forced* carrier split (**v14**), not the unique clean one — a scan finds $\{(2, 1) \rightarrow (3, 5, 11), (3, 2) \rightarrow (7, 13, 55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only $(3, 2)$ spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while $(2, 1)$ is all compiler-core. Cited in **origin_theory**; Wolfram-mirrored ($318 \rightarrow 321$).

- **Surfaces synced.** **run_all.py**, **script_registry.csv**, **status_ledger.csv**, a triple paragraph in **origin_theory**, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: **v417**)

- **New module **v417** (DIAMOND.EISEN.01).** The hexagonal dual of **v415**: the $\tau=\omega$ flavor side put on operators and unified with the seam into one order-30 Coxeter clock. **[E]** the family rotation $P = (1\ 2\ 3)$ is the order-3 Eisenstein deck ($P^2 + P + I = \text{ONES}$, so $\omega^2 + \omega + 1 = 0$ on the plane $\perp (1, 1, 1)$), the order-3 dual of $J^2 = -I$; **[E]** it realises the hex CM norm as $(3I+2P)(3I+2P^2) = 7I+6\text{ONES}$, $\text{Spec}\{7, 7, 25\}$ ($7 = N_{\mathbb{Z}[\omega]}(3+2\omega) = \text{scalaron}$, $25 = g_{\text{car}}^2$) — the dual of **v415**'s $\{13, 13, 9\}$; **[E]** the CP clock $W = -P^2$ (ρ of **v233**) has $W^2 = P$, $W^3 = -I$, $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$, giving $\delta_{\text{CKM,lead}} = \arg \zeta_6 = \pi/3$ and $\delta_{\text{PMNS}} = \arg \zeta_6^4 = 4\pi/3$ (**v316/v320**); **[E]** all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15} = -1$, $\zeta_{30}^{10} = \omega$, $\zeta_{30}^6 = \zeta_5$, $\zeta_{30}^5 = \zeta_6$) while the seam $\mu_4 = i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30) = 8 = \text{rank } E_8$ — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants ($h=30$ and $\text{rank}=8$). **[C]** honest gap: the carrier μ_5 has no 3×3 operator (the golden φ is an output-side trace, **v313/v349**); negatives: no U, V -diamond operator carries ζ_6 , $[D, P] \neq 0$ (no μ_{12}), $N(3+2\zeta_6) = 19 \neq 7$. Cited in **tfpt_2**; Wolfram-mirrored ($315 \rightarrow 318$).
- **Surfaces synced.** **run_all.py**, **script_registry.csv**, **status_ledger.csv**, the Sheet-Diamond keybox of **tfpt_2**, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: **v415**, **v416**)

- **New module **v415** (DIAMOND.GAUSS.01).** The square CM-norm dictionary of **v222/v230** is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. **[E]** the integer commutator $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$ has $\text{Spec}(J) = \{i, -i, 0\}$, $J^2 = -I$ on the rank-2 image and $J^3 + J = 0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (**v410/v95**); **[E]** $3I + 2J$ and $5I + 4J$ have spectra $\{3+2i, 3-2i, 3\}$, $\{5+4i, 5-4i, 5\}$, so the Gaussian integers $3+2i$ (norm $13 = \Delta_Q$) and $5+4i$ (norm $41 = 10b_1$) are *literal eigenvalues*; **[E]** $(aI+bJ)(aI-bJ)$ reads (a^2+b^2) on the μ_4 -plane and the real part² on the kernel ($\{13, 13, 9\}$, $\{41, 41, 25\}$); **[E]** the intrinsic order-4 deck $D = -I + J - J^2$ ($D^4 = I$, $\chi_D = (x+1)(x^2+1)$) has χ_2 -line $\ker[U, V] = a - 1$. **[C]** the geometric-deck identification sharpens (does not close) the **v140** $\mathbb{Z}_3$ residue. Cited in **tfpt_2**; Wolfram-mirrored.
- **New module **v416** (ARITH.TRICHOTOMY.01).** The atoms $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings. **[E]** in $\mathbb{Z}[i]$ (square/seam) $2 = |\mathbb{Z}_2|$ ramifies, $3 = N_{\text{fam}}$ is inert, $5 = g_{\text{car}}$ splits as $(2+i)(2-i)$; **[E]** in $\mathbb{Z}[\omega]$ (hex/flavor) $3 = N_{\text{fam}}$ ramifies, 2, 5 are inert; **[E]** the ramified prime of each ring is its own atom; **[E]** the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$

of $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403) ramify over exactly one atom each, product $2 \cdot 3 \cdot 5 = 30 = h(E_8)$, two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flux/golden reading. Cited in `origin_theory`; Wolfram-mirrored.

- **Wolfram extension 308 $\rightarrow$ 315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns $-\chi$ for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis $V = Q \text{diag}(0, 1, 1)$ of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$ (proved by the inductive step $F(n)V = F(n+1)$), so $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$; [E] the four bilinear families $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$, $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$, $a^T V^n a = 11 \cdot 2^{n-1}$, $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$ give $(6, 7, 9, 11)$ at $n=1$ and $\mathbf{1}^T V^2 \mathbf{1} = 13 = \Delta_Q$, $\mathbf{1}^T V^3 \mathbf{1} = 27 = \mathbf{1}^T R a$, $\mathbf{1}^T V^4 \mathbf{1} = 55$, $\mathbf{1}^T V^4 a = 56 = \dim \mathbf{56}_{E_7}$; [C] the binary spine is FORCED by $\text{Spec}(V) = \{0, 1, 2\}$, so the carrier/Lie matches (rank $E_8 = 8$, $\dim S^+ = 16$, $248/8 = 31$) and the theta cross-link ($\sigma_3(2) = 9 = a^T V \mathbf{1}$, $\sigma_3(3) = 28 = \mathbf{1}^T V^3 a = \det(I+R)$, $\sigma_3(5) = 126$) are audit-typed (the v95 G_2 -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure V -power readout, $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1}) / ((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$, identical to the established $g_{\text{car}} \|\text{Pl}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$ (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the (U_{wall}) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the $\mathbb{Z}_2$ wall ($t = -2$, $\det = 2$, v135), $J = M(1, -2) = C - 2V$, completing the sheet axis $J, K, C, F = M(1, -2..1)$. [E] $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det) = (6, 3, 2)$; $a^T J a = 30 = h(E_8)$, $\det(I+J) = 12 = \dim \mathfrak{g}_{\text{SM}}$ and $\det(2I+J) = 40 = |R(D_5)|$; [C] the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On $M_t = C + tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1 = 3t + 12$, $e_2 = 2t^2 + 15t + 25$, $e_3 = 4t^2 + 14t + 14$ give $\Delta e_1 = 3 = N_{\text{fam}}$, $\Delta^2 e_2 = 4 = |\mu_4|$, $\Delta^2 e_3 = 8 = \text{rank } E_8$ (the e_3 curvature is the v218/v224 sheet quadratic; the full triple is new); [E] the anchor energy $a^T M_t a = 52 + 11t$ ($30 \rightarrow 41 \rightarrow 52 \rightarrow 63$, step $11 = a^T V a$); [C] atom/Lie readings audit-typed.
- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center $C = M(1, 0)$ is a resolvent portal: [E] $\det C = 14 = \dim G_2$, $\det(I+C) = 52 = \dim F_4$, $\det(2I+C) = 120 = |R^+(E_8)|$, the exceptional shell ladder $G_2 \rightarrow F_4 \rightarrow E_8^+$ as $\det(C+kI)$, $k=0, 1, 2$ (with $\text{tr } C = 12 = \chi_{E_8}(i)$, $\sum C = 31 = 2^{g_{\text{car}}} - 1$, v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \ver citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror 303 $\rightarrow$ 308.

2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHIO.PUNCTURE.01).** Names the φ_0 puncture term $48c_3^4$ as the order- $|\mu_4| = 4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture

divisor, summed over $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$ admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split $\varphi_0=1/(6\pi)+48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.

- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma=\text{diag}(1, -1, -1)$ has spectrum $\{\pm 1\}$ and projector $(I+\sigma)/2$ spectrum $\{0, 1\}$, none in $(0, 1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0, 1)$ and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4|=2^2$ the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O] $\rightarrow$ [E]/[C].
- **Axion branch consistency (tfpt_4_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$ (untuned, in band), while the seam hilltop $\theta_i\approx 170^\circ$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2=1$) is ± 1 , a projector eigenvalue ($r^2=r$) is 0 or 1, neither lies in the open interval $(0, 1)$, while $2/3$ does — so *no prime-2-only attractor exists*. Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms = propext, Classical.choice, Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11=\|Pl(K)\|_1=\det(1|a|\sigma)$ is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06–09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain v114–v117 keeps .06–.09 (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal `depends_on` carried along). No claim content changed.
- **Website.** `papers.ts` document count $8 \rightarrow 9$ (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E* $\rightarrow$ *A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7* $\rightarrow$ *Doc 8* for the research contracts and the residual interfaces re-typed v_{geo} [O], G_{net} [C], F_{transfer} [C]; *objections Paper 1* $\rightarrow$ *Doc 1*.
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [E] $\rightarrow$ [E]/[C] (given RP) and θ_{12} “genuinely open” $\rightarrow$ “conditionally derived”; `tfpt_2` Starobinsky A_s [E] $\rightarrow$ [E]/[C] (M exact, A_s a band over $N_\star \in [50, 60]$) and the Part-II abstract “opens the one remaining gap” $\rightarrow$ “closes the last structural gap”; `tfpt_research_contracts` the QFT4D “nonperturbative measure stays open” line and the “only U_{point} stays open” headline reconciled with the [C] redundancy / v_{geo} reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [C] redundancy; `introduction` (U_{wall}) $\rightarrow$ v_{geo} .

Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).

- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model (v_{geo} [O]; $G_{\text{net}}/\text{SEAM.EQUIV.01}$, F_{transfer} , N_* , QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the `qft_skeleton` “Stelle ghost” open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] `v405_seam_equiv_omega.py` (SEAM.EQUIV.02): the *dual keystone* at $\tau=\omega$ – the family/flavor sector as the order-3 Eisenstein/ A_2 face of E_8 , dual to the seam $= (E_8)_1$ at $\tau=i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; [E] A_2 is the $\tau=\omega$ Eisenstein lattice (Cartan $\det = 3 = N_{\text{fam}}$, $h = 3 = N_{\text{fam}}$); [E] R lives in $E_6 \times A_2$ ($248 = 78 + 8 + 2 \cdot 27 \cdot 3$, $\det R = 8 = \dim A_2 = \text{rank } E_8$); [E] $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau=\omega$ CM phase; [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\text{fam}} = \det Q$. [C] the family sector is the order-3 Eisenstein/ A_2 deck; the two keystones sit at the only two elliptic points of $\text{PSL}(2, \mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo} -class scale. Cited in `origin_theory`; Python-only.
- [E]/[C]/[O] `v406_pmns_phase_origin.py` (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the $\tau=\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo} -class unit, so PMNS has *zero free flavor dials beyond* v_{geo} . [E] the three angles (v9/v268) $\rightarrow$ unitary U_{PMNS} (v270); [C] $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau=\omega$ hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C] $J_{\text{PMNS}} = -0.0297$ derived; [E] the absolute scale = one seesaw ratio = v_{geo} -class (v272/v153). [O] residual = the seesaw operator realisation (M_R , v184) + the Σm_ν floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- [E]/[C]/[O] `v407_dn_pairings_omega.py` (FLAV.SELECTOR.CLOSE.01): the $R4'$ selector residue folds into the $\tau=\omega$ keystone – the three (d, n) pairings ARE the $\tau=\omega$ family-slice atoms. [E] the frame $(1, a, \sigma)$ has $\det = 11 = \|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n \cdot 1 = 2$, $n \cdot a = 8$, $n \cdot \sigma = 121 = 11^2$ ($\Rightarrow (d, n) \Rightarrow R$, v136/v139); [E] $n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6 \times A_2$, v405); [E] the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$. [C] deriving (d, n) folds into SEAM.EQUIV.02 (d anchor-derived, v139); [O] the lone residue “why $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8 -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times / (\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\text{fam}}$ (RM/static); [E] ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; [E] $2^{\omega(h)} = \text{rank}$ is E_8 -special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). [C] K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no

new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).

- **[E]/[C] v404_e8_char_cm_duality.py** (E8.CMDUAL.01): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending v282 ($\chi_{E_8}(i) = 12$) to the second elliptic point. **[E]** $\chi_{E_8}^3 = j$ (q-series); **[E]** SEAM $\tau=i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, v214/v282); **[E]** FAMILY $\tau=\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2 - \lambda + 1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, v220/v267); **[E]** $\tau=i, \tau=\omega$ are the *only* elliptic points of $\text{PSL}(2, \mathbb{Z})$, so the values are forced. **[C]** the seam/flux rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone SEAM.EQUIV.01 closes) while the flux/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: $U_{\text{wall}}, v_{\text{geo}}$, the CP texture), a candidate dual keystone at $\tau=\omega$. Cited in `origin_theory`; core values $j = 1728/j = 0$ already mirrored (v214/v220/v267/v282), Python-only.

2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- **[E]/[O] v398_seam_state_rigidity.py** (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. **[E]** RP-definability hinge (OS canonical transfer $T=e^{-H}$ from RP + reflection alone, v54, quasi-free v155, no μ_4 normal form); **[E]** band-limited character-block-diagonal core $[\rho, |k|]=0$ swept $N=4.64$ (beyond the 3-dim H^1); **[E]** holomorphy $\Rightarrow (E_8)_1$ ($c=8$, $|\det \text{Cartan}|=1$ vs 4, index $4=|\mu_4|$). **[O]** the one residual: $[\rho, \Lambda_\Sigma]=0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- **[E]/[C]/[O] v399_gravity_complete.py** (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete = Boundary-complete + Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). **[E]** the five gravitational readout classes (Newton 8π , Λ , scalaron c_3^7 , horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$; **[O]** residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- **[E]/[O] v400_grav_nonlocal_action.py** (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local $R+R^2$ as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). **[E]** entire $a=e^u \Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces $R+R^2$ with the atom-fixed scale $M_{\text{scal}}^2/M_{\text{Pl}}^2=c_3^7$ (v36/v253). **[O]** the entire- a assumption + perturbative-only.
- **[E]/[O] v401_metrology_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set $\{a, \pi, v_{\text{geo}}\}$ certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$; **[E]** the G -vs- Λ over-determination 0.11% (v274); **[O]** v_{geo} theorem-forbidden ($\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$).
- **[E]/[C] v402_ftransfer_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor F_{transfer} , the composition $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$ (consolidating the FR.*.SOLVE solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: v396)

- [E]/[C] **v396_phi0_icosahedral.py** (HYP.PHI0.01): the tree-level seed $\varphi_0^{\text{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\text{Aut}| \pi) = (F/(g_{\text{car}} N_{\text{fam}})) c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\text{Aut}| = 120$); $F/h = |Z_2|/N_{\text{fam}} = 2/3$ (the same survival ratio as v327). [E] Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (v216/v342). [E] honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. [C] φ_0^{tree} is a combinatorial reinterpretation, not a new independent injection; the puncture stays [O].

2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- [E] **v395_hypergraph_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a 9×3 labelled grid (affine- E_8 nodes $\times$ family channels) unifies v299 (carrier growth), v327 (branching rule M) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column $+ M$ on each node’s family vector ($T_{\text{micro}} = T_{\text{net}} \otimes M$); joint attractor marks $\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; v324’s $T_{\text{net}} \otimes T_{\text{cusp}}$ emerges as the cusp-readout basis. [O] the 3-arm seed (P_2) and analytic φ_0 remain irreducible (v312).

2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — do the “external” frontier scheme-numbers land on the $\{2, 3, 5\}$ clock? A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). [E] the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\text{car}}$ is ON the clock (v212); [E] the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ is ON the clock (v303); [E] the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ is ON the clock (v211/v373); [E] the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its $\pm 5.3\%$ band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). [C] verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number C_p . Cited in **tfpt_4_frontier**; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)

- **v394 (COXETER.ATOMS.01)** — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets. A bird’s-eye [E] re-reading tying v390 (the prime facets) to the anchor $a = (1, 1, 2)$ and v319 (the 5×6 clock). [E] $a = (1, 1, 2)$ has $e_3=2=|Z_2|$, $p_0=3=N_{\text{fam}}$, $e_2=5=g_{\text{car}}$, product $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ — the clock’s primes *are* the anchor’s atoms. [E] atoms = facet fields (v390): $2 \rightarrow \mathbb{Q}(i)$, $3 \rightarrow \mathbb{Q}(\sqrt{-3})$, $5 \rightarrow \mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] the QG susceptibility is pure $\{2, 3\}$: $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$, $(2/3)^6 = 2^6/3^6$, exponent $6 = 2N_{\text{fam}}$. [E] both dynamic rates carry prime-2: $2/3 = |Z_2|/N_{\text{fam}}$ and $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$ with $|\mu_4|=4=2^2$. [E] $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$ (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01** — a new registered open research question. Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector’s gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383’s “every sector is the same gapped operator” to “the same *seam-driven* operator” (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing

dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction’s correction, this gives the magnitude, so a parameter-free central value can be quoted as “value $\pm$ computed first correction”. [E] FIXED-POINT band (the φ_0 -derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; = θ_{12} ’s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$); [E] SEAM-GAPPED band (Koide F_{pole} , recovery) = $(2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337); [E] EXACT-IDENTITY band = 0 structurally ($\det R = 8, N_\Phi = 1, \theta_{\text{eff}} = 0$); [E] EXTERNAL-RATE band = external physics, quoted ($m_p/m_e \pm 7.5\%$ v374; n_s/r the $N_\star \in [50, 60]$ band; $\eta_B 1.07 \times$ v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01’s one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$); [E] a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin $(E_8)_1$ over the same- c $SO(16)_1$ (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01 — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): [E] `component_tendsto` (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \rightarrow 0$); [E] `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [E] `deviation_unique` (two starts give deviation vectors both $\rightarrow 0$, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [E] `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

2026-06-24 (Website: an animated “Universal Gap Lab” visualising v383–v391)

- **Website — the UniversalGapLab animation on /verification.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$ (seam-gapped, v303/v387/v388) or the golden $(\varphi + 2)/4$ (v312), with a gapless $r = 1$ track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: v391 reduces it, not a closure)

- **v391 (ALPHA.QUILLEN.PROGRESS.01) — an honest attempt at the external target ALPHA.QUILLEN.EXACT.01 (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays `[O]` (external math, type A, v384). `[E]` a solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle of circumference β equals $(2 \sinh(\beta m/2))^2$, whose m -variation $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). `[E]` so a ζ -determinant’s coupling-variation is structured heat-kernel data, and v382’s counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at $\alpha^{-1} = 137.0359992$. `[E]` the reduction: the open step now reduces to the named sub-target “the seam $U(1)$ operator’s Seeley–DeWitt coefficient $= 8b_1c_3^6$ ”. `[O]` *not closed*: the actual seam-operator ζ -determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, v336) stays the other external `[O]`; the value α^{-1} stays `[E]`. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01 — a Lean 4 proof of the gap $\Rightarrow$ unique-attractor $\Rightarrow$ parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction $x \mapsto x_\star + r(x - x_\star)$ with multiplier $r = \lambda_2/\lambda_1$. `[E]` `iter_sub`: the closed form $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$ (general n); `[E]` `iter_tendsto`: $0 \leq r < 1$ (the gap) $\Rightarrow \text{iter}_n \rightarrow x_\star$ (the unique attractor); `[E]` `starts_agree`: two starts differ by $r^n(a - b) \rightarrow 0$, so the attractor is independent of the initial state (parameter-freeness); `[E]` `no_gap_no_forget`: $r = 1$ (no gap) leaves $\text{iter}_n - x_\star = x_0 - x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; `[E]` the seam rate $(2/3)^6$ and the golden $(\varphi + 2)/4 = (5 + \sqrt{5})/8$ both lie in $[0, 1)$. Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in `origin_theory`; ledger row `FORM.SPECTRALGAP.01`.

2026-06-24 (Completing the Coxeter clock: v390 the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam) facet of $30 = 2 \cdot 3 \cdot 5$)

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ — the golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6 \in \mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333). `[E]` $h(E_8) = 30$ factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime:

$\text{disc } \mathbb{Q}(i) = -4$ (2 ramifies), $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$ (3 ramifies), $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$ (5 ramifies); the ramified primes are $\{2, 3, 5\}$ with product $30 = h(E_8)$. [E] the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly $x^2 + 1$, $j(i) = 1728 = 12^3$, $\tau = i$ fixed by the modular S of order $4 = 2^2$, $|\mu_4| = 4 = 2^2$ — the static, even, seam-orientation facet. [C] so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2, 3, 5\}$ is forced by $h(E_8) = 30$, not chosen. Cited in `origin_theory`; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the loop level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [E] super-polynomial falloff: $p^k e^{-p^2/M^2} \rightarrow 0$ for every degree k . [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2 dp$ diverges quadratically. [E] the superficial degree $\rightarrow -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\# \text{internal})p^2/M^2}$; dressed cutoff-ratio $\rightarrow 1$ vs GR $\rightarrow 4$). [E] the regulator is the seam scale $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C][O] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in `tfpt_4_frontier`; Python-only.

2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (v387) is a *typed* budget.** A machine-checked answer to “does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. [E] *seam-gapped* (Koide F_{pole} v303, recovery v221, the QG bound capped by $\chi = 729/665$ v337): rate $(2/3)^6$; the discrete compiler: golden $(\varphi + 2)/4$ (v312). [E] *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$): no λ_2 , so the correction is 0 structurally and a band would contradict the [E] status. [E] *external-rate* (η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only F_{pole} has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g. $\sin^2 \theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). [E] the Koide first correction (door 2): F_{pole} is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), the source quotient 0.33% below $2/3$ — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ($\sin^2 \theta_W = 3/8$ vs 0.231); [C]

the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, v253); [E] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \rightarrow \mu e$), *not* $p \rightarrow e^+ \pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\text{bound}} \sim 10^7$ – proton-safe, with *no* fake τ_p window (v265). Cited in `tfpt_5_redteam`; Python-only.

- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\text{scal}}$, v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ($\text{correction}_n \sim (\lambda_2/\lambda_1)^n$). [E] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi + 2)/4$ (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for F_{transfer} ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6 \ln(3/2)$ (v56/v82); the affine- E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2 \cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (v312/v313); the decoupling margin $\Delta_{\text{eff}} \approx 1.648$ (v76/v337). [E] the two number-field facets (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$. [C] the other sectors are the same structure — gravity ($\delta S=0$ equilibrium, v358/v359), the QG measure (Gaussian saddle, v365), the boundary QFT (holomorphic $c=8$ RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221), $\tau=i$ (v333). [C] the meta-theorem: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.
- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden (v_{geo} , the No-Unit theorem v153/v364), [E] (C) external physics (F_{transfer} , v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for S_{pert} , and v382 the Quillen determinant-line variation as a tracked target)

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \leq 4$ and the counterterm space is finite at every order; [E] BRST nilpotency $s^2=0$ verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [E] the seam gap $\Delta = 6 \log \frac{3}{2} > 0$ (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and S_{pert} is not the ambient measure QG.AMB.01 (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this* $F_{U(1)}$ functional” obligation (previously tracked only as EM.WARD.01’s residual, v341/v342) is elevated to its own citable target: $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [E] the Quillen split holds at the $\alpha^{-1}=137.0359992$ root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5)+q(A_3)=2$); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of SEAM.EQUIV.01 (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **v369 (QGAMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy $\Rightarrow \text{DHR}((E_8)_1) = \text{Vec}$: $\#\text{primaries} = |\det \text{Cartan}| = 1$ for E_8 vs 4 for $SO(16) \Rightarrow$ no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ($|\det K| = 1$); [E] finite Petz recovery rate $(2/3)^6$ (v221); [E] the gap-decoupling margin $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_\Sigma^{\mu_4}$ is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicity of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: `SEAM.EQUIV.01` is *closed modulo a cited theorem* (lattice v367/v368 + the S_3 stack v376–v379 + Lean `FORM.SEAM.MMST.01`; residual `[O]` = the cited MMST/OS existence only); `QG.AMB.01` is a `[C]` *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; `AUDIT OK`.

2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **v380 (`GRAV.KMS.HESSIAN.01`).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. `[E]` the seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). `[E]` each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \rightarrow u=-1$; $T_2 \rightarrow -1\pm i$); the entire a has none. `[E]` the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$ — the ghost decouples, “entire” = “do not truncate”. `[C]` $\Rightarrow$ perturbative graviton unitarity; `[O]` the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — `FORM.SEAM.MMST.01`)

- **`FORM.SEAM.MMST.01` — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, $\text{rank} = c = 8$, the range $8 \leq 8 \leq 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$ ’s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack v376–v379; so `SEAM.EQUIV.01` is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge $c=8$ from the lattice (v376), the $(E_8)_1$ character (v377), the genus-1 torus count = 1 (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)

- **v376 (`SEAM.S3.CENTRALCHARGE.01`).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the $L=4m+2$ sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\text{car}} + N_{\text{fam}}$, chiral ($c_- = 8$, v367/v368). Numerical $c=8$ from the lattice. Python (`numpy`).

- **v377 (SEAM.S3.E8CHARACTER.01).** The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16 \cdot 5 \cdot 3$) pins $(E_8)_1$ over the same- c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- **v378 (SEAM.S3.MODULAR.01).** The genus-1 discriminator: KLM condensation $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$, $|\det \text{Cartan}(E_8)|=1$ vs $D_8=4$, and the single character's modular T -eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the “hard part” v344 isolated. Python (sympy).
- **v379 (SEAM.S3.RP.01).** Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With $c=8 + (E_8)_1$ in hand, the OS inputs are complete and C7/QG.AMB.01 is then discharged by the redundancy theorem v369. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem v336), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (v375))

- **v375 (OBSERVATORY.REGISTRY.01).** A CI over `freeze_file.csv`: [E] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [E] the headline values re-derive from $\{\varphi_0^{\text{ret}}\}$ ($\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$, $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}$, $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$); [C] a live scorecard records θ_{12} vs the JUNO first run (0.28σ , compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ , the flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ , a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

2026-06-23 (Track 3 — the F_{transfer} functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371), η_B Boltzmann (v372), axion finite- T relic (v373), m_p/m_e uncertainty budget (v374))

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes Q above $2/3$; the required ϵ is negative while QED's is positive), so the transfer is the non-QED $53/54$ operator / $(2/3)^6$ Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen $M_1=M_{\text{sca}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ (v212) gives η_B within a factor ~ 1.1 of 6.1×10^{-10} with no M_R dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2 \sim 0.12$ (in $[0.08, 0.16]$, lands on Ω_{DM} untuned) while the hilltop 170° over-produces ($\sim 5.4\times$); replaces the v326 toy. [C]/[X](lattice $\chi(T)$ input).
- **v374 (FR.QCD.BUDGET.01).** The m_p/m_e uncertainty budget: with $b_3 = -7$ [E] and the two declared externals ($\alpha_s(M_Z)$, C_p , $\sim 5.3\%$ each), the central ~ 1824 (v262) carries a propagated $\pm 7.5\%$ band $[\sim 1690, \sim 1960]$ CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ($p+ip$) lattice model with numerical Chern $|C|=1$ realises the $c_-=8$ edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the $p+ip$ Bloch model is gapped at $M=1$ with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|=1$ ($C=0$ trivially at $M=3$); [E] $N_{\text{Maj}} = \dim S^+ = 16$ copies give $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x} |E| \sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ($c_- = 8$). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar $1/(p^2(p^2+M^2))$ pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ($\text{tr } P_2 = (d-2)(d+1)/2$, $\text{tr } P_1 = d-1$, $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$ sum to $d(d+1)/2$; $d=4$ split $5+3+1+1=10$), the EH propagator is $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour (v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304’s entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE: $\text{tr } P_2 = 5$ is dimension counting, not g_{car} (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST’s free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode convergence, finite model log-det $\text{Tr}' \log M = 6 \log \frac{9}{2}$; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ($\chi = 729/665$, v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.

- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST’s free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E] $D = \dim S^+ = 16$ Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8$, $\text{rank } E_8 = 8$ so $\text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in range); [E] gapped ($\Delta = 6 \log \frac{3}{2} > 0$, derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ($c_- = D/2 = 8 \neq 0$). [C] the MMST scaling-limit theorem then applies, and [E] the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K = 1$ vs $SO(16) \det K = 4$, v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the $R + R^2$ regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit v_{geo} . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume $\Rightarrow$ Einstein tensor, Lovelock $\Rightarrow$ conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 + v_{geo} residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, v_{geo} sharpened: after the parameter-free gravity, the *gravity* sector adds NO new dimensionful input either, so v_{geo} is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\text{geo}}, \pi\}$, ZERO dimensionless dials: v364)

- **v364 (VGEO.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made v_{geo} a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$ reduce to v_{geo} times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio) $\times$ (a power of v_{geo}); TFPT’s free content is $\{v_{\text{geo}}, \pi\}$ with ZERO dimensionless dials, a $\sim 26 \rightarrow 1$ reduction vs the SM. [O] v_{geo} stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier’s gravitational anomaly is forced $c_- = 8$, and the matter vacuum-energy backreaction is finite ($\rightarrow$ the forced Λ from α , not M_{Pl}^4) — the loop closes; no new SM-quantum-number read-out (declined): v361)

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT’s carrier matter is computed. [E] the carrier’s gravitational anomaly is forced: $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE $\rightarrow \Lambda$ from α : the matter vacuum energy through $G_{ab} = c_3^{-1} T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2)) e^{-2\alpha^{-1}}$ (v60), the 123 orders = 119.03 + 3.92 — *not* the M_{Pl}^4 catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit: $\delta \langle K_B \rangle = (8\pi^2 R^4/15) \delta \langle T_{00} \rangle$, carrier T_{ab} = the SM’s (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_- = 8$ and Λ — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap $\{1, (2/3)^6, (1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the R^2 scalaron (spectral-action $a_4 = R^2/72$, v36; $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^7$, exponent $7 = g_{\text{car}} + N_{\text{fam}} - 1$) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law $\rightarrow f(R) = R + R^2$, v28/v359). [E] it is ALREADY falsifiable via $n_s = 1 - 2/N_*$, $r = 12/N_*^2$ (live CMB-S4 kill test). [E] the gap $\Delta = 6 \ln \frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature² term sits at $\xi v_{\text{geo}} \sim$ the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent $7 = g + N - 1$ and the gap exponent $6 = 2N_{\text{fam}}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from v358, Λ from α via v60); B6 closed at the local level: v359)

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab} G_{ab} = (1 - d/2)R = -R$ in $d=4$, the structure that carries Λ . [E] Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a (G_{ab} + \Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. [E] coefficient 1: $8\pi G \rightarrow 8\pi = 1/c_3$ fixed (v358). [E] coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2)) e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions $\rightarrow$ the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate,

gap-decoupled); v_{geo} the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the Λ here matches the α -readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K \rangle$ with TFPT's atoms gives the LINEARISED Einstein equation with G from c_3 ; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement $\Rightarrow$ Einstein equation” with TFPT's atoms. [E] PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K \rangle$ gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3 -roles consistent (Unruh $1/(2\pi)=4c_3$, EH $1/(16\pi)=c_3/2$, Einstein $8\pi=1/c_3$, Bekenstein $1/4=1/|\mu_4|$) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law's coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — the thermodynamic and geometric origins of c_3 are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux $\delta\langle K_B \rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00} \rangle$. [E] ENTROPY DENSITY ATOM-FIXED: $1/4=1/|\mu_4|$ (v57) + central charge $c = g_{\text{car}} + N_{\text{fam}} = 8$ (form v59). [O] residual: linear $\rightarrow$ full non-linear (the original B6) + the v_{geo} unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ($\Delta=6\ln(3/2)>0$, v302) quasi-free 16-Majorana collar (v155/v160) $\rightarrow$ **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301) $\rightarrow$ **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) $\rightarrow$ **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly reduces the v336/v351 “does the collar have a chiral edge?” residual $\rightarrow$ **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, $8 \leq 8 \leq 16$) $\rightarrow$ **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3=1/(8\pi)$ a one-sided Gauss-Bonnet ($|\mathbb{Z}_2| \nmid K$, v58). [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.
- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta-Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs $\Phi=0$) and $\det R=8$ is read off — not a PDE. Only the absolute amplitude U_{point} (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum \exp = 120 = \# \text{pos roots}$; $\prod \deg = |W(E_8)| = 696,729,600$; exponents = the $\varphi(30)=8$ totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set; $\max \deg = 30 = h(E_8)$. [C] SUGGESTIVE, not a theorem: $12=h(E_6)$, $18=h(E_7)$, $30=h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219), but “ $\deg(E_8)$ contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches $14=\dim G_2$, $20=\det L$, $24=|W(A_3)|$, $12=|R(A_3)|$, $18=p_4(a)$ (v66 already flagged 12, 18 as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14}3^55^27$ “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8 ’s 8 Casimir invariant degrees only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of E_8 ’s eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents +1), exactly THREE feed a readout — 2 (metric), 8 ($\text{rank} \rightarrow c_3=1/(8\pi)$), 30 (Coxeter $\rightarrow g_{\text{car}}=5$ and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set ($\text{rank } 8$, $h=30$, $\det K=1$, $D_5 \oplus A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED $\Rightarrow$ NOT NUMEROLOGY: $\varphi = 2 \cos(\pi/5)$ appears only in internal structure (the affine- E_8 attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold ($h=30=2 \cdot 3 \cdot 5$), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly,

not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

2026-06-22 (the bird’s-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms $\rightarrow$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop: $g_{\text{car}}=5 = \text{max prime of the output } h(E_8)=30$, and the 8 in $c_3 = \text{rank } E_8$ – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K=4\pi$), not a dial. [C] this refines v352’s “framework + π ”: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the $c=8$ which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π : v351, v352)

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$ ambiguity” ($E_8 \det 1$ vs $SO(16)=D_8 \det 4$, both $c=8$, flagged open in v277/v344) is RESOLVED, non-circularly, by the seam’s ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5 \oplus A_3$ is E_8 (v154), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, v216) selects E_8 (full μ_4 condensation $16 \rightarrow 4 \rightarrow 1$) not $SO(16)$ (partial $\mathbb{Z}_2$ $16 \rightarrow 4$). The bootstrap agrees ($h(E_8)=30$ max prime $5=g_{\text{car}}$ vs $h(D_8)=14$ max prime 7). [E] so $\det K=1$ is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E] $g_{\text{car}}=5$ is forced three ways (v6); the 8 in c_3 four ways ($\text{rank } E_8 = h(D_5) = \phi(30) = \det R$, v54); the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework + π . Wolfram mirror +2 (297 $\rightarrow$ 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – $g_{\text{car}}=5$ IS the icosahedral 5 of $h(E_8)=30$, and the golden ratio is emergent from the μ_4 -glue, not external: v350)

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6): $g_{\text{car}}=5$ is over-determined three ways – rank-fill ($g_{\text{car}}+N_{\text{fam}}=8$), Coxeter-match ($g_{\text{car}} = \text{max prime of } h(E_8)=30$), Pascal ($2^4=C(5,0)+C(5,1)+C(5,2)$). [E] so $g_{\text{car}}=5$ IS the icosahedral 5-fold (the 5 in $h(E_8)=30=2\cdot 3\cdot 5$), not the D_5 rank; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue: D_5 ($h=8$)/ A_3 ($h=4$) are non-golden alone, but $D_5 \oplus A_3 \rightarrow E_8$ (μ_4 index-4, v154) gives $h(E_8)=30$ (golden) – the seam’s own μ_4 produces the 5-fold, so v349 checked the un-glued pieces, the wrong object,

and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 $\rightarrow$ 297).

2026-06-22 (the honest test – does the raw seam carry φ ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to “is $g_{\text{car}}=5$ a pentagon or a count?”: v349)

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has $h=8$ (eigenvalue $\sqrt{2}$) and A_3 has $h=4$ – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. [E] $5 \mid h$ holds only for $A_4=SU(5)$ ($h=5$), H_3 ($h=10$), E_8 ($h=30$) – the output/icosahedral side, so φ is the icosahedral INPUT. [E] φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). [O] the resolution: the keystone reduces to one question – is $g_{\text{car}}=5$ a PENTAGON (a 5-fold rotation $\Rightarrow$ golden $\Rightarrow$ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of $g_{\text{car}}=5$, not derivable from $g_{\text{car}}=5$ -as-a-count); there is no hidden φ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 $\rightarrow$ 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). [O] so the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. [O] honest subtlety: TFPT’s φ is currently E_8 -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 $\rightarrow$ 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom P_3 : v347)

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base $S^2(2, 2, 2, 2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2, 3, 5)$ ($\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier $(2, 3, 5)$ input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline

(v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 → 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam $\Rightarrow \det K=1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the (2,3,5) atoms: v346)

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam $\Rightarrow \det K=1$ as six links and shows exactly one is open. [E]/[C] L1 raw seam $\rightarrow \mu_4$ + four marks (Gauss–Bonnet, v216); [O] L2 μ_4 + the (2,3,5) atoms $\rightarrow$ the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3 $2I \rightarrow$ affine E_8 McKay (v219); [E] L4 $2I \rightarrow E_8$ du Val (v232); [E] L5 $\mathbb{C}^2/2I \rightarrow$ Poincaré link $H_1=0$ (v345); [E] L6 $H_1=0 \rightarrow \det K=1$. [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional $SU(2)$ axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \rightarrow 2T/2O/2I$; (2,3,5) unique to $2I$, and $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$ ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the (2,3,5)-rewrite attractor link computed to be the Poincaré homology sphere ($\det K=1$), the combinatorial core complete, only the geometric bridge open: v345)

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has $H_1 = \text{coker}(\text{Cartan})$. [E] the Smith normal form of the E_8 Cartan matrix is the identity, so $\text{coker}(E_8) = 0$ – the E_8 -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8=1$); among ADE only E_8 does this ($A_n \rightarrow \mathbb{Z}/(n+1)$, $D_n \rightarrow \mathbb{Z}/4$, $E_6 \rightarrow \mathbb{Z}/3$, $E_7 \rightarrow \mathbb{Z}/2$). [E] $\pi_1 = 2I = SL(2,5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4| h(E_8)$. So R3’s combinatorial half is [E]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K=1$), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 → 293).

2026-06-22 (the one keystone bit $\det K=1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)

- **v344 (SEAM.DETK.01).** A bird’s-eye, full-spectrum map of SEAM.EQUIV.01’s only open residual, the holomorphy bit $\det K=1$. [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link $H_1=0$ / one 1-dim irrep / $\det \text{Cartan} = 1$ / full μ_4 condensation / $\tau=i$ CM), unified by $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$ across ADE ($A_n \rightarrow n+1$, $D_n \rightarrow 4$, $E_6 \rightarrow 3$, $E_7 \rightarrow 2$, $E_8 \rightarrow 1$) – only E_8 gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87); $\det K$ is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3

hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2, 3, 5)$ rewrite attractor is the affine- E_8 graph (v312) and the E_8 du Val link is the Poincaré sphere (v232), so $\det K=1$ becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi = 729/665$) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ c ($c=8$ consistent, but the Carlip ambiguity and no holomorphy $\det K=1$) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E] $c_3 = 1/(8\pi)$ is the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $\chi=2$, v216). [E] the α^2 (Calderón) term is the Gilkey gauge-curvature coefficient ($30\Omega^2/360 = \frac{1}{12}\Omega^2$, $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$), so it is forced, not an ansatz. [E] the c_3 -orders $\{0, 3, 6\}$ are an arithmetic ladder and $2c_3^3 = 1/(256\pi^3)$ carries π^3 (three boundary insertions). [C] the exact coefficient needs the seam F -normalisation. [O] the cubic α^3 Maxwell moment (a topological/Chern level) + the full “this is the seam $U(1)$ ζ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01. α^{-1} stays [E] (v3). Wolfram mirror +1 (290 $\rightarrow$ 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the *form* of α , not more decimals”. The EM closure $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\text{seam}}) = 0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$. [E] every ingredient is a NAMED object, not a knob: $M=41$ is a $U(1)$ index ($\frac{4}{5}M = 8b_1$, $b_1=41/10$ the SM $U(1)_Y$ beta, EM Ward v48); $c_3=1/(8\pi)$ the Gauss–Bonnet boundary coefficient ($8=\text{rank } E_8$); $-5/4 = -q(D_5)$ a discriminant form and $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\text{base}}=3/4=q(A_3)$; $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$. The Quillen split is verified at the $\alpha^{-1}=137.0359992$ root. **No new gate:** α^{-1} stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that

residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289 → 290).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have A_s / the old versions derived it” and “can a ratio define mass itself”. [E] A_s is predicted and *dimensionless*: $A_s = N_\star^2 (M_{\text{scal}}/\bar{M}_{\text{Pl}})^2 / (24\pi^2) = N_\star^2 c_3^7 / (24\pi^2)$ with $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\text{Pl}}$ — it cannot define an absolute mass (No-Unit, v153, stands). [C] the -11.4σ at the slow Higgs channel ($N_\star=51.4$) is the slow-vs-fast reheating dichotomy; the A_s -preferred $N_\star=56.2 \sim$ the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124 \times 10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (v84) untouched.

2026-06-22 (the first-principles boundary, mapped: can M_{Planck} be computed? can F_{transfer} be first-principles? both NO, with the exact reason: v339)

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* — $v_{\text{geo}}/M_{\text{Planck}}$ is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E] F_{transfer} *cannot be made first-principles* either: for m_p/m_e the structure is in place (v262, carrier $b_3=-7$ + closed electron) but two inputs stay external. [X] $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ~ 8.8 at 2×10^{16} GeV, v246). The lattice $C_p \sim 2.83$ is parameter-free but lattice-only (v35); η_B /axion ride on cosmological initial conditions; Koide $Q=2/3$ is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: v336, v337, v338, FORM.CARTAN.DET.01)

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct c); WZW range $\text{rank} \leq c \leq D$, and $(E_8)_1$ ($c=8$, rank 8, $D=16$ Majoranas, $SO(16)_1 \rightarrow (E_8)_1$) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$. [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic $(E_8)_1$ net” ($\det K=1$), not a from-scratch construction.

- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1 - (2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure $\Rightarrow$ QG.AMB.01 decoupled; negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [E] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same $c=8$) but $4 = \det \text{Cartan}(D_8)$ is genuinely not a unit while $\det \text{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a **COROLLARY** of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4=1$) fixes it $\Rightarrow \omega \circ \rho = \omega$. So SEAM.EQUIV.01 $\Rightarrow$ QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ($c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] **QG.AMB.01 decoupled, not a gate:** margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) $\Rightarrow$ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items $2 \rightarrow 1$; live residual = $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$. [O] the one residual = SEAM.EQUIV.01’s continuum OS reconstruction.
- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance` ($= \text{QGEO.SYM.01}$) follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, `FAQ`, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)

- **v333_cm_selection.py** (QGE0.CMSELECT.01, [E]/[O]). The CM-selection mechanism for the last keystone residual: $\tau=i$ (the square) is the *unique* order-4 modulus — $S=\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $S^4=I$ and fixes i , and an order-4 deck selects $j=1728$ (vs the order-3 hexagon $j=0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau=i$ is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGE0.SYM.01). NOT a closure.
- **v334_conformal_resolution.py** (QGAMB.CONFORMAL.01, [E]/[C]/[O]). The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \rightarrow i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2}=\sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R+R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|=1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel **decide** / the integer-inverse argument. The remaining content (Witt’s uniqueness: E_8 is the only even unimodular lattice in $\dim < 16$) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|=1$ and the μ_4 commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)

- **v331_necessity_of_H.py** (QGE0.NECESS.01, [E]/[O]). The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam $\Lambda=|k|+M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau=i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho, \Lambda] \neq 0$. The square has cross-ratio 2 ($j=1728$), the unique 4-config with an order-4 automorphism (v214/v267), so $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$ and “necessity of H ” sharpens to: the raw dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.
- **v332_qgamb_metric_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D-1)(D-2)/4$ ($= -\frac{3}{2}$ at $D=4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial $R+R^2$ ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator $|\det \text{Cartan}(E_8)|=1$ (E_8 has one anyon, vs D_8 ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by **decide** on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4=1$, $i^2=-1$, and (on

concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)

- **v329_os_gap_reduction.py** (QGEO.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6 \ln \frac{3}{2}$, *independent of system size*. Together with $H \Rightarrow \omega \circ \rho = \omega$ and $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.
- **v330_qgamb_admissible_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n-1) \rightarrow (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\text{OS}} \geq 0$ with gap $6 \ln \frac{3}{2}$. The full QG_{amb} (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det \text{Cartan}(A_3)|=4=|\mu_4|$ (3×3 , `det_fin_three`) and $|\det \text{Cartan}(D_5)|=4$ (5×5 , `decide`), giving the carrier $|\det \text{Gram}|=4 \cdot 4=16$ (the v281 anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum` $\rightarrow$ `GeometricModularFlow` $\rightarrow$ `Mu4ModularSymmetry` $\rightarrow$ `StateInvariance` ($=\text{QGEO.SYM.01}$) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325_pillowcase_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau=i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho = \omega$, holomorphic $c=8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\text{orb}}=0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326_fttransfer_suite.py** (FR.SUITE.01, [E]/[C]). The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam

rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B , H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e , RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (v187/v303).

- **v327_hypergraph_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight $2/3$ is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6 = 64/729$ is the recovery gap; with a PROOF that $2/3$ is not a root of the affine- E_8 charpoly ($p(2/3) = -3520/19683 \neq 0$, sharpening v312). v324's injected datum is reduced to the anchor arity $\{2, 3\}$.
- **v328_theta13_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified: $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent $5/6$ exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.
- **Wolfram mirror** 285 $\rightarrow$ 288. v325 ($\chi_{\text{orb}}=0$) and v327 (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6 = 64/729$, and the non-graph-spectral proof) added to **tfpt_readouts_extension.wl**; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates (U_{wall}) and G_{metric} ” framing was replaced by the current $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, **tfpt_4_frontier**, **origin_theory**, **tfpt_5_redteam**, **tfpt_research_contracts**, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs**. Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the **origin_theory** “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the **tfpt_5_redteam** Target-A Outcome; and on the website the FAQ “What is still open?”, the **OpenGates** closing-condition block, the **papers.ts** research-contracts/red-team entries, the verification-DAG **qft** node, the glossary **G_net** entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected**. Version label 5.0 $\rightarrow$ v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886 $\rightarrow$ 3357 jobs (FORM.SEAMEQUIV.01); README Python suite v213/212 $\rightarrow$ v324/322 modules; Wolfram extension 269/269 / 249/249 $\rightarrow$ 285/285 (README, website replication page, **zenodo_description.html**); residual “Paper-3” old-paper attributions in **tfpt_2** neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)

- **Bug fix — δ_{CKM} display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD = 1.198232` rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to 68.65° . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202_rare_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an F_{transfer} bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new F_{transfer} bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ($\pi/3$) component of the quark phase”: the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$, so $\delta_{\text{PMNS}} = 240^\circ$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305_witness_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (`v314/v315`).
- **v203_eht_achromatic.py — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \pmod{4}$ (the μ_4 fingerprint of the four marks, `v201`); a μ_4 -orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.
- **Seed-hyperplane figure (Fig. ??).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed φ_0 , the error-weighted mean matching the axiom to 0.04%, with $\sin^2 \theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, `275 → 285`, *all passing*): `v313` (the affine- E_8 characteristic polynomial with the golden factor $x^2 - x - 1$; the $(2, 3, 5)$ atoms as $2 \cos(\pi/k)$; the $31/30$ selection), `v314` (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), `v315` ($30=5 \cdot 6$, $\varphi(30)=8$, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), `v316` ($\rho=\zeta_6=\zeta_{30}^5$, $\rho^4=-\rho$, $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$), `v317` ($N_{\text{fam}}=3$ as the μ_3 orbit, the $1+2$ Galois split), `v318` ($\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$ a pure function of π) and `v320` (the CP lock $\delta_{\text{PMNS}}=240^\circ$). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to `285/285`.
- **v322_cp_lock_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The `v320` CP-lock sharpened into a quantitative, pre-registered kill test. `[E]` the leading phase sits on

the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$, and *which* one is selected is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$ (node 4); [C] the sub-leading correction is bounded by the quark analogue $3\lambda^2 \approx 8.7^\circ$, so the prediction is the band $240^\circ \pm \sim 9^\circ$; [C] against NuFIT 6.0 (NO, $\delta_{CP} = 212^{+26}_{-41}$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; [X] the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim 5\text{--}15^\circ$) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.

- **v323_bw_geometric_modular.py** (`QGE0.BW.01`, [E]/[C]/[O]). The keystone pushed one honest step via Bisognano–Wichmann. [E] the μ_4 clock is a geometric rotation $\rho = \text{diag}(i^n) = \exp(i\frac{\pi}{2}L)$, $\mu_4 \subset U(1)_{\text{rot}}$; [E] for a rotation-covariant covariance $C = f(L)$ the modular Hamiltonian $K = \log((1-C)/C) = g(L)$, so $[K, L] = 0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K, L] \neq 0$); [C] given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGE0.SYM.01` ($\omega \circ \rho = \omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGE0.BW.01`.
- **v324_hypergraph_fiber.py** (`HYP.FIBER.01`, [E]/[C]). The constructive v312 follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. [E] the carrier network $T_{\text{net}} = (A + 2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\text{cusp}} = \text{diag}((1-w)^{2N_{\text{fam}}})$, $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (spectrum $\{1, (2/3)^6, (1/3)^6\}$); the fibred $T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes (w=0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes (w=\frac{1}{3})$) – the $N_{\text{fam}}=3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \times 6$ of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of $2/3$ from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321_killtest_board.py** (`KILL.BOARD.01`, [E]). The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r > 0.01$), #3 DESI/Euclid w (kill $w \neq -1$), #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row `KILL.BOARD.01`.

2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (`FORM.SEAMEQUIV.01`, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$ assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Muger / Kawahigashi–Longo–Muger holomorphy; Conway–Sloane E_8 uniqueness) as named axioms and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a *decide*-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the $|\det \text{Cartan}|$ that separates the holomorphic E_8 from the same- c rival $D_8 = SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning

the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$: v320)

- **v320_galois_cp_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ (the $\mathbb{Z}_2$ sheet flip); [C] this upgrades the previously assigned $\delta_{\text{PMNS}} = 240^\circ$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a δ_{PMNS} robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6 , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319_translation_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, v315). [E] the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\text{fam}}$ (ticks $0, \dots, 5$): the recovery exponent is exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, v316). [E] position 0 is the *conserved law* — the resummed clock rate(n) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from $n=1$, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \dots, 29\}$), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is $\mathbb{Z}/5 = g_{\text{car}}$ (ticks $1, \dots, 5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2 \cos(\pi/5)$), field-obstructed (no rational dynamic image, v314) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (v55). [C] verdict: the discrete $\rightarrow$ dynamic translation *is* the order-30 clock = (static 5-ring) $\times$ (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\text{fam}}$), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row TRANSLATE.CLOCK.01.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ — zero dimensionless free parameters: v318)

- **v318_arithmetic_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 = \text{rank } E_8$, $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$), forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] the magnitude seed $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (ARCH.CORE.01), v_{geo} the unique dimensionful input (No-Unit

Theorem) — complete input $\{a, \pi, v_{\text{geo}}\}$. [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGE0.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in `origin_theory`; ledger row ARITH.CAPSTONE.01.

2026-06-21 (are the 3 generations a μ_3 /Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **v317_galois_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\text{Spec}(Q_+) = \{1, 2, 3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w + \frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$); [E] the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the v316 CP conjugation) refines the orbit to 1+2 — fixing $w=0$ (rational) and swapping $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row GALOIS.FAMILY.01.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: v316)

- **v316_galois_readout.py** (GALOIS.READOUT.01, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-6 = $2N_{\text{fam}}$ factor c^5 of v315 (the carrier being $\zeta_5 = \zeta_{30}^6$); [C] the leading CP phases are its cyclotomic arguments $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$ and $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$ (power 4 = $|\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits v231/v233); [E] the family Galois $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) is CP conjugation $\delta \mapsto -\delta$; [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in `origin_theory`; ledger row GALOIS.READOUT.01.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: v315)

- **v315_coxeter_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2 \cdot 3$, rational) sectors, or just factorize? [E] the cyclic group *factorizes*: $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the v314 field obstruction); [E] the E_8 exponents = totatives(30) split as totatives(5) $\times$ totatives(6), $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$; [E] but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree $8 = \text{rank } E_8$; [E] the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining v223); [E] $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row COX.COUPLE.01.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: v314)

- **v314_rate_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- E_8 network, carrying the golden ratio) and the dynamic part (the

seam transfer, carrying the recovery rate $(2/3)^6$) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$ plus the golden part $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$ and $2 \cos(\pi/3)=1 = N_{\text{fam}}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$, same field, same atoms), but [E] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction ($2/3$ is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2 - \phi) \approx 6.37$ not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, and the ϕ -quasicrystal $E_8 = H_4 + \phi H_4$ ($240 = 2 \cdot 120$) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the $(2, 3, 5)$ atoms are the affine- E_8 network spectral angles, $\phi = 2 \cos(\pi/5)$ the $g_{\text{car}}=5$ signature: **v313**)

- **v313_golden_atoms_spectral.py** (`GOLD.ATOMS.01`, [E]/[C]). A follow-up to v312’s observation, made exact. [E] the affine- E_8 adjacency characteristic polynomial factors as $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, so the golden minimal polynomial x^2-x-1 divides it and $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$ is an *exact* eigenvalue; [E] the non-trivial eigenvalues are $2 \cos(\pi/k)$ for exactly $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ ($2 \cos(\pi/2)=0$, $2 \cos(\pi/3)=1$, $2 \cos(\pi/5)=\phi$, $2 \cos(2\pi/5)=1/\phi$), so the golden ratio is *specifically* the $g_{\text{car}}=5$ (5-fold) signature and $g_{\text{car}}=5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the $(2, 3, 5)$ spherical excess $1/2+1/3+1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the v63/v76 gap-margin numerator $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [C] direction: ϕ is the E_8 -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in `origin_theory`; ledger row `GOLD.ATOMS.01`.

2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: **v308–v312**)

- **v308_seam_equiv_chain.py** (`SEAM.EQUIV.CHAIN.01`, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic $c=8$ $(E_8)_1$ net” assembled into ONE well-typed logical chain $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$, with the exact discriminators $|\det \text{Cartan}(E_8)|=1$ vs $|\det \text{Cartan}(D_8=SO(16))|=4$, the carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ and $c=g_{\text{car}}+N_{\text{fam}}=8$ all machine-checked; the input gap is the derived recovery gap $6 \ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row `SEAM.EQUIV.CHAIN.01`.
- **v309_modular_bedrock.py** (`QGEO.MODHAM.01`, [E]/[O]). TOE attack point 2: the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of `QGEO.SYM.01`. Cited in `tfpt_research_contracts`; ledger row `QGEO.MODHAM.01`.
- **v310_carrier_sm_anomaly.py** (`CAR.SM.ANOM.01`, [E]). TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$ is exactly one anomaly-free Standard-Model generation

($SU(5) \subset SO(10)$: $10+\bar{5}+1$), with $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1 , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ β sub-question; the interacting Epstein–Glaser S -matrix and the Yukawa sector stay $[C]/[O]$. Cited in `tfpt_research_contracts`; ledger row `CAR.SM.ANOM.01`.

- **v311_gap_decoupled_measure.py** (`QGAMB.CLUSTER.01`, $[E]/[C]/[O]$). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6 \ln(3/2))$, finite susceptibility $729/665$), and the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. `QG.AMB.01` stays $[O]$ (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row `QGAMB.CLUSTER.01`.
- **v312_hypergraph_substrate.py** (`HYP.INJECT.01`, $[E]/[O]$). TOE attack point 5: the hypergraph substrate sharpened — the affine- E_8 Kac marks are the Perron eigenvector ($A m=2m$) and the subleading eigenvalue is exactly $2 \cos(\pi/5)$ = the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$ and the analytic seed φ_0 . The open question is precisely delimited, not closed. Cited in `introduction`; ledger row `HYP.INJECT.01`.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305_witness_independence.py** (`WITNESS.ECON.01`, $[E]$). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ($|\mu_4|=4$, $g_{\text{car}}=5$, $|\mathbb{Z}_2|=2$, $N_{\text{fam}}=3$, $\dim S^+=16$, $\text{rank } E_8=8$, 240, 248, $|2I|=120$, $h(E_8)=30$, $|W(D_5)|=1920$, $\Omega_{\text{adm}}=48$, the gap exponent 6) is a documented image of the single anchor $a=(1, 1, 2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, $6.5 \times$ compression made explicit), so the recurring $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by v100, never multiplied); the negative control shows α^{-1} ’s “137” is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, v3). Cited in the `tfpt_5_redteam` body; ledger row `WITNESS.ECON.01`. Python-only.
- **v306_seed_crossval.py** (`SEED.CROSSVAL.01`, $[E]$). A leave-one-out cross-validation turning the “many observables, one seed φ_0 ” worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$, β , Ω_b , λ_C ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2 \theta_{13}$, $\sim 2\sigma$); a data $\leftrightarrow$ formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row `SEED.CROSSVAL.01`. Python-only.
- **v307_data_watchdog.py** (`DATA.WATCHDOG.01`, $[E]$). The operational decision pipeline over the frozen registry (v84): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2 \theta_{13}$ (2.0σ , the honest current pressure point); $\sin^2 \theta_{12} = 0.306747$ is compatible (0.02σ , the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and $w=-1$ is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.

- **Target-A holomorphy kill test (freeze_file.csv).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same- c rival $SO(16)_1$) would break holomorphy and kill “seam net = $(E_8)_1$ ” — since holomorphic $c=8 \Leftrightarrow E_8$ is machine-proven (v83/v143) and $\det K=1 \Leftrightarrow$ invertible/SRE (v237/v301/v302).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of c ” framing of SEAM.EQUIV.01: v304)

- **v304_idg_nonlocal_ghost.py (QGAMB.IDG.01, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F/v278` Stelle-ghost attack on the perturbative gravity sector. [E] the local $R+R^2/\text{Weyl}^2$ four-derivative propagator $1/(p^2(p^2+M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close QG.AMB.01: the trace regulator is not yet shown to resum to an entire $a(\square)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row QGAMB.IDG.01.
- **Double-pushout figure (tfpt_1_architecture_e8).** A new shared TikZ macro (TFPTdoublepushout in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order- $|\mu_4|=4$ glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (v1) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (introduction, tfpt_research_contracts).** SEAM.EQUIV.01 is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. v267). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on F_transfer: one shape, one seam-rate, v303)

- **v303_ftransfer_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $\rightarrow$ the E_8 marks, gap $6 \ln(3/2)$) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [E] `F_pole` (Koide) *is* the seam dynamics (Möbius multiplier $F'=(2/3)^6=\lambda_2$, the seam-clock eigenvalue $\Rightarrow$ a unique attractor at the seam rate, v82/v54); [C] `F_Boltzmann` (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); [C] `F_relic` (axion) freezes at an adiabatic fixed point; [E] `F_QCD` (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; `F_pole` has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory’s simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite v303 5/5, AUDIT OK.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$, v302)

- **v302_seam_gap.py** closes the chain on the single statement v301 had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum

is $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1 - (2/3)^6 = 665/729$; $(2/3)^6$ the μ_4 -deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, SEAM.EQUIV.01’s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6 \ln(3/2)$) — *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite v302 5/5, AUDIT OK.

2026-06-19 (Route A’s last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301_route_a_invertible.py** attacks the single remaining *unconditional* gap of SEAM.EQUIV.01. Since v300 collapsed Route B into Route A’s rationality, that gap is Route A’s open hypothesis (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev’s periodic table, class D). [E] the carrier is 16 Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$ (v148); [E] $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant $|\det K| = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce. [O] LIT-A’s ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of SEAM.EQUIV.01 reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. [E] suite v301 6/6, AUDIT OK.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, v300)

- **v300_flataway_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum (“the spectral mismatch grows with ε ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint — $\text{spec}(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy $2 \times 2 \llbracket [2k, g/2], [g/2, 2k] \rrbracket \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy ($2 \rightarrow 1+1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* [C] the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q -coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches E_8 as the last finite point, v299)

- **v299_hypergraph_0d_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): **[E]** the icosahedron is a 3-uniform hypergraph ($|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8 ; 30 edges $= h(E_8)$); **[E]** the $1 \rightarrow 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith ($\rho \leq 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \leq 2 \Leftrightarrow$ a local witness $Am \leq 2m$); **[E]*** the autonomous rule (diffusion-generated witness + *local* gate $Am \leq 2m$, Perron only as audit) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$; **[E]** the negative controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8 . Honest type: **[O]** the $(2, 3, \cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* **[E]** build green, AUDIT OK.

2026-06-18 (E_8 as a network attractor — the Wolfram-program reading, v298)

- **v298_e8_network_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. **[E]** the affine- E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4m$ on the $(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. **[C]** the Wolfram reading “ E_8 is the universal attractor of a minimal $(2, 3, 5)$ network” holds at the Coxeter-skeleton level. **[O]** a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. **[E]** build green, AUDIT OK.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **v296_flataway_a2_closed.py (FLATAWAY.A2.CLOSED.01)** evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$ -term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small- t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.
- **v297_route_a_literature_stack.py (SEAM.EQUIV.A.LIT.01)** writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) $\rightarrow$ LIT-B (Müger; Kawahigashi–Longo–Müger) $\rightarrow$ LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). **[E]** build green, AUDIT OK.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295_flataway_a2_exact.py (FLATAWAY.A2.01)** upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\text{Tr } e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon=0$ is the *global* minimum — so $\Delta \text{Tr} \geq 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean FORM.FLATAWAY.01 (SeamDeckClosure.lean Part 5)**: the formal $\Delta=0 \Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (**heatDeviation_eq_zero_iff**); built and axiom-checked (only propext, Classical.choice, Quot.sound). So the heat route’s

positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [E] build green, AUDIT OK.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292_flataway_heat_reduction.py (FLATAWAY.HEAT.01)**: the heat route, made precise. The heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact “the seam fixes the a_2 heat coefficient”.
- **v293_flataway_spectral_hessian.py (FLATAWAY.SPEC.01)**: the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294_flataway_rp_energy.py (FLATAWAY.ENERGY.01)**: the variational route. With the rigid mark divisor (4 cone angles π , Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290_z4_smooth_curvature_adversary.py (SEAM.ADVERSARY.01)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon \cos(4\theta)$ [E] passes the v288 commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [E] shifts the DtN spectrum ($\max |\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]=0$ — which is exactly why Flat-Away cannot be read off the commutator.
- **v291_flataway_contract.py (FLATAWAY.RP.01)** states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon=0$) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289_marklocal_raw.py (MARKLOCAL.RAW.01)** decomposes the one remaining Route-B residual (“why is the raw seam sub-principal term mark-local?”) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks $\Rightarrow$ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]=0$ (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- L^2 QGEO proved’.

- **v287 literature matrix:** L3 (‘invertible bulk $\Rightarrow$ single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286_seam_equivalence_contract.py (SEAM.EQUIV.01)** names the one open theorem — *the raw RP seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ E_8 into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287_free_rp_bulk_to_holomorphic_boundary.py (SEAM.EQUIV.A01):** Route A (AQFT) dependency checker — the 5-lemma chain $\text{RP}+\text{gap}+\text{chirality}+\text{Gaussian}+\text{invertible} \Rightarrow \mu=1 \Rightarrow \text{holomorphic} \Rightarrow (E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary* (the AQFT form of $\det K=1$). 4/5 discharged.
- **v288_full_12_subprincipal_z4.py (SEAM.EQUIV.B01):** Route B (DtN) — *proves* the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod{4}$, so $\Lambda = |n| + M_f$ commutes with $\rho = \text{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$; generic curvature $\rightarrow 4.44$), lifting v201/v284 to L^2 . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284_route_i_rp_uniqueness.py (QGEO.ROUTEI.01)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 $\text{RP} \rightarrow \text{DtN}$ (v194), L2 $\text{marks} \rightarrow \text{pillowcase}$ (v195/v216/v214), L3 flat Steklov $= \sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov ($\chi_{\text{orb}}=0 \Rightarrow$ unique flat $\tau=i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.
- **v285_route_ii_seam_condensation.py (QGAMB.ROUTEII.01)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector $\Leftrightarrow \det K=1 \Leftrightarrow$ holomorphic $\Leftrightarrow E_8$; the tower $16 \rightarrow 4 \rightarrow 1$), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the $(E_8)_1$ -at- $\tau=i$ object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. Build green, AUDIT OK.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E_8 -at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280_pillowcase_steklov.py (QGEO.STEKLOV.01)** runs the QGEO chain on the actual flat $\tau=i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281_holomorphy_modular_data.py (QGAMB.MODULAR.01):** $\# \text{anyons} = |\det \text{Gram}|$ ($E_8 \rightarrow 1$, $\text{SO}(16) \rightarrow 4$), the anyon-condensation tower $16 \rightarrow 4 \rightarrow 1$, Gauss–Milgram $\rightarrow c=8$; holomorphy ($\det K=1$) is the single Tier-B discriminator.

- **v282_e8_tau_i_unification.py (QGAMB.UNIFY.01):** the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \rightarrow 1$.
- **v283_live_killtest_scorecard.py (PRED.LIVE.01):** the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279_qgeo_obligation_lemma.py (QGEO.OBLIG.01) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Trojanov-unique, $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow \text{mark-locality} \rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278_lszy_bridge_unitarity.py (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** [E] LSZ bridge: the EG T -products of S_{pert} (v269) are the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta = 6 \log(3/2) > 0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$ *exactly* = the two-body phase space, so $2 \text{Im } M = \sum \int d\Pi |M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector's Stelle ghost (v269/rt_F) is non-unitary $\rightarrow$ QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277_seam_calderon_e8_match.py (QGAMB.TIERB.01) reduces Tier-B to ONE bit.** [E] the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248 = \dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det \text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. [C] the seam side matches via v276 (flat $\tau=i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276_qgeo_flat_closes_commutator.py (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]=0$. [E] if $H = f(\Delta_{\text{flat}})$ and the order-4 deck ρ ($z \mapsto iz$) is a flat- $\tau=i$ isometry, then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ *exactly*, to all orders (verified on the discretised square torus: $\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H . [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat $\tau=i$ pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **redteam/rt_F_qft4d.py (REDTEAM.F.01) stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the non-perturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the Λ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the concrete one-loop quartic v271 and gauge β -coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν -mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, `run_all.py` green, AUDIT OK, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272_nu_mass_scale.py (FLAV.NUSCALE.01) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo} , v153); [C] the observed $m_3=0.050$ eV with the TFPT PS→GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; [X] the NO floor $\Sigma m_\nu=0.0586$ eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **v273_eg_gauge_running.py (QFT4D.SPERT.03) derives the SM one-loop b_i via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same

41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.

- **v274_scale_overdetermination.py (SCALE.OVERDET.01) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to $\bar{M}_{\text{Pl}}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438 \times 10^{18}$ GeV, to 0.11%; the seam being a conformal $c=8$ CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- **v275_qgamb_roadmap.py (QGAMB.ROADMAP.01) gives the open gate QG.AMB.01 a module.** [E] Tier A gap decoupling ($\Delta_{\text{eff}} = 1.648 > 0$, v36/v76); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, run_all green, AUDIT OK, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **v271_eg_oneloop_quartic.py (QFT4D.SPERT.02) takes the v269 S_{pert} skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product T_2 ; the massless propagator has scaling degree $\text{sd}=2$ in $d=4$, the one-loop bubble has $\text{sd}=4=d$, so $\omega=\text{sd}-d=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d, d)=1$, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270_pmns_jarlskog_assembly.py (FLAV.PMNS.03) assembles the four TFPT channels** ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, $\sin^2 \theta_{23} = \frac{1}{2}$, $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$, $\delta=4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ ($J_{\text{max}} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative CP-violating sign reproduced. [O] δ is the $\mu_6/\text{triality}$ phase (assembled, not from M_ν/D_F) and the absolute ν -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269_spert_paqft_skeleton.py (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: S_{top} (2d DHR braiding $|M|=1$, statistics, v243), S_{pert} (4d Epstein–Glaser/BV S-matrix of the spectral action), S_{phys} (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators $\dim \leq 4$; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6 \log(3/2)$ gives an IR-safe adiabatic limit ($S_{\text{pert}} \rightarrow S_{\text{phys}}$ via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268_theta13_carrier_trace.py (FLAV.TH13.01) closes the θ_{13} exponent origin.** The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \text{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$ -breaking $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) “ θ_{13} from M_ν ” route flagged by the new script atlas. [O] the CP magnitude and the absolute ν -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make_script_atlas.py generates verification/script_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (status_ledger.csv for claim→status→dependencies→supersedes, script_registry.csv/script_clusters.csv for the theme grouping, docs_map.csv for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into bash build.sh gen. [E] build green, AUDIT OK.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267_qgeo_rigidity_minimal_axiom.py (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks” ($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a ; cross-ratio 2 $\Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on $y^2=x^3-x$; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]=0$ and $\omega \circ \rho = \omega$ all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal $j=0$ ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with $j=1728$ (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$) — the *reduction* is closed (axiom $\Rightarrow$ everything), the axiom is not. [E] build green, AUDIT OK.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266_ps_threshold_proton.py (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the v265 fork.** The explicit two-step Pati–Salam thresholds give $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ (writes ps_threshold_proton.json): [X] the minimal 16-Higgs content ($M_{GUT} \sim 2.4 \times 10^{15}$) gives $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K and is *killed* even at the high hadronic band; [C] the E_8 -allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$ yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ($\sim 1.4 \times 10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be* $+(15, 1, 1)$), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265_qft4d_fork_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (qft4d_fork_freeze.json), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). **[X]** SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1=\alpha_2$ scale α_3^{-1} off by ~ 5.6 , spread ~ 8.8 at 2×10^{16} GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8 -allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2 \theta_W = 3/8$ (v248), the $O(1)$ $\kappa = M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test (**[X]**, no fake τ_p window; carrier gauging stays the contract). **[E]** a prose guard (QFT4D.NO_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to tfpt_research_contracts; **[E]** build green, AUDIT OK.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262_fqcd_mp_me.py** (FR.MPME.02) implements the one contract-only F_{transfer} pipeline. A real two-loop α_s solver runs $\alpha_s(M_Z) = 0.1179$ with n_f thresholds (carrier $b_3 = -7$, **[E]**) to $\alpha_s(2 \text{ GeV}) \sim 0.30$ and $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A **[C]** transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 (**[O]**) to FR.MPME.02 (**[C]**).
- **v263_mnu_from_df.py** (FLAV.PMNS.02) makes M_ν a seesaw operator of D_F . **[E]** $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$ -symmetric inputs) $\mu\tau$ -symmetric; **[C]** it reproduces $\theta_{23} = 45^\circ$ and $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; **[O]** $\theta_{13} = 0$ in the leading texture, the CP *magnitude* (240°) is the μ_6 /triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **v264_raw_seam_marklocal.py** (QGEO.ENERGY.03) sharpens the bedrock. Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$ -invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda] = 0 \Rightarrow \omega \circ \rho = \omega$ on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, **[O]** — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in tfpt_4_frontier, tfpt_2_standard_model and tfpt_research_contracts. **[E]** AUDIT OK, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt_2 “ambient closure” section now states

that the cutoff of its S_{rel} is the seam KMS weight ($f_2/f_0=1$, v259) and D_{rel} the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGE0.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGE0.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)

- **v258_dirac_covariance_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, $\beta=1$) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ .
- **v259_modular_cutoff_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff f is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa \approx 1.3$ pins $c_{PS}/c_{\text{grav}} \approx 1.7$).
- **v260_k3_kummer_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T_{\tau=i}^2/(z \mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1 , even, signature $(3, 19)$); [E] the seam is $T^2/(z \mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes = $\dim S^+$ (v197), $16=4 \times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 “glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.

- **v261_modular_spectral_closure.py** (QFT.MSC.01) is the assembly capstone. The whole QFT round (v238–v260) is certified as *one* relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6 \log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (AUDIT OK).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257_quasiorient_modpd.py** (PS.NCG.ORIENT.02) corrects the slightly too-generous **v256 typing**. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); [E] *quasi-orientability* holds (the left/right A -irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- L/R -neutrino case); [C] modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254_inner_fluctuations.py** (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation. The inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1, 2)_{1/2}$ — no junk, no triplet, no $(10, 1, 3)$ /126. [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E] σ is an SM singlet $(1, 1, 0)$ with $B-L = 2$. [C] $\sigma = \text{scalaron}$.

- **v255_spectral_action_expansion.py (PS.SPECACT.01) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: [E] structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl² + R^2 +Higgs-quartic, $N = 96$); [E] $b/a^2 = 1/3$ (top-dominated); [E] $\sin^2 \theta_W = 3/8$; [E] the R^2 spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic $\rightarrow \sim 125$ GeV with σ ; [C] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ explicit; [O] exact decimal needs the cutoff moments.
- **v256_orientability_poincare.py (PS.NCG.ORIENT.01) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null $\sim (1, 1, -1)$ on the SM K-theory $\mathbb{Z}^3$; [E] the orientability local condition $[\gamma_F, \pi(a)] = 0$. [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, v252–v253)

- **v252_full_finite_triple.py (PS.DIRAC.02) advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle $\oplus$ antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2=+1$, $JD=DJ$), grading ($\gamma^2=1$, $[\gamma, a]=0$, $J\gamma=-\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}}=m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **v253_kappa_scalaron_ps.py (PS.KAPPA.01) discharges residual 6(b) of v249.** κ in $M_{PS} = \kappa M_s$ ($M_s = \kappa_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ σ — the CCvS Pati–Salam mechanism, v251)

- **v251_first_order_sigma.py (PS.DIRAC.01) advances the Dirac contract v250 from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2=+1$, $\gamma^2=1$, $D\gamma=-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R \nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ . Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] σ = scalaron (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** v247_e8_branching_no126.py (PS.E8BRANCH.01): the E_8 hull branches under $SO(10) \times SU(4)$ as $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$, so the $SO(10)$ content is exactly

$\{1, 10, 16, 45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and $10+16+45$ is E_8 -supplied (only one $45 \Rightarrow$ proton-marginal). `v248_ps_algebra.py` (PS.ALGEBRA.01): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (PS.RGTEST.01): the two-step $PS \rightarrow SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0\text{--}1.7$. `v250_dirac_triple_contract.py` (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ($H_F = 3 \times 16$, the seven NCG axioms, σ = scalaron) that would turn “carrier is gauged” from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- `v246_unification_data_crosscheck.py` (QFT4D.RGTEST.01) — **the first hard data confrontation of the spectral-action prediction**. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}\text{--}10^{17}$ GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed **[X]/[O]**: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- `v245_spectral_matter_content.py` (QFT4D.MATTER.01) **discharges the computable core of the 4d contract v244**. The carrier half-spinor (the $SO(10)$ 16) is verified, in exact rational arithmetic, to be one Standard-Model generation: **16** = $SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions + ν_R), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2 \theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3} g_1$), and the Higgs is the $(1, 2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of CONTRACT.QFT4D.01 is now **[E]**; only the seam $\rightarrow$ Dirac realisation and the run-down couplings stay **[O]**. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238)**. `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{OS} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \rightarrow 4 \rightarrow 1$ (v235/v237). `v242_spin_statistics_modular.py`

(DHR.MODULAR.01): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the `v238` mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is `v36`; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, `v238`)

- **New `v238_modular_lindblad_dynamics.py` (DYN.SEMIGROUP.01, cluster registry).** A companion to the modular round (`v196/v198`) and the recovery code (`v221/v64`) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the `v198` state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel’s generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (`v64`) — the static recovery bound (`v221`) and the dynamical mass gap (`v64`) are one number, one exponentiated. **[E]** the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real ≤ 0 spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (`v198`) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data_limited, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ($w_{\text{in}} = -1$) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) **gravastar-compactness** — the Nariai $Q_{\text{geom}} = 3/8$ equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C} = 3/8$ (shared de Sitter endpoint $1/2$); since $1/3 < 3/8 < 4/9 < 1/2$ this is a light-trapping ECO, yielding an echo template (round-trip delay ~ 0.7 ms at $62 M_\odot$, amplitude $\leq (2/3)^6$) — a **[C]** structural echo, no $\mathcal{C} \leftrightarrow Q_{\text{geom}}$ map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES $\sim 3.3\sigma$ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **η_B — full BDP Boltzmann ODE solve confirms the route at the frozen scale** (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, **[C]**). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10} , with *no* free M_R dial. This moves the leptogenesis route from “solve pending” to a consistent **[C]** readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.

- **The Petz recovery map and the rank-one baby universe, realised numerically** (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221). The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected $\lambda=1$ (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain** (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts` `search-targets`). The Stage-1 machinery (Kerr-ringdown subtraction $\rightarrow$ matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`).** The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with $c = 8$ ” ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$ glue realised at Lie level (v143) and its odd sectors twisted/Ramond ($E_8 = 120_{\text{NS}} + 128_{\text{R}}$, v148), while the $q(A_3)$ normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ($120+128 = 248$; glue index $= |\mu_4| = N_{\text{fam}}+1 = 4$); status stays [C]/[O] (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/ (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet** (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [E]/[C]). Sharper than v220/v225 (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j=0$ Eisenstein CM unit): $\delta_{\text{CKM}}^{\text{leading}} = \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$. The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of

Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the v88 misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [C] (CP.MU6.01).

- **The seam as the E_8 Kleinian singularity (v232_e8_kleinian_seam, origin_theory, [E]/[C]).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise QGEO.REALIZE.01 a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}} + N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2) -curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1=0$, $\pi_1=2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8 ; TOP0.E8.01, bedrock stays [O]).
- **CP is the universal family/triality phase, only sheet-split (v233_cp_triality_phase, tfpt_5_redteam, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the $2/3$ cusp weight). Since $\rho^4 = \rho^1 \rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (v234_seam_holomorphy_selection, tfpt_research_contracts, [E]/[O]).** The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing E_8 : (i) holomorphy (μ -index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8 ; and holomorphic $c=8=g_{\text{car}}+N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8 . So P2, G_{net} , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is “free bulk $\Rightarrow$ holomorphic boundary” (GATE.HOLO.01).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ (v235_seam_chern_simons, tfpt_research_contracts, [E]/[O]).** A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K -matrix with $\#\text{anyons} = |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|=1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5 \oplus A_3$ (det 16, 16 anyons) $\rightarrow SO(16)_1 = D_8$ (det 4) $\rightarrow (E_8)_1$ (det 1, holomorphic) — all rank 8, $c=8=g_{\text{car}}+N_{\text{fam}}$ — so holomorphic = E_8 = the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order- $|\mu_4|$ Lagrangian glue (det 16 $\rightarrow$ 1, vs. a partial order-2 isotropic $\rightarrow 4=D_8$). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem (GATE.HOLO.02).
- **The $(2, 3, 5)$ Brieskorn singularity is the one generator of the skeleton (v236_brieskorn_capstone, origin_theory, [E]).** Capstone of the icosahedral round: the singularity $x^2+y^3+z^5$, whose exponents are the atoms ($|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$, has Milnor number $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ (a *fifth* origin of the “8”), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8 , link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h \rangle = \mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (TOP0.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam is SRE (v237_seam_sre_closure, tfpt_research_contracts, [E]/[O]).** Sharpens GATE.HOLO.02's residual from definitional to falsifiable: the K -matrix ground-state

degeneracy on a genus- g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so “no topological degeneracy on any closed surface” $\Leftrightarrow \det K=1$; among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2, 3, 5\}$ (v219_icosahedral_mckay, origin_theory, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$, sum $30=h(E_8)$, sum of squares $120=|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex) (v222_cm_norm_duality, origin_theory, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j=1728$) gives $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3+2i)=13=\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j=0$) gives $N(3+2\omega)=7=\text{scalaron}$. The $(3, 2)$ split generates $(5, 6, 7, 13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (v223_coxeter_totative_clock, origin_theory, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7 \rangle = \{1, 7, 13, 19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h)=\text{rank}$ and $h=2 \cdot 3 \cdot 5$ (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227_degree_exponent_channel_split, tfpt_5_redteam, [E]).** Exponents sum $120=|R^+(E_8)|$ (magnitude channel), degrees sum $128=\text{rank} \cdot \dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228_rr_index_gate, origin_theory, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D=4=|\mu_4|$ on $\mathbb{P}^1$ forces $h^0=5=g_{\text{car}}$, $\text{rank } H_1=3=N_{\text{fam}}$, $h^0+H_1=8=\text{rank } E_8$, $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGEO.RR.01).
- **The seam as a finite recoverability code (v221_seam_qecc, origin_theory, [E]/[C]).** Code dimension $\dim S^+=16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1, (2/3)^6, (1/3)^6\}$, recovery bound $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading $[C]$, gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220_cp_hexagonal_modulus, tfpt_5_redteam, [E]/[C]).** The seam deck is the square modulus $j=1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j=0$ ($\mathbb{Z}/6$, $\arg \rho = \pi/3$). The frozen $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$ has leading term $\pi/3 = \arg \rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (CP.MOD.01).
- **The dual normal frame (d, n) with CP as orientation (v225_dual_normal_frame, tfpt_5_redteam, [E]/[C]).** $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$ Nariai; $n = (5, -9, 6)$ reads $\det$ on the first-generation column; $\det(\mathbf{1}, d, n) = 21 = 3 \cdot 7$; $\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$ — CP as orientation (DUAL.FRAME.01).

- F_{transfer} as one path on the Sheet Diamond (v224_diamond_ftransfer_path, tfpt_2_standard_model, [E]/[C]). $M(s, t) = R + Q \text{diag}(s, t, t)$: K, C, F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps (1, 8, 10), (1, 8, 16); the Koide source $\rightarrow$ pole is the 1-D projection (FTR.PATH.01).
- The charged leptons as an étale Frobenius algebra (v229_lepton_frobenius_algebra, tfpt_2_standard_model, [E]/[C]). $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$, product $\frac{32}{9}$; $A = \mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (LEP.FROB.01).
- The center budget (7, 11, 13) as three local norms (v230_center_budget_norms, tfpt_2_standard_model, [E]). $C = R + Q \text{diag}(1, 0, 0)$ row sums (7, 11, 13): $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$ (hex), $13 = N_{\mathbb{Z}[i]}(3 + 2i)$ (square), $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- The Sheet Diamond is a discrete geometry with two axes (v218_diamond_axis_geometry, tfpt_2_standard_model, [E]). Sharpens v94/v95 with three exact [E] lemmas (no new numbers). (1) **Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C + xU) = 14 + 6x$ (slope $6 = |R^+(A_3)|$) and quadratic on the sheet axis $\det(C + yV) = 14 + 14y + 4y^2$ (2nd difference $8 = \text{rank } E_8$); the anchor block has $\det B(C + xU) = 2 \det(C + xU)$ and sheet curvature $6 = |R^+(A_3)|$ — two curvatures $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$. (2) **Plücker transfer ladder:** $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$ lifts in steps $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$ then $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$ — the decuple then the full generation (identity [E], source $\rightarrow$ pole reading [C]). (3) **Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2 \text{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3 \cdot 5 \cdot 7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28 = 2 \dim G_2$, $52 = \dim F_4$ stay audit-only, not spine, as in v94/v95); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- The four marks emerge from Gauss–Bonnet — count derived, not assumed (v216_marks_gauss_bonnet, tfpt_research_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), $\text{rank } H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling v217_marks_emergence_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension $\rightarrow 248/248$); v217 numerical (Python-only). Suite $\rightarrow$ 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- QGEO.SYM.01 as a kill-test, not a proof-promise (v215_seam_deck_killtest, tfpt_research_contracts, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry

cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4 -deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0 \mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f \mathbb{Z}_4$ -invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side **[E]** (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 **[O]** (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via v214/v54/v56). Suite $\rightarrow$ 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01 (v214_seam_pillowcase, tfpt_research_contracts, [E]/[O]; ledger QGEO.PILLOW.01).** The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2, 2, 2, 2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the v201 “flat away from the marks”. **(2)** New link: $\text{cross-ratio}(\mu_4) = 2$ (v168) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ j -invariant/CM arithmetic); Klein- J values mpmath-numerical. Suite $\rightarrow$ 214 modules.

2026-06-15 (the F_{transfer} transfer functor contract CONTRACT.F.01)

- **F_{transfer} formalised as a typed functor with four axioms (v213_ftransfer_functor, tfpt_research_contracts, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4 -deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\|\text{Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity ($\text{spec } T$ positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are F_{pole} (v183), $F_{\text{Boltzmann}}$ (v212), F_{relic} (v211), F_{QCD} (v164). CONTRACT.F.01 is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211_axion_spine_angle, tfpt_4_frontier, [C]).** A more *robust* misalignment angle than the determinant-line hilltop

$\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite- T solver reaches Ω_{DM} at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.

- **Leptogenesis scalaron-decouple branch (v212_leptogenesis_decouple, tfpt_4_frontier, [C]).** Both Boltzmann inputs share the decouple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than v184's $M_R \varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10} . The coupling mechanism is posited not derived, so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised (CohomologyGrading.lean, ledger FORM.QGEO.03, [O]).** A new module machine-proves the COHO node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ ($k=1, 2, 3$) have pullback character $\rho^* \omega_k = i^k \omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since $i^4=1$), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3) =$ the A_3 exponents $= \text{Spec}(Q_+)$, rank $3 = N_{\text{fam}}$. The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^* \omega_1 = \omega_3$, $\sigma^* \omega_2 = \omega_2$, $\sigma^* \omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in AuditContract.lean. The geometric COHO/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised (Möbius Uniformisation.lean, ledger FORM.QGEO.02, [O]).** A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the clock $\rho : z \mapsto iz$ has $\rho^4 = \text{id}$ and $\rho^2 \neq \text{id}$ (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma \rho \sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes 1, -1 , swaps $i, -i$); and the multiplier classification “ $z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in AuditContract.lean. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised (SeamDeckClosure.lean, ledger FORM.QGEO.01, [O]).** A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no sorry/admit, no domain axioms — #print axioms = propext, Classical.choice, Quot.sound only): (i) the 4th-root character orthogonality $\sum_{j < 4} \zeta^j =$ (if $\zeta=1$ then 4 else 0) for $\zeta^4=1$ (so a μ_4 -mark sum is supported only on modes $\equiv 0 \pmod{4}$); (ii) the clock generator $-i$ has order 4; (iii) markLocal_blockDiagonal: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed structure SeamDeckPremise (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as CalderonProjector encodes the Paper-1 analytic input; (v) SeamDeckPremise.clock_invariant: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise*

(the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (v210_mark_local_dtn, tfpt_research_contracts, [O]).** A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (v204_muon_seam_g2, tfpt_4_frontier, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.
- **An independent gravitational $3/4$ (v205_xi_threequarter, tfpt_4_frontier, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative H_0 from the Λ branch (v206_h0_lambda_branch, tfpt_4_frontier, [C]).** $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5\text{--}67.1$ km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207_asymptotic_safety, tfpt_4_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric} . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208_bh_thermodynamics, tfpt_horizon_readouts, [E]/[C]).** Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209_bh_regular_core, tfpt_horizon_readouts, [C]).** No singularity ($\varphi \rightarrow \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt_3, [O]).** Added a paragraph to the “Möbius origin of π ” section: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via

stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2, \mathbb{C})$ rotation (v180) and the $\text{RP} \rightarrow \text{OS}$ causal cone is the same Lorentz structure — so “Möbius origin of π ” and “one Lorentzian cone” are two faces of one boundary $\text{PSL}(2, \mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to c_3 (tfpt_4_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\text{top}} = 48c_3^4$ that fixes the α -kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\text{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude $Q_e Q_m$ is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]:** the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C]).** The closed TFPT CKM point ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\text{BR}(K^+ \rightarrow \pi^+ \nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\text{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test [X]:** a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O]).** The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4 -character-block-diagonal iff f is $\mathbb{Z}_4$ -invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4 -mark sum $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$

has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g , hence is automatically $\mathbb{Z}_4$ -invariant ([E]). So the μ_4 -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGEO.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion_relic/relic_solver.py, [C]).** The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170^\circ$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6\text{--}2.3$ (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120\text{--}130^\circ$ or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is $O(1)$ -uncertain, so a full finite- T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite- T solve confirms it (axion_relic/full_finiteT_solve.py).** A converged nonlinear misalignment solve ($\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170^\circ$ — $\sim 5.5\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106^\circ$. The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199_seam_state_invariance.py [E]/[O] (claim QGEO.STATE.01, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4 -character classes $\{n=r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200_seam_variational_scan.py [C]/[O] (claim QGEO.VARI.02, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4 ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the F_{transfer} four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb_tfpt_v1.yaml) and its status semantics already exist.* Suite now 199 modules, highest ID v200; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198_modular_commutator_reduction.py** [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$ and the clock $\rho: z \mapsto iz = \text{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω) , so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4 -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is μ_4 -invariant” ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in **tftpt_research_contracts**; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **v195_marks_lefschetz_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited): $\rho: z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ D_4 marks” to the milder premise “ H^1 carries the A_3 -exponent rep” — less circular, still [O]. Cited in **tftpt_research_contracts**.
- **v196_seam_energy_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the QGEO.SYM.01 conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4 -breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in **tftpt_research_contracts**.
- **v197_rr_carrier_clifford_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$. [E] arithmetic, [C] identification (as v189). Cited in **origin_theory**.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194_raw_seam_dtn_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] the residual relocates: the full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow

of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193_qgeo_energy_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (QGEO.COHOH.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7)**. `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189_riemann_roch_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already QGEO.COHOH.01) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by rank(D_5) = 5. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **v190_nariai_entropy_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191_universal_branch_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known $2/3$ ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192_qgeo_energy_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). run_all.py green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **v184_etaB_anchored_boltzmann.py [C] (claim FR.ETAB.03).** An honest test of the external-review η_B proposal ($M_1 = M_R \varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{P1}$ both miss it (c_3/φ_0 -powers off $\sim 75\%/ \sim 40\%$), so $M_1 = M_R \varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in tfpt_4_frontier.
- **v185_axion_relic_solver.py [C] (claim FR.DM.03).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ($m_a \approx 23.8 \mu\text{eV}$) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim 21\%$ at $\theta \sim O(1)$), so the $\theta_i \approx 170^\circ$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in tfpt_4_frontier.
- **v187_ftransfer_laws.py [E] (claim FR.TRANSFER.01).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B , axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in tfpt_4_frontier.
- **v188_frontier_wording_guard.py [E] (claim AUDIT.WORDING.01).** A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording $\rightarrow$ v184 [C]; the old “open relic scale” dark-matter line $\rightarrow$ v185; the old “near, not exact” Koide line $\rightarrow$ v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the tfpt_4_frontier Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, v_{geo} as unit gauge, G_{net} a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183_koide_f_corner_transfer.py** [E]/[C] (**claim FR.KOIDE.07**). An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an F_{transfer} special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present**: the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182_reviewer_residual_map.py** [E]/[C] (**claim REVIEW.MAP.01**). A careful external review raised four concerns (A mapping $2D \rightarrow 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv \text{QGEO.SYM.01}$), D (Möbius forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer} , v_{geo} , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ($\sin^2 \theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.IS0.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single

source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.

- **G_{net} typed in three explicit levels** (website “What is open” card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~ 181 scripts (foundations $\rightarrow$ SM readouts $\rightarrow$ seam = horizon $\rightarrow$ horizon/flux geometry $\rightarrow$ R1–R5/premise-(A) $\rightarrow$ PyR@TE cross-checks $\rightarrow$ AQFT bridges $\rightarrow$ AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows `AX.P1.01/AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181_clock_is_conformal_symmetry.py [E]/[O] (claim QGEO.SYM.01).** The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the μ_4 -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain **REALIZE(v175)** → $\dots$ → **SYM.01(v181)** ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, **tfpt_5_redteam** Target A, the **tfpt_research_contracts** central-theorem keybox, **origin_theory**; and on the website **OpenGates**, **VerificationDag**, **papers.ts** (Target A $\times 2$, G_{net}), the **glossary** G_{net} entry and the **faq** live-residual answer — all now end at QGEO.SYM.01.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180_clock_is_möbius.py [E]/[O] (claim QGEO.ISO.01).** Cracks QGEO.CONF.01 one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, *non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) *isometry* $\Rightarrow$ *conformal* $\Rightarrow$ *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). Hence QGEO.CONF.01 is *implied* by the milder premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in **tfpt_research_contracts**; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **v179_conformal_realisation.py [E]/[O] (claim QGEO.CONF.01).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so “genus-0” is the same $\chi = 2$ that gives the “8” in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). **KERNEL** then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So

the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism”. The two doors of `v177/v178` are one geometric realisation, left honestly `[O]`. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations `v178` — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` `[E]/[O]` (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite `[E]` core of each and reduces each to *one sharper* premise. *Marks core* `[E]`: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ , the marks are forced to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so `MARKS` reduces to the single topological/clock premise. *Kernel core* `[E]`: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (`v162`) and the unique residue-normalised basis (`v177`) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So `KERNEL` reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/`v156`; no drift `v158`). Net: neither obligation closed; both sharpened to milder premises with their finite cores `[E]`. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree `v177` — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` `[E]/[O]` (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`). A second external residual-class audit pointed out that the `v176` statement takes the marks/ D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` `[E]` — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful- D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening `v168`’s cross-ratio to the full uniformisation); `COHOM` `[E]` — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; `MODULE` `[E]` — a *unique* μ_4 -equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); `NET` `[E]` (`v154/v175`). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4 -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via `v168/v137/v162/v154/v175`). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem `v176` — the one central proof obligation)

- `v176_seam_collar_realisation.py` `[E]/[O]` (claim `QGEO.REALIZE.02`). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented

one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c=8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4 -equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$ (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \rightarrow \frac{53}{54}u$ source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT_RECORD_ID was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175_net_existence_full_cone.py** [E]/[O] (claim QFT.NETRP.01). The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m* : the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\text{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174_seam_fock_readings.py** [E] (**claim QFT.FOCK.01**). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four μ_4 -corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL ($\deg \leq 2$) keeps $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in **tfpt_2** (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170_diamond_slice_compression.py** [E] (**claim E8.SLICE.01**). E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3, 4, 6, 10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3, 4, 8, 14, 20, 32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in **tfpt_3** (slice atlas). Exact; reading is audit (anti-numerology).
- **v171_os_moment_cluster.py** [E]/[C] (**claim QFT.OSMOMENT.01**). Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster $729/665$, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$. Cited in **tfpt_research_contracts** (G5).
- **v172_trace_anomaly_seed.py** [E] (**claim SEED.ANOMALY.01**). The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 = \text{scalaron exponent } 48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ factorisation. Cited in **tfpt_2** (seed).
- **v173_pfaffian_cp_car.py** [E] (**claim FLAV.STRONGCP.02**). Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5 -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in **tfpt_2** (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo**. Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Also fixed **build.sh zenodo** on macOS bash 3.2 (empty-array expansion under **set -u**).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **v168_qgeo_rigidity.py** [E]/[C] (**claims QGEO.RIGID.01, QGEO.REALIZE.01**). Hardens the *finite* half of the one remaining Tier-1 hinge **GATE.QGEO** (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz / (z^4 - 1)$ ($k = 1, 2, 3$) are μ_4 -eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4 , four parabolic marks and this μ_4 -decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ — the QGEO Rigidity Theorem (**QGEO.RIGID.01**, [E]). The seam-collar realisation stays the one premise (**QGEO.REALIZE.01**, [C]). Mirrored in the Wolfram extension (226/226). Cited in **tfpt_research_contracts** (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **v169_etaB_boltzmann_interface.py** [C] (**claim FR.ETAB.02**). Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240^\circ$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10} , untuned) — the route is *viable*. But M_1 /washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in **tfpt_4_frontier** (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166_higgs_free_seam.py** [E]/[C] (**claim HIGGS.FREESEAM.01**). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral $c=8$ fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 at one loop to ~ 0.002). The double condition predicts $m_H \sim 129\text{--}134$ GeV; the same condition at the scalaron scale gives ~ 107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in **tfpt_4_frontier**.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ , reduced to top-only (**verification/pyrate/sm_higgs_betas.txt**). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163_rg_stability_flavor.py** [E]/[C] (**claim FLAV.RGSTAB.01**). Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2 -suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2} Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2 -driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in **tfpt_2** (solar/reactor angle box).
- **v164_qcd_scale.py** [E]/[O] (**claim QCD.LAMBDA.01**). The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop,

n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/ Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.

- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β ’s, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module v159_pyrate_gauge_crosscheck.py [E]/[C] (claim EM.BUDGET.02).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}^{2g_{\text{car}}-2} + 1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_{transfer} + irreducible core, v167)

- **New module v167_inventory_post_closure.py [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (GATE.METRIC.16–19) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1 = N_{\text{fam}} = 3$); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer} , one functor with four typed conditional interfaces (Koide, η_B , axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger SYNTH.INVENTORY.03). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A_2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module v165_premise_a_closure.py [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 (net existence) closes by assembly of estab-

lished mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$, $248=120+128$, character $E_4/\eta^8=j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via $DtN = |k|$ (v156/v157). R_1 (**seam clock**) **reduces to GATE.QGEO** [C]: a μ_4 -equivariant flat gapped generator is *unique* ($= A_0^*$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification “seam boundary $= \mathbb{P} \setminus \mu_4$ ” = **GATE.QGEO** (cohomology established, v137, $b_1=3=N_{\text{fam}}$). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) = (A_2 assembly) + ($R_1=\text{GATE.QGEO}$) + the $\{\pi, v_{\text{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger **GATE.METRIC.19**). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module v162_seam_transport_identification.py** [E]/[O]. Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4 -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$ for $U = \text{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2=6=|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of $\omega_k=|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)

- **New module v161_one_particle_reduction.py** [E]/[O]. *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\text{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m=0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo

class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger GATE.METRIC.17). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ — verified by the *signed* set-partition/Pfaffian moment $\leftrightarrow$ cumulant inversion on a concrete pure Majorana covariance (the v113 “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap}>0$, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}=1820$ and $\binom{16}{6}=8008$ higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection \times \{YYYY-MM-DD...\}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral $c=8$ fixed point is provably stable (v158, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current $h=1$, the quartic $h=2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the $h=1$ currents are chiral (v157, not (1,1)-marginal), and the lowest interaction (quartic, $h=2$) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the $b_1=41/10$ cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; `contracts / origin_theory / next.txt / website` moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*

$|k|$ (Lee–Uhlmann — order-1 Ψ DO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic $c=8$ theory is *rigid* — every field has $\bar{h}=0$, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving $\mu=1$ ($248=120_{\text{NS}}+128_{\text{R}}$). With $(E_8)_1$ unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ($\Delta=6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5+3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q -series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual*: exact given (i) the free-bulk premise (DtN = $|k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the $248=120+128$ split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- G_{net} ’s last premise reduces to a single shared physical premise (v155, Python-only). The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression PTP of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net} , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- v_{geo} **re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals

collapse onto it ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π } (ledger `ANCHOR.VGEO.02`).

- **G_{net} stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has $[B:A]=|L|=4$, $c=5+3=8$, $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ ($[\mathbf{E}]/[\mathbf{C}]$), and F_{transfer} as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/origin_theory/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d=2$ that ratio *is* the induced Newton constant $1/G$ (v68). So v_{geo} (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π }, and `SEAM.THEOREM.01`’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin_theory/next.txt/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO_TOKEN secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, tex-artefacts/, figures/, verification/, experiments/; excludes _archive/, website/, all *.pdf, .cursor/). ~270 files / ~3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references; stale literal [A]/[I]/[L] markers in running text replaced by the 4-class [O]/[E].

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references in all three documents; the horizon “Scope/typing” marker line and a stale literal [A] in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` on failure, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón

jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. SEAM.THEOREM.01 narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger SEAM.EHMODEL.02).

- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin_theory/next.txt moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot 1$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0 -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and t -independent (Cheeger image sum $(N^2 - 1)/(12N)$); for a *gapped* operator the ζ -regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m / 12\pi) \int \sqrt{g} R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, origin_theory, contracts, next.txt, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 (4+5=9, \sigma \perp w_0(a))$ and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{PI}(K)\|_1$. R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z \mapsto iz, z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is $\det'$ -clean (v147).** Variance ratio $(1 - \alpha)$ (forced by norm = area) gives the Born-squared ratio $(1 - \alpha)^{p_2}$, the $\ln$ series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\text{NS}} + 128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.

- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{P1}$ vs M_{P1} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies” $\rightarrow$ “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + $c=8 \Leftrightarrow$ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net} : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A \Sigma$ (v97/v98) pairs the $Q_+=1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \rightarrow -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading = $Q_+ \circ \rho$. GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings $\rightarrow$ two (v142, frame integrality).** Integer covectors in the $(1, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \pmod{11}$; the index is $\|\text{Pl}(K)\|_1$);

with $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$ the σ -pairing is forced $\equiv 0 \pmod{11}$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).

- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$ -graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1’s det-ratio step derived in-family (v144).** e_2 -rigidity of the traceless SdS cubic gives $r_{br_c} = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays `[C]` with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant $\rightarrow$ EH step (`SEAM.THEOREM.01`); the gate keeps its `[O]` typing untouched.
- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script $\rightarrow$ check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite $\leftrightarrow$ `run_all.py` $\leftrightarrow$ registry $\leftrightarrow$ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.

- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal [P] in `tfpt_4` (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$, A_s -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations

(e.g. $[I]/[L] \rightarrow [E]$, $[N]/[P] \rightarrow [E]/[C]$). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.

- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$ rows are made consistent with `v86/COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star=51.4$ ($n_s=0.9611$, $r=0.0045$), replacing the stale point values $N_\star=55$ (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed $[E] \rightarrow [C]$ (conditional on $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$), matching the other two tables and the ledger $[P]$ typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry author, date and `v5.2 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`).

`changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.

- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven **keyboxes** are reduced to the single headline “In one sentence” card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a **winbox** (a result, not a claim), and the explanatory *plausible/follows next* **winboxes** become prose. The loud boxes now carry only genuinely load-bearing callouts (one **keybox**, five **winboxes**, one **gapbox**) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only **keybox** (claim), **winbox** (result) and **gapbox** (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex` (`\input` by `tfpt_style`).** reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the $2/3$ *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (leading c_3 , fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result (0.3092 ± 0.0087 , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence

hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.

- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$); F_{transfer} is named as one missing functor with Koide/ η_B /axion/ m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv \text{tfpt}_N$). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\text{spec } A_0^*$); $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law $\rightarrow$ zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).

- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\text{rowsum}(V) = \text{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the A_0^* zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents and cusp weights $= \text{spec } A_0^*$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the $R4'$ residue is three pairings; the $H^1 \rightarrow$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the GATE.QGEO residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ~ 110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tftpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s, t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank = c) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue A_0^* (diag = $a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff$

$a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.

- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.